



BASICS OF SUSTAINABILITY

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Preface

This introduction to the foundations and challenges of sustainable development is aimed at students and lecturers dealing with the most pressing issues of our time from different disciplinary perspectives. It arose from the desire to promote a common understanding – across disciplinary boundaries – in teaching, in research, and in social practice.

The world as we know it today is characterized by a multitude of interwoven crises: Climate change, biodiversity loss, social inequalities, economic instability, and political polarization – in short, a polycrisis. These challenges are neither linear nor easy to solve. They call for a profound rethink of our economic, social, and ecological systems, as well as our actions and our idea of development and progress.

Against this background, this textbook sees itself as a shared learning space: it offers systematic access to central concepts, scientific perspectives, normative questions, and specific fields of action for sustainable development. The aim is not only to impart knowledge, but also to promote the ability to reflect, think critically, and be creative.

This book was written in the context of the study programmes of the Centre for Development and Environment (CDE) at the University of Bern. It reflects the many years of experience in research and teaching on sustainable development and brings together contributions from various disciplines – from the environmental sciences, geography, and economics to sociology, ethics, and law.

Special thanks go to all the authors who have contributed to the creation of this book with their expertise, passion, and commitment, and to our translator (each Chapter was written in German and subsequently translated into English). We would also like to thank the students, whose critical questions and perspectives are a key driver for the further development of the content. After all, sustainable development is not only a scientific project, but above all a social project – and this begins with education.

We hope that this book will provide a sound basis for your own engagement with sustainability – and encourage you to take responsibility and play an active role in shaping a fair and sustainable world.

How to cite

Bader, C. (Ed.). (2025). Basics of Sustainability (Version 1.3) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

Citing specific chapters:

Chapter 1:

Bader, C. (2025). Understanding and concepts of sustainability. In C. Bader (Ed.), Basics of Sustainability (Version 1.0) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

Chapter 2:

Bader, C. (2025). Approaches to sustainability. In C. Bader (Ed.), Basics of Sustainability (Version 1.0) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

Chapter 3:

Fitz, J., & Bader, C. (2025). Environmental Sustainability. In C. Bader (Ed.), Basics of Sustainability (Version 1.0) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

Chapter 4:

Poelsma, F., & Bader, C. (2025). Social Sustainability. In C. Bader (Ed.), Basics of Sustainability (Version 1.0) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

Chapter 5:

Bader, C., & Bezzola, N. (2025). Economic Sustainability. In C. Bader (Ed.), Basics of Sustainability (Version 1.0) [Online book]. CDE, University of Bern. Retrieved from <https://cdeteaching.github.io/Basics-of-sustainability/>

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Reading guide

Definitions

Examples and reflections

Further readings

Introduction



Figure 1: “Wrapped Globe (Eurasian Hemisphere)” by Christo (2019)

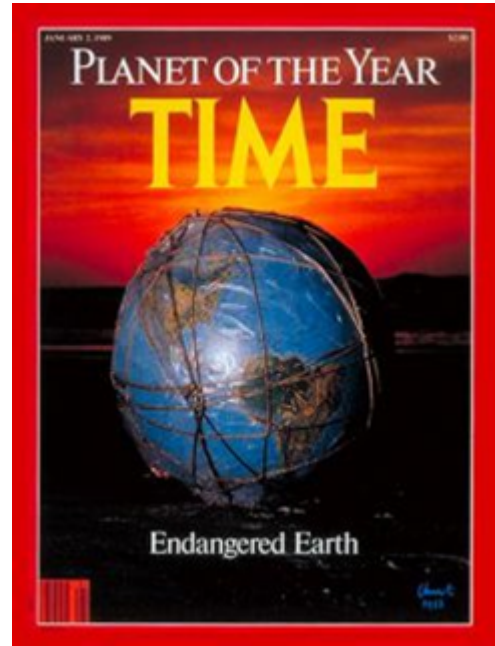


Figure 2: “Planet of the Year” Times Magazine Cover (1989)

Amid the mounting challenges of sustainable development, Christo and Jeanne-Claude’s “Wrapped Globe” is a powerful symbol of humanity’s responsibility towards our planet and its resources. The artwork depicts a globe wrapped in transparent plastic and a filigree net. Meanwhile, in real life, the world is facing a “polycrisis” – the word used to describe the many serious crises our Earth is facing, including ecological crises, growing inequality, excessive national debt, and the effects of the Covid-19 pandemic, to name but a few. In a polycrisis, crises are increasingly intertwined and mutually reinforcing (Tooze 2022), and they are mainly caused by a structural dependence on growth (as measured by gross domestic product, GDP) (Hickel et al. 2022); a vicious circle of ever-increasing concentration of economic and political power in the hands of a few (Piketty 2014); and persistent inequalities between and within countries (Chancel et al. 2021; Milanović 2016). And yet, we keep striving for GDP growth in our society and our economy, in order to maintain and create jobs, finance our social security systems, secure tax revenues, and fulfil the needs of companies and industries that depend on growth to exist. As these expectations become increasingly unrealistic, the idea of decoupling economic growth from resource consumption has gained traction. However, there is no empirical evidence that doing so will achieve anywhere near the scale required to halt multidimensional ecological collapse (Hickel and Kallis 2020; Parrique et al. 2019; Wiedmann et al. 2020).

Living in a “full” world

The plastic cover and the net wrapped around Christo's globe thus represent the interconnectedness and interdependence of the Earth's various elements, and emphasize the need to maintain and preserve these relationships. A similar idea was described by the economist Herman H. Daly (2015), who put forth a concept of the “empty” and the “full” world. The empty world describes a situation in which human activities and resource use have only a minor impact on the environment. In this world, natural systems are still intact and untouched, and resources are sufficient to meet human needs. This contrasts with the full world, in which human activities and resource use overload the ecosystem and pollute the environment. Since at least the Second World War, we have pursued an industrialized society and a growth economy. As a result, we now live in a “full” world, where natural resources are scarce and the balance of ecosystems is under threat.

Daly's epiphany came in 1962 upon reading Rachel Carson's book, *Silent Spring*, which called for a life in harmony with nature. Daly was already sceptical about the hyper-individualism of economic models, and Carson's work highlighted the conflict between a growing economy and a fragile environment. Following a lecture by the economist Nicholas Georgescu-Roegen on his magnum opus, *The Entropy Law and the Economic Process* (1971), Daly adopted the idea that the economy was more like an hourglass than a pendulum, with valuable resources turning into waste and thus largely irreversibly lost.

There is no master plan to counter the polycrisis and make our economic and social system more sustainable. To contribute to sustainable development, we need to analyse the causes of symptoms such as biodiversity loss or global warming, and learn to understand how they are connected and how they interact. In our study programmes, we will therefore look at current problems and challenges such as global warming, pollution, biodiversity loss, social inequality, and economic disparities, to understand how they can be tackled at both the local and the global level.

The aim of this textbook is to encourage readers to think critically about the role of individuals, communities, businesses, and governments in the context of sustainable development – and thus to identify and pursue approaches to support sustainable development. In addition, we want to develop visions of the future, especially in the Master's study programmes, that make it possible to imagine a high quality of life in a sustainable modern age, and that make changing the present seem attractive rather than daunting. We want to envision a different food culture, a different economic system, a different type of land use, and a different way of building and living. To make progress towards sustainability, it will be critical to engage stakeholders and foster collaboration across sectors and disciplines. As this textbook will emphasize, education, communication, and participation will also be indispensable in shaping a more sustainable future.

With Christo's Wrapped Globe in mind, and Daly's understanding of the empty and the full world as a foundation, we will embark on a journey through the many aspects of sustainable development. We hope that you will find this journey both informative and inspiring, and that it will give you an understanding of both the urgency and the opportunities for sustainable development.

Finally, we hope that this textbook will inspire readers to reflect on their own roles and responsibilities in relation to sustainability, and that it will equip you to make a difference to ensure that the Earth remains a place worth living in for future generations. Only together can

we bring about the changes needed to create a sustainable, just, and environmentally friendly world.

Part I

Foundations

1 Understandings and Concepts of Sustainability

Where do the concept, mission statement, and guiding principles of sustainable development come from? How did the concept emerge, how did it evolve – and why? The following explanations provide an overview of the history of sustainability, the political background, and key conferences and documents. In short, what has made sustainability what it is today. It's about seeing the big picture. Because only those who know the past can assess the future.

Sustainability has been the buzzword of recent years, from science to politics and business to the media. Initially, the term has a positive connotation because it is associated with the long term, the durable, and many things are “sustainable” today: coffee, corporate philosophy, even tuna pizzas. These are the demands of a consumer society with a considerable – but inconsiderate – appetite for abundance. First thing in the morning, the shampoo removes our dandruff “sustainably”. Then we drive to work at a company that prides itself on its “sustainable corporate philosophy”, and over lunch we discuss sustainable investments. At home, we stick a tuna pizza in the oven – sustainably sourced and produced, of course, as it says on the box.

Today, the term sustainability is everywhere: it's used in connection with energy, mobility, building renovation, nutrition, population development, corporate environmental management, and climate protection. It's also used in art, culture, design, and advertising.

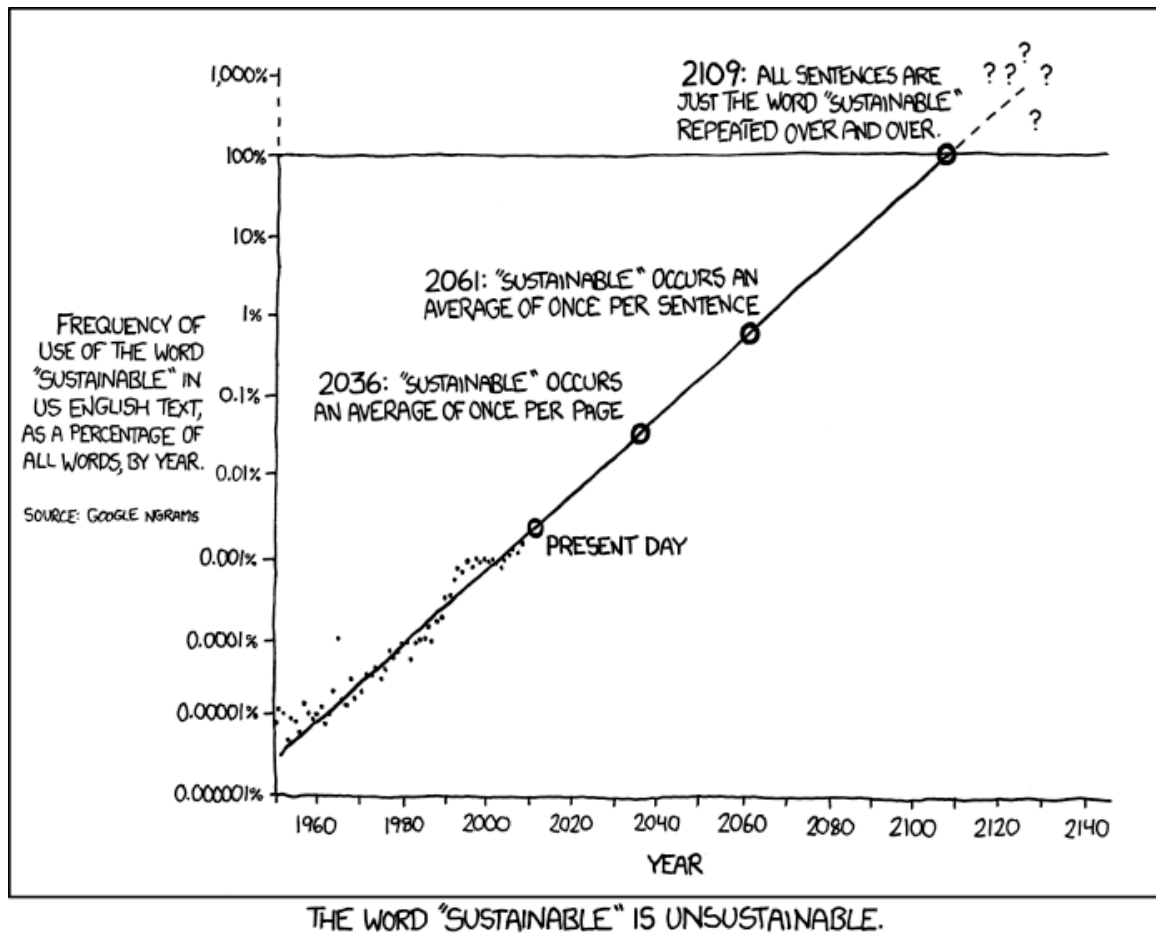


Figure 1.1: Frequency of use of the word "sustainable" in US English texts, as a percentage of all words per year, based on data from Google Ngrams. Source: <https://xkcd.com/1007/>.

Is the term sustainability so overused as to slowly become meaningless? Even if this is the case, we shouldn't abandon key terms like this lightly. For example, it's still important for companies to have a sustainability strategy, even if many such strategies are inadequate or amount to greenwashing. We need to clarify the meaning of "sustainability" and hold those who use it to account. And we should base our interpretation on science and historical developments. The historical precursors to the sustainability model explained in this textbook are:

- Start of the discussion about sustainability: Carlowitz's forest management principle of 1713
- Clash of economy and ecology First and second UN Development Decades (1960s and 1970s)
- The Limits to Growth (1972)
- Brundtland Report (1987)
- Rio Earth Summit 1992 Agenda 21 UN Millennium Development Goals (MDGs) (2000–2015)
- 2030 Agenda with the Sustainable Development Goals (SDGs) (2015–2030)

1.1 A normative concept

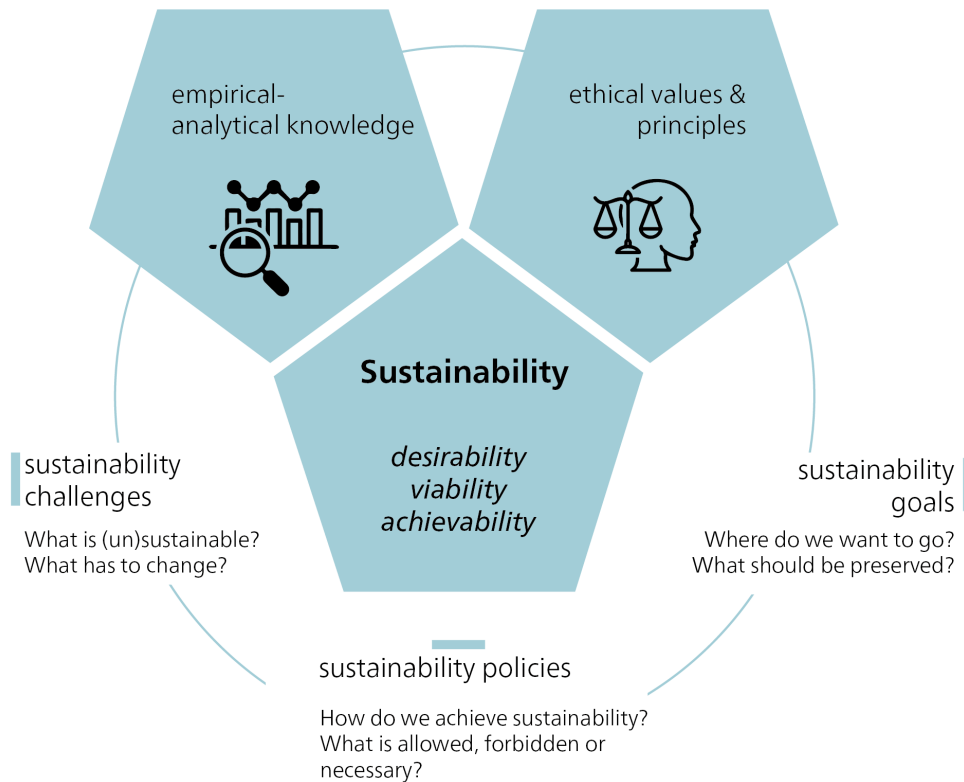


Figure 1.2: Sustainable Development as a normative concept. Source: Own illustration.

To achieve sustainability, we need **sustainable development**: a strategic process that requires changes in our socio-ecological systems and institutions. “Development” implies controlled improvement, and the ethical question of what exactly “better” means is key. The use of the term “development” has been criticized by some (e.g. Lang and Mokrani 2013; Kothari et al. 2019), as it is often equated with unbridled growth. The unbridled growth of our “ecological footprint” – the proportion of the biosphere used for human production, consumption, and waste – is unsustainable in the long term. Nonetheless, there are different interpretations of development, ranging from economic growth to improving quality of life. Sustainable development therefore remains a stimulating and controversial concept. Sustainability and sustainable development play a key role in today’s political discourse. The term **unsustainability** refers to conditions or developments that are considered negative, while **sustainability** represents a positive state.

The concept of sustainable development commits us to certain values and norms that define ethical goals and rules of behaviour. These values and norms shape our ideas of what we consider

a desirable or “positive” change. A neutral point of view is not possible, as our perceptions are shaped by a variety of influences. Our personal experiences, upbringing, social environment, and cultural background all influence how we see and interpret the world, including our perceptions of what “should be” and what actions “should be taken”. Ethics play a key role in determining how the current situation can be improved to achieve sustainability, by defining normative goals and limits. Sustainable development is therefore a conceptual framework that is strongly driven by ethical considerations.

It follows that sustainability challenges such as poverty reduction and climate change are ethical challenges. The identification of situations as sustainability problems (and therefore as “negative”) and the choice of solutions are based on ethical values. Sustainability goals are also based on ethical values, as well as on knowledge of cause-and-effect relationships. Sustainability issues are often referred to as **wicked problems**, because of their complex and pluralistic nature. Global warming, in particular, has been described as a **super wicked problem**, as finding solutions is urgent, political institutions are often inadequate, and decision-making processes suffer from short-sightedness.

1.2 Global challenges as wicked problems

Mike Hulme, author of *Why We Disagree About Climate Change* (2009), suggests that we view climate change as a “wicked problem” of enormous scale and likely longevity. This approach helps us see climate change not as a problem that needs to be solved, but as a condition in which we are directly involved. The categorization of problems as “wicked”, i.e. those that do not lend themselves to clear-cut solutions, originated with the planning theorists Horst Rittel and Melvin Webber (1973). They argued that planners have to deal with unpredictable human behaviour, and that some problems are too complicated to be solved completely. Some “wicked problems” may never be fully solved, but we can learn to cope with them and not let them dominate us.

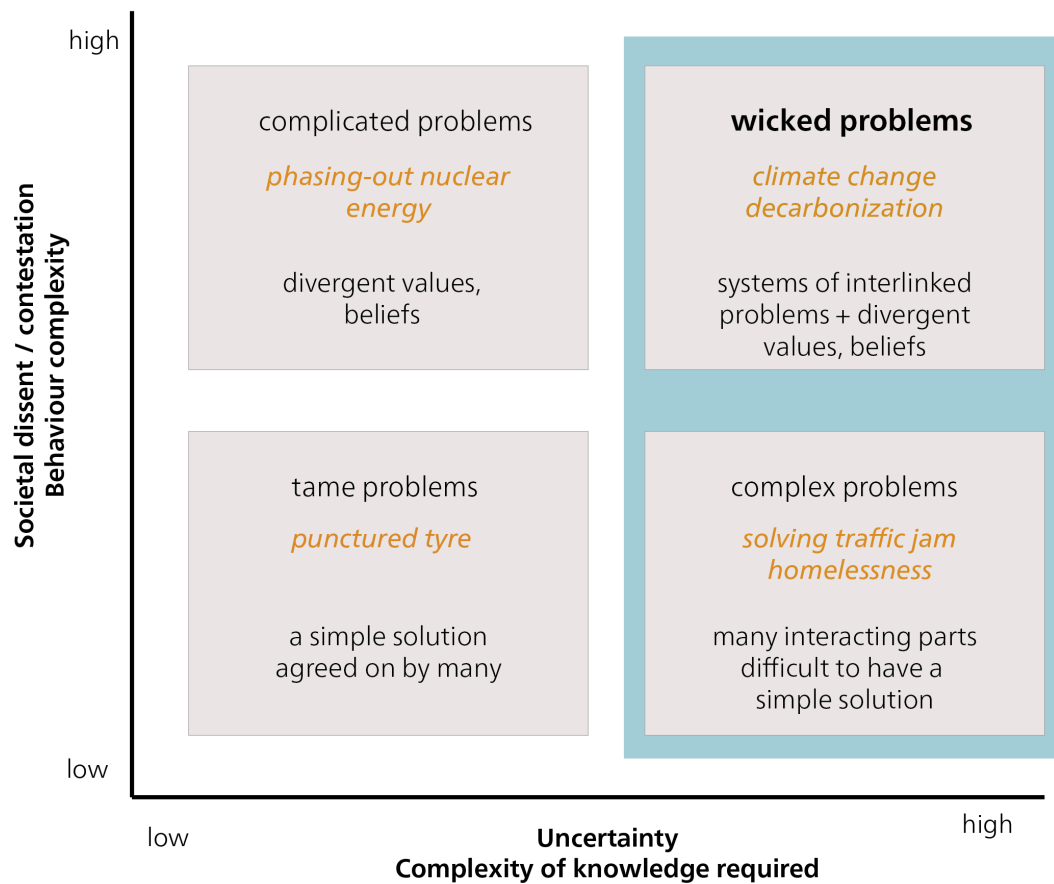


Figure 1.3: Sustainability challenges as wicked problems. Source: Own illustration based on Roth and Senge (1996).

Wicked problems

According to Bannink and Trommel (2019), wicked problems arise at the intersection of factual uncertainty and a heterogeneity of preferences and interests. Wicked problems are characterized by (Alford and Head 2017; Rittel and Webber 1973; Sediri et al. 2020):

- **Complexity:** Wicked problems are characterized by many interrelated factors and interactions. There are no clear cause-and-effect relationships, and changes in one area can have unforeseen effects in other areas.
- **Normative conflicts:** Wicked problems are perceived and interpreted differently by different stakeholders and interest groups. There is no clear definition or consensus on what exactly the problem is, or how it should be solved.
- **Interdisciplinarity:** Wicked problems require an interdisciplinary approach as they involve different topics, perspectives, and stakeholders. Finding a solution often requires the cooperation and coordination of different disciplines and experts.
- **Uncertainty:** Incorrect, missing, or inaccessible information about the problem situation and about the continuity of the values of the variables involved.
- **No definitive solution:** Wicked problems defy clear-cut solutions. They are dynamic, change over time, and require continuous adjustment and iterations.

Examples of wicked problems include climate change, poverty alleviation, global health, and sustainability. The complexity and interaction of different factors in these areas make it difficult to find simple and clear-cut solutions. Dealing with wicked problems requires a high degree of reflection, collaboration, and the use of systemic thinking methods.

How have we maneuvered ourselves into the current situation, where wicked problems pose such a challenge to sustainable development? In this textbook, we will analyse and learn to understand three key global challenges:

- the emergence of human-induced global warming;
- the persistence of global poverty and rising inequalities;
- and the threat of species extinction and overexploitation of natural resources.

The debate about human influence on global warming and climate change has intensified since the publications of the Intergovernmental Panel on Climate Change (IPCC). The driving force behind global warming is our dependence on oil and other fossil fuels for transport, (electricity) generation, agriculture, and many everyday products. Another wicked problem is global poverty. Although there have been global efforts to reduce poverty for decades, the globalized market economy often leads to increased poverty in certain regions, and the gap between rich and poor seems to be widening overall (Chancel et al. 2021). High-income countries have maintained their prosperity and living standards at the expense of others. These are “externalized costs”, and they include environmental degradation. For example, the wealth and living standards of high-income countries are largely responsible for the threat of species extinction and the overuse

of natural resources (Lessenich 2018). In addition to the above mentioned “big three” – climate change, poverty, and biodiversity loss – there are many other global sustainability challenges, such as deforestation, desertification, declining soil fertility, dwindling fish stocks, pollution, wars, conflicts, and increasing migration.

In analysing these comprehensive challenges we must also consider the temporal dimension. In 2004, graphs depicting socioeconomic and Earth system trends from 1750 to 2000 were first published, revealing a dramatic upsurge and aptly termed “The Great Acceleration” (Steffen et al. 2004).

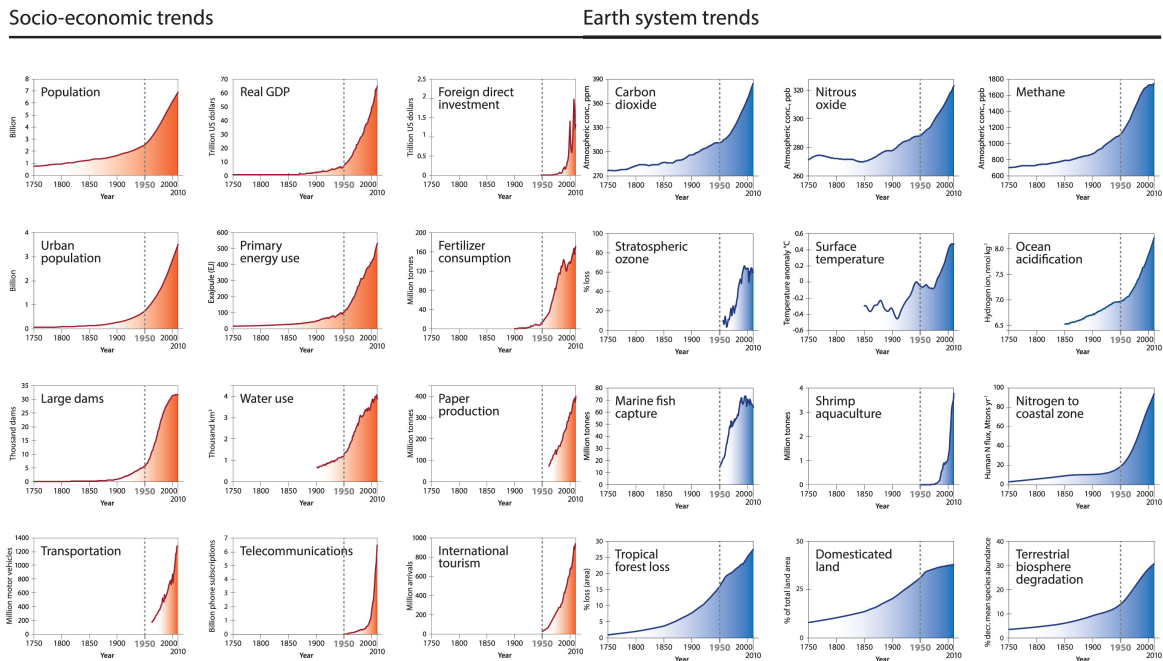


Figure 1.4: 12 socioeconomic and 12 Earth system trends from 1750 to today are a strong indication that the Earth system has entered a new state. Source: <https://futureearth.org/the-great-acceleration/>.

The Great Acceleration described the exponential growth dynamics that occurred as a result of the Industrial Revolution, which led to a surge in productivity in the mode of production and a significant increase in material wealth. At the same time, but to a much lesser extent, there was a massive increase in natural resource consumption and emissions. Socioeconomic growth went hand in hand with the acceleration of biophysical trends. Figure 1.4 describes some examples of important biophysical and socioeconomic indicators, all of which start to increase with the Industrial Revolution. From the middle of the 20th century, the trend towards exponential growth becomes apparent.

The Wheat and Chessboard Problem: an illustration of exponential growth

The Wheat and Chessboard Problem is a mathematical problem often used to illustrate the concept of exponential growth. In the story, a servant asks the king to fill every square on a chessboard with grains of wheat, doubling the number on each square. Growth starts slowly, but with each doubling, the number of grains increases exponentially. By the 50th square, the number of rice grains would be large enough to cover Berlin's 365-metre-high television tower on Alexanderplatz. This story illustrates the immense power of exponential growth and its impressive results. Understanding the concept of exponential growth is crucial to analysing phenomena such as population growth, technological progress, environmental change, and the spread of disease. It illustrates how even small changes or developments in a system can have a significant impact. History reminds us that we need to think carefully about how we manage such growth and the long-term consequences it can have.

Resource extraction has increased significantly in recent decades, from 22 billion tonnes in 1970 to 70 billion tonnes in 2010 (UNEP 2016). Global warming is another pressing issue. The latest IPCC report (2023) predicts an increase in global average temperature of between 1.5 and 5.8 degrees Celsius, depending on the scenario and future emissions. Another alarming phenomenon is deforestation. Every two seconds, an area of forest the size of a football field is cut down – an area the size of New York City every day. And then there is the decline in biodiversity, marked by the extinction of animal and plant species, which threatens the ecological diversity of our planet.

The Great Acceleration (Figure 1.4) describes the observable and measurable (negative) developments of key socioeconomic and biophysical indicators since 1950. The causes of these negative developments can be far removed from the place where the problems arise or become most evident. For example, job losses in one country may be caused by a multinational company's decision to relocate part of its operations to another country, in order to maximize profits. We cannot hope to fix such issues unless we understand how the system – in this case, the globalized economy – works. Similarly, we need to understand the Earth's global climate systems to imagine what the consequences of global warming might be in a particular area or region.

This is why many of the methods and concepts of sustainability science require systemic thinking. A systemic thinking approach often begins with brainstorming to create a “rich picture” of all the factors you need to consider in order to understand the current behaviour of the system in question. For example, in the case of a multinational company closing a particular business unit, these would be the factors that might influence the decision to close the unit rather than to continue or expand it. While an initial mapping may look overly complex or even messy, creating a linear flowchart can help clarify the flows of inputs and influences. This kind of mapping may already reveal opportunities to reassess the relevant influences and redesign the system to avoid unwanted outcomes. However, further work may be required to develop “conceptual models” of the system that identify unforeseen opportunities for improvement.

Incorporating systemic thinking into the methods and concepts of sustainability science enables us to tackle the complexity of problems and develop sound strategies for sustainable development. Systemic thinking thus provides an important basis for understanding and shaping the different understandings and concepts of sustainable development, as the next section discusses in more detail.

1.3 Sustainability: When did we start talking about it?

In 1713, Saxony's chief mining officer, Hans Carl von Carlowitz, demonstrated the necessity and possibility of "sustainable forestry" in a book entitled *Sylvicultura oeconomica, oder hauswirthliche Nachricht und Naturmässige Anweisung zur wilden Baum-Zucht* (*Sylvicultura oeconomica, A Guide to the Cultivation of Native Wild Trees*). Another sustainability pioneer was Duchess Anna Amalia, the mother of Duke Carl August. In the late 1700s, she initiated the world's first forestry reform, which was aimed at ensuring a continuous and steady supply of timber. At that time, Europe had an insatiable appetite for *materia prima*, or raw materials needed for building ships and houses, and for cooking and heating. This demand threatened to deplete resources and endanger their long-term survival.

In the 18th century, scholars in Germany as well as in other parts of Europe addressed the finite nature of natural resources. Unlike von Carlowitz, however, no one spoke about sustainability. One important aspect was the provision of food for a growing population. Before industrialization, economic growth was largely determined by nature as a factor of production, such as the number of mines or ships built, or the amount of land available for farming. However, Thomas Robert Malthus, a British economist, warned that food production would not be able to keep up with the rapid population growth following the Industrial Revolution. Malthus believed that the population would grow at an exponential rate, while food production would increase at a linear rate. Although Malthus's thesis did not come true in the dramatic way he predicted, his work is often regarded as the first systematic treatise on the limits to growth in a finite world.

For a time, Malthus's ideas fell out of favour, as unlimited growth seemed possible due to the Industrial Revolution and the rise of capitalism. Talk of natural limits didn't resurface until the mid-20th century, when a "neo-Malthusian" perspective re-emerged in the context of the Club of Rome (founded in 1968), debates on planetary boundaries, and critiques of growth. While Malthusianism traditionally focuses on external, physical limits to growth, the ecological economist, Giorgos Kallis, proposes a different approach. Kallis (2019) advocates personal moderation and voluntary restraint, an approach to life that has deep roots in Romanticism and ancient Greek philosophy. He argues for an inner limit to our wants, to free us from the economic assumption of scarcity and the associated obsession with growth. While acknowledging the reality of external limits, Kallis believes they should not dominate the environmental argument against limitless growth. Kallis therefore asks whether it makes sense to frame climate change primarily as a problem of external limits, of scarcity, or of a finite atmosphere that cannot absorb any more of our emissions.

In the Malthusian logic, we would have to ever increase our production to meet the needs of a world with limits. As long as this mindset persists, the focus is on how we can exceed those limits, and how we can push the capacity of the climate to absorb a growing population and its demands.

1.3.1 The clash of economy and ecology

Originally, sustainability was a principle of resource economics that combined the economic goal of maximizing long-term forest use with ecological conditions for regrowth. In the 19th century, the concept of an "eternal forest" emerged with the aim of putting an end to overexploitation, and

to ensuring a long-term supply of wood, a vital raw material. Forestry scientists such as Heinrich Cotta developed mathematical methods for calculating timber stocks and growth. Over time, however, the focus shifted towards the pure yield theory, which focused on short rotation periods and high-yield monocultures, as well as on maximizing financial benefits. The focus was suddenly on the highest possible direct cash yield instead of a steady high timber yield. Natural cycles were replaced by capitalist dynamics; exchange value took precedence over utility value. This devalued the guiding principle of sustainability, and it took over a hundred years, until the 1960s and 70s, for the scientific disciplines of ecology and sustainability to regain prominence.

1.3.2 Club of Rome and The Limits to Growth

“If the present growth trends in world population, industrialization, pollution, food production, and resource depletion continue unchanged, the limits to growth on this planet will be reached sometime within the next one hundred years.” (Meadows et al. 1972, 23)

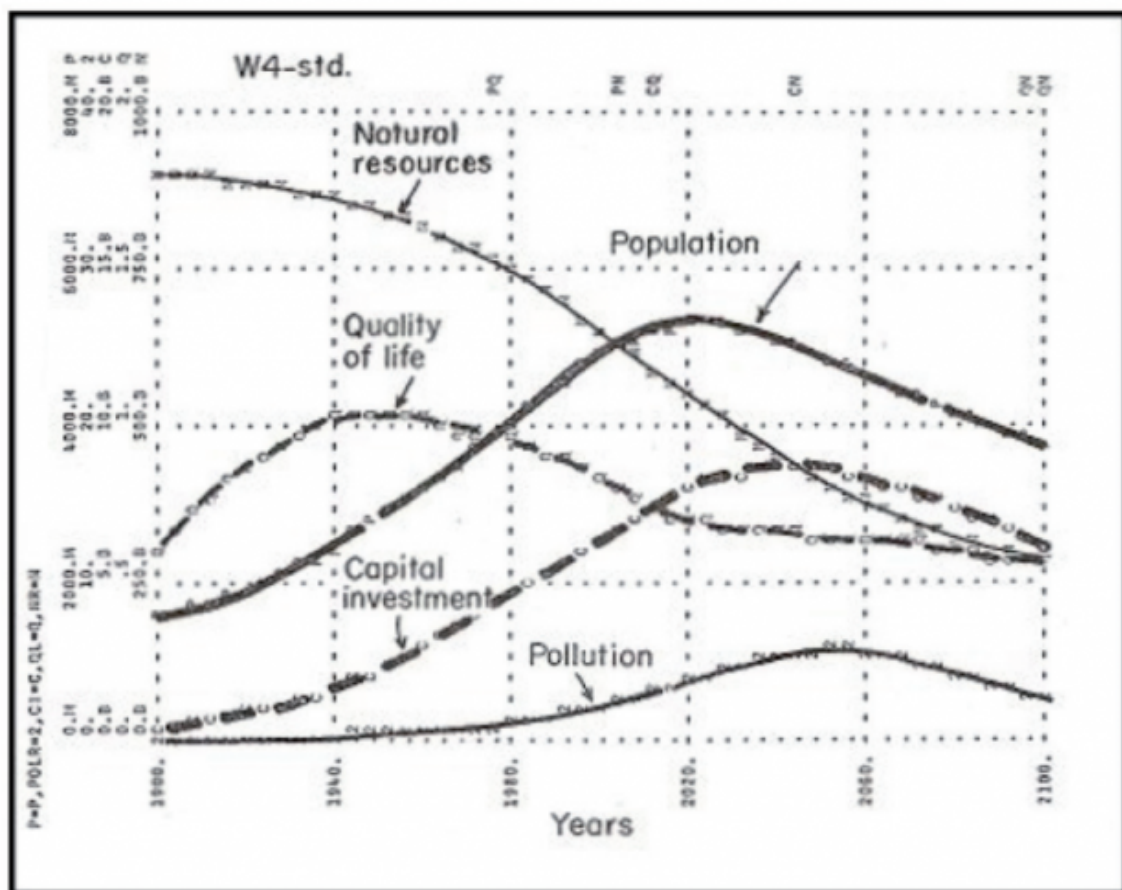


Figure 1.5: The “Basic model” of the simulation carried out at MIT, published in the autumn of 1970. Source: <https://donellameadows.org/>.

In 1972, the Club of Rome published a report, *The Limits to Growth*, which introduced a much broader understanding of “sustainability”. Scientists began to call for a global state of equilibrium, or homeostasis, that would only be possible through coordinated international action. They integrate economic, ecological, and social aspects of sustainability and use the model of the dynamics of complex systems (“system dynamics”) to understand the world. They take into account the interactions between population density, food resources, energy, materials, capital, environmental degradation, and land use. Computer simulations of different scenarios show similar results: a catastrophic decline in the world’s population and living standards within 50 to 100 years, if current trends continue. The problem with the resource- and emission-intensive industrialized societies is that growth is not linear but exponential. In the long run, this kind of growth leads to collapse. And ecological collapse can only be prevented through a course correction, argued the report.

The *Meadows Report*, as it is also known, was heavily criticized for its predictions and methods. Some critics argued that the system dynamics model was too simplistic and failed to consider important factors such as technology and innovation. Others said insufficient account was taken of human adaptability and the possibility of political solutions and change. Others yet criticized what they called a Malthusian outlook and a pessimistic view of the future. Despite these criticisms, however, the report sparked an important debate about sustainability and the need to protect the environment.

A year later, E.F. Schumacher published *Small is Beautiful: Economics as if People Mattered* (1973). Schumacher challenged the prevailing notion of limitless economic growth and technological progress. Instead, he advocated for a sustainable, human-centred economy that respects local communities and the environment. Schumacher argued that a decentralized economy, based on human values, not just profit, would create a better future for all. He also emphasized the importance of education and cultural development in creating a sustainable society.

1.4 From the development debate to the sustainability debate

Worsening air and water pollution helped raise the profile of environmental issues in politics and the media. Greenpeace was founded in 1971. In the 1960s and 1970s, Paul Crutzen and his research team studied the impact of nitrogen oxides on the ozone layer, predicting that this layer would be greatly depleted by human-made CFCs. The use of CFCs in refrigerators and air conditioners was subsequently banned. In response to the growing importance of environmental issues, the United Nations organized the first ever major environmental conference in Stockholm in 1972. The United Nations Conference on the Human Environment, as it was called, led to the creation of the United Nations Environment Programme (UNEP) and independent environment ministries in many countries.



Figure 1.6: Demonstration against air pollution, Zurich, December 1986. Source: Schweizerisches Sozialarchiv/Gertrud Vogler.

The emergence of environmental problems in the 1970s and 1980s coincided with the first “development crises” after the Second World War. In those days we still spoke of “underdeveloped countries”. Following the success of the Marshall Plan in rebuilding Europe after World War II, it was assumed that similar programmes, such as a Marshall Plan for Africa, would lead to a rapid catch-up in socioeconomic development of the poor countries of the Global South. But the 70s and 80s proved otherwise. Even exemplary countries such as Mexico and Brazil succumbed to debt crises, demonstrating that achieving socioeconomic development and overcoming poverty were far more complex challenges.

From then on, it was clear that issues related to development and the environment had to be considered together at the intergovernmental level. In 1983, the United Nations set up a World Commission on Environment and Development, chaired by the Norwegian Prime Minister, Gro Harlem Brundtland.

1.4.1 Brundtland Commission

Gro Harlem Brundtland appointed 22 commission members, $\frac{3}{4}$ of whom were from the Global South. During this time, the Soviet Union, facing economic stagnation, was beginning to change political course. As the arms race with the US had contributed to a substantial budget deficit, Mikhail Gorbachev, the newly appointed General Secretary of the Communist Party of the Soviet Union, introduced major reforms. At the same time, a new world view and global consciousness were starting to emerge. It was against this background that the Brundtland Commission was tasked with drawing up a report on the perspective of global, sustainable, environmentally friendly development up to and beyond the year 2000. The report presented in 1987 was entitled *Our Common Future*, but it's often referred to as the *Brundtland Report*.

The Brundtland Report found that global environmental problems were mainly caused by unsustainable consumption and production patterns in the North and severe poverty in the South. The overexploitation of nature and the depletion of natural resources were exacerbating inequalities (of income and wealth), increasing absolute poverty, and posing a threat to peace and security. The Report sought to develop a fair and just definition of sustainability:

“Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.” (WCED 1987)

This definition introduced an ethical perspective into the sustainability debate, placing the principle of responsibility – for both present and future generations – at its centre. By focusing on human needs, the Brundtland Commission adopted an anthropocentric position. The Commission thus identified three key principles for analysing problems and guiding action: a global perspective, the interconnectedness of environment and development, and the pursuit of justice or equity. The concept of justice/equity was further divided into two distinct perspectives:

1. The **intergenerational perspective**: Responsibility for future generations.
2. The **intragenerational perspective**: Responsibility for people living today, particularly in developing countries, and ensuring equity within countries.

Sustainability and sustainable development

The concepts of “sustainability” and “sustainable development” differ in focus. Sustainability is static; it emphasizes a consistent state, while sustainable development is the dynamic process of achieving that state, implying movement and referring to something that is emerging.

1.5 From Rio 1992 to today

After the Brundtland Report’s call for international action, the focus turned to translating demands and proposals into binding treaties and conventions. The UN chose a conference as an instrument for this, holding it exactly 20 years after the first global environmental conference, the 1972 Stockholm Conference. The June 1992 UN Conference on Environment and Development – held in Rio de Janeiro, Brazil, and also known as the Earth Summit – was the largest international conference to date, with delegates from over 170 nations.

The Rio Earth Summit adopted the following five documents:

- A Forest Declaration, aimed at the ecological management and protection of the world’s forests;
- A Climate Protection Convention committing states to reducing greenhouse gas emissions worldwide to 1990 levels;
- A Biodiversity Convention to combat the decline in biodiversity through steps that are binding under international law;
- The Rio Declaration on Environment and Development; and

- Agenda 21, probably best known of the five agreements, according to which national governments are responsible for implementing the sustainability model in their countries.

The Rio Declaration on Environment and Development emphasizes that long-term economic progress is only possible using an ecosystem approach. This, in turn, requires a new and equitable global partnership among governments, people, and key societal groups. To achieve this, international agreements are necessary to protect the environment and the development system. The Rio Declaration established key principles of sustainable development, including precaution and polluter pays. For example, Principle 15 states:

“In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation ” (UN 1992b).

In December 1992, the Commission on Sustainable Development (CSD) was established to ensure effective follow-up of the Summit.

1.5.1 Agenda 21: Think globally – act locally

Agenda 21, which contains 40 chapters, addresses all critical policy areas and actions for sustainable development. According to the Preamble,

“...integration of environment and development concerns and greater attention to them will lead to the fulfilment of basic needs, improved living standards for all, better protected and managed ecosystems and a safer, more prosperous future. No nation can achieve this on its own; but together we can – in a global partnership for sustainable development.” (UN CSD 1992)

1.5.2 The Millenium Development Goals (MDGs)

The core principles of Agenda 21 – empowering women, protecting the environment, and promoting an equitable and inclusive society – laid the foundation for the Millennium Development Goals (MDGs). Adopted by the UN in 2000, the MDGs comprised eight specific goals to be achieved by 2015. Although not all of these goals were met, the MDGs raised significant awareness of the need for sustainable development. The MDGs were succeeded by the Sustainable Development Goals (SDGs), which were adopted by the UN in 2015.

1.5.3 Rio+20

Two decades after the historic 1992 Earth Summit that adopted Agenda 21, the international community gathered again for Rio+20, the UN Conference on Sustainable Development. The main objective of this 2012 conference in Rio de Janeiro was to reaffirm global political commitment to

sustainable development. Participants, including government leaders and NGO and private sector representatives, discussed a wide range of issues, including poverty eradication, environmental protection, sustainable energy, food security, and resource management. The Summit's key outcome was the adoption of a declaration entitled "The Future We Want". This declaration renewed the international community's commitment to sustainable development and to the Rio Principles and past action plans, and outlined further measures to achieve these goals. In addition, Rio +20 paved the way for a universal development framework to define global sustainability goals and priorities beyond 2015.

1.5.4 The 2030 Agenda and the Sustainable Development Goals (SDGs)

"We are the first generation that can put an end to poverty and we are the last generation that can put an end to climate change, so we [must] address climate change." Ban-Ki Moon, UN Secretary General 2007–2016, on the 2030 Agenda

Rio +20 laid the foundation for the 2030 Agenda, which the UN adopted in 2015. The 2030 Agenda builds on the results of Rio+20 and sets out a comprehensive framework for sustainable development. The 2030 Agenda is based on five key principles or pillars (the 5 Ps) and introduces the 17 Sustainable Development Goals (SDGs), which aim to create a sustainable and just world by 2030.

The 5 Ps of the SDGs

The 5 Ps represent five key principles of sustainability: People, Planet, Prosperity, Peace, and Partnership.

People: This P stands for the aim to promote social justice, equal opportunities, good health, and education for all people. It also seeks to end poverty, promote gender equality, and improve people's well-being.

Planet: The aim of this P is to protect natural resources and to use them sustainably, to preserve biodiversity, to protect the climate, and to reduce pollution. Overall, it seeks to preserve our planet and promote sustainable environmental practices.

Prosperity: This refers to economic growth, sustainable production and consumption patterns, and the creation of jobs and economic prosperity for all. This P aims to promote a strong and sustainable economy that benefits all people.

Peace: This is about promoting peace, justice, good governance, and strong institutions. It's also about preventing conflict, reducing violence, and building inclusive societies.

Partnership: This refers to the importance of global cooperation, partnerships, and solidarity among all stakeholders. It's about working together to implement the 2030 Agenda by sharing resources, experience, and knowledge.

1.5.5 2015 Paris Climate Agreement

The Paris Climate Agreement of 2015 is a landmark international agreement adopted at the 21st Conference of the Parties (COP 21) to the UN Framework Convention on Climate Change (UNFCCC)

in Paris. The Agreement aims to limit the global temperature rise to well below 2 degrees Celsius compared to pre-industrial levels, and to endeavour to limit it to 1.5 degrees Celsius. The Paris Agreement is based on the principle of common but differentiated responsibilities and respect for national circumstances. It encourages all countries to take action to reduce their greenhouse gas emissions and to set themselves voluntary climate targets, known as Nationally Determined Contributions (NDCs).

The 2015 Paris Climate Agreement applies to every country that has ratified it, including Switzerland. As a Party to the Agreement, Switzerland has committed to contributing to the global reduction in greenhouse gas emissions and to implementing the Agreement's objectives.

1.5.6 Conclusion – there is progress, but it's too slow

While there has been some progress in institutionalizing and disseminating sustainability approaches in business, policymaking, and other areas, a large number of studies show that the world as a whole is on an unsustainable course (see the [Voluntary National Review of Switzerland 2022](#)). A similar picture is painted by various studies on the global environmental situation, such as UNEP's Global Environment Outlook 5 (GEO-5) published in 2012, the 2010 Millennium Development Goals Report (MDGs Report 2010), and reports on the 2030 Agenda and climate change (IPCC 2022). As these studies show, the international community is a long way from sustainable, intergenerationally equitable ecological, social, and economic development.

For example, international climate policies have yet to halt the rise in CO₂ emissions. As the main driver of anthropogenic climate change, these emissions have increased by around 50% since 1990. Species loss continues to accelerate in many places. The global fight against poverty is falling short of the desired goals, and economic globalization over the past three decades has worsened socioeconomic in-equality in many countries, often linked to sociopolitical and sociocultural disparities.

1.6 The guiding principle of sustainable development as a normative concept

Turning the concept of sustainability into action and developing implementation strategies is a major challenge for our society. There are different understandings and concepts of sustainability and sustainable development, with no consensus on how to achieve sustainability and which measures to prioritize. Some of these understandings are presented below. The diversity of approaches reflects the different perspectives and interests of different stakeholders. To navigate this complexity, we need to reconcile economic, social, and environmental considerations. This requires an integrative approach. The challenge is to define common goals and develop strategies to achieve sustainability at all levels – global, national, regional, and local. We need broad societal participation and dialogue, as well as cooperation between governments, business, civil society, and research institutions, to find solutions and drive the necessary change.

“When powerful metaphors become fashionable buzzwords, we risk that diversity is accompanied by vagueness, i.e. the phenomenon of a term that has several meanings that ‘have so much in common that it is difficult to separate them’.” (Strunz 2012, 113; as cited in Feola 2015, 377)

While the concept of sustainability typically carries a positive connotation, it is subject to diverse interpretations. For it to function as a robust guiding principle across ecological, social, and economic dimensions, clear criteria are essential. However, this raises a fundamental question: on what grounds can these criteria be established?

According to Hirsch Hadorn and Brun (2007), “sustainable development” should:

- Fulfil *needs*...
- ...in a *just* way,
- ...with a view to people living today and in the future, and
- taking into account the diversity of values and the limits to which nature can be used.

The guiding principle of sustainable development is therefore not merely a product of scientific research; it is, first and foremost, a normative, ethically grounded concept. It brings together “ethical and analytical ideas” and formulates “norms that express what is desirable and what should happen” (Renn et al. 2007, 39). As a result, sustainable development is a social process of negotiation and decision-making – of searching, learning, and gaining experience – that is guided by ethical considerations. Accordingly, sustainability research must always be aware of its involvement in social processes of perception and evaluation.

1.6.1 What values and norms should we be guided by, and why?

Moving from the theoretical framework of sustainable development to its practical implementation requires the consideration of ethics. Ethics provide evaluation criteria, methodological procedures, or principles for “justifying and criticizing rules of action or normative statements about how we should act” (Fenner 2008, 5). Thus, ethics provide a framework for structuring and justifying the complex decisions that arise in dilemmas. Unlike problems with a single, optimal solution, a dilemma is more complex and involves the careful weighing of trade-offs and different courses of action. The core function of ethics is therefore not to solve monocausal problems, but to structure and categorize complex dilemmas.

Put simply, “morality” refers to personal or social beliefs about right and wrong, while “ethics” is the philosophical study of these beliefs. Ethics studies the rules and principles that “ought” to govern human behaviour. Ethics can be divided into general and applied ethics (Figure 1.7).

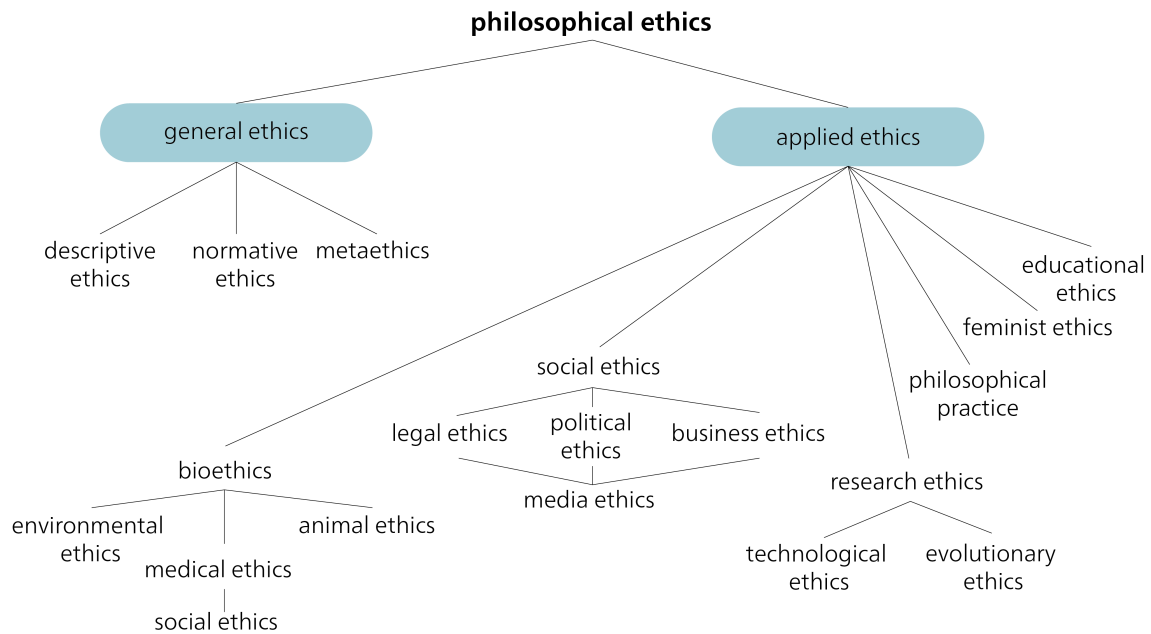


Figure 1.7: The different branches of philosophical ethics. Source: Own illustration based on Pieper and Thurnherr (1998, 9).

General ethics provides us with methods and concepts to discuss fundamental moral problems. It consists of three branches: normative ethics, descriptive ethics, and metaethics. Normative ethics focuses on determining what is morally right or wrong. For example, questions about what constitutes a good life can have different answers, depending on people's values.

For this reason, normative ethics is further divided into teleological and deontological approaches. Teleological ethics (rooted in *telos*, the Greek word for “end”, “purpose”, or “goal”) evaluates actions based on the good they produce. Utilitarianism, for example, is a teleological theory that evaluates actions based on the resulting utility or happiness. One of the first systematic elaborations of utilitarianism is Jeremy Bentham's (1748–1832) *An Introduction to the Principles of Morals and Legislation* (1780). Deontological ethics, on the other hand, evaluates actions on the basis of their characteristics rather than their consequences. Its root is *deon*, Greek for “duty”. An example of a deontological ethical theory is Kantian ethics, developed by the German philosopher Immanuel Kant.

The distinction between teleological and deontological approaches is often discussed using the “trolley problem” (see further reading box below).

Further reading

- *Justice: What's The Right Thing To Do? Episode 01 "THE MORAL SIDE OF MURDER."* 2009. With Michael Sandel. The Moral Side of Murder. <https://www.youtube.com/watch?v=kBdfcR-8hEY>.
- Sandel, Michael. 2013. *Gerechtigkeit*. Ullstein.

1.6.2 The imperative of responsibility – Hans Jonas

Hans Jonas's "imperative of responsibility" (1979) offers an important contribution to the sustainability debate by emphasizing our ethical responsibility for future generations and for nature. Jonas's approach, which has been described as a kind of "ethics of the future", aims to address the new challenges posed specifically by a technological civilization. He views the concept of ethical responsibility towards future generations as an asymmetric, non-reciprocal relationship (Jonas 1987, 177). The asymmetry is derived from the power of a "moral subject" over someone or something that requires care; Jonas describes the vulnerability of life (ontological vulnerability) and responsibility as our duty of care to protect other beings from harm (Jonas 1987, 391).

Jonas's approach differs from other ethical theories in that it doesn't take the existence of humanity for granted: instead, it includes duties towards future generations as well as to nature. Jonas argues that the first duty of an "ethics of the future" consists in recognizing the distant effects of human action (Jonas 1987, 64). Given the uncertainty about the long-term consequences of human action, he proposes a decision-making principle based on the Latin term, *in dubio pro malo*, which he interprets to mean: "[...] when in doubt, give the worse prognosis precedence over the better, as the stakes have become too high for the game" (Jonas 1987, 67). Jonas thus develops a new categorical imperative, emphasizing the need to preserve both nature and humanity. It is based on the idea that nature has an inherent value and purpose: "Act in such a way that the effects of your action are contractual with the permanence of genuine human life on earth", or "Act in such a way that the effects of your action are not destructive to the future possibilities of such life" (Jonas 1987, 36).

Already in the late 1970s, therefore, Hans Jonas articulated a fundamental uncertainty about the risks and harm that could arise for future generations from the technologies that were being used, such as nuclear energy. Decision-making around issues such as new technologies requires a continuous evaluation of risks and opportunities, often amid incomplete information and uncertain future forecasts. These choices not only impact the environment, but also raise ethical questions about our obligation to ensure sustainability for future generations. Jonas' imperative of responsibility emphasizes the link between responsibility and duty as key concepts in the sustainability debate. According to Jonas, current generations have a forward-looking responsibility to future generations. He leaves open, however, the extent of this duty, and whether future generations are entitled to absolute or comparative standards.

Jonas' imperative of responsibility can be considered in connection with various areas of responsibility that represent different ethical approaches: anthropocentrism, pathocentrism, biocentrism, and physiocentrism.

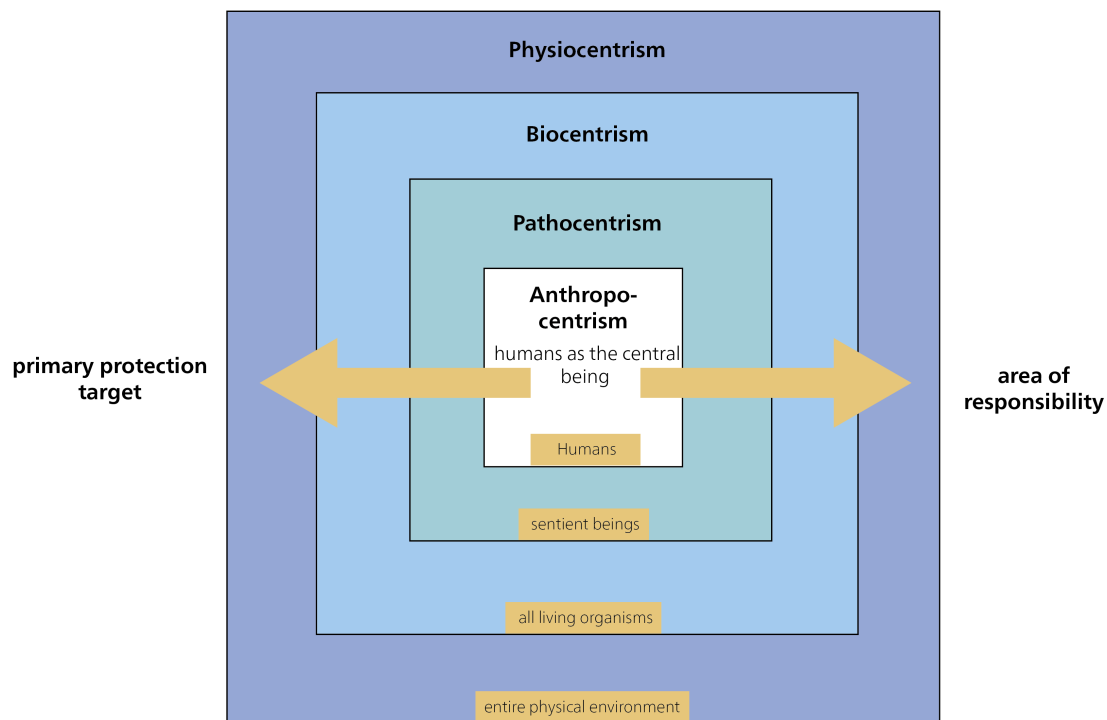


Figure 1.8: Different ethical approaches. Source: Own illustration based on Carnau (2011, 143).

Anthropocentrism is the world view that places humans at the centre of importance, emphasizing a primary responsibility towards human society and its well-being. This includes considering the needs and interests of both present and future generations. Pathocentrism, on the other hand, extends responsibility towards all sentient beings and their well-being. It recognizes that not only humans, but also animals and other living beings, have the right to be free from suffering and harm. Biocentrism, in turn, recognizes the intrinsic value of all living organisms and focuses on the protection and preservation of biodiversity.

Hans Jonas' imperative of responsibility reconciles all of these different areas of responsibility. It urges us to take responsibility for people as well as for other living beings, nature, and the environment. In doing so, it is important to adopt a balanced and comprehensive perspective that takes into account long-term effects and the well-being of all.

By integrating these different areas of responsibility, we can cultivate a holistic and sustainable ethic that takes due account of the needs and interests of all aspects of life and nature.

1.7 Sustainable development in the context of development debates

Is “sustainable development” a development theory? Yes and no. No, because sustainable development is a broader social model and acts as an overarching framework that encompasses

various development theories. Yes, because sustainable development is a normative theory that de facto competes in scientific and social practice with other theories of social and, in particular, socioeconomic development.

Development theories analyse social development, either retrospectively or prospectively, and either as a whole or through specific aspects (e.g. economic development). The retrospective study of social development seeks to explain past or current social developments and to draw lessons for future planning and development policy. The prospective study of social development aims to inform social development management and policy. Development theories can be categorized into different types: normative, strategic, and explanatory theories. Each type focuses on different aspects and objectives of development research.

Normative theories focus on the question of what development should look like, and what normative goals it should achieve. They examine values and goals associated with development, often referring to concepts such as social justice, sustainability, or poverty alleviation. Normative theories emphasize the definition of criteria for good development, and provide a normative framework for policy decisions and measures. Examples include the theory of sustainable development, John Rawls' theory of social justice, or Amartya Sen's capability approach.

Strategic theories investigate which strategies and measures are needed to achieve desired development outcomes. They focus on questions of policy design, resource allocation, and the implementation of measures, analysing how specific approaches, such as neoclassical growth theory, can promote development.

Explanatory theories aim to explain the causes and mechanisms of development. They analyse factors that drive development processes in societies and study the links between the different variables. This involves examining economic, social, political, or cultural factors to identify patterns and connections. Examples include modernization theory and dependency theory.

Note: All theories are based on normative foundations – i.e. underlying values and beliefs – and therefore imply, even unintentionally, certain ideas about how society should develop.

Table 1.1: Overview of Development Debates. Source: Extended table based on Bieri (2022, 378).

Category	1950s/60s	1970s	1980s	1990s	Since 2008
Narrative	Reactive growth, modernization, and integration into the world economy	Fight against extreme poverty and inequality	Lost decade: collapse of commodity prices; increase in debt	Sustainable development; decade of hope	Polycrisis
Basic idea	Catch-up development through industrialization and large-scale projects	Satisfaction of basic needs, growth, and effective distribution	Overcoming the debt crisis through structural adjustment measures, reduction of state services	Global sustainable environmental policy, economic growth, peacekeeping, continuing basic needs strategy, participation	Bringing together various crisis phenomena (financial crisis, climate change, biodiversity crisis, etc.)

(continued)

Category	1950s/60s	1970s	1980s	1990s	Since 2008
Theories	Modernization theories (stage theories)	Dependency theories versus modernization theories	Market liberalism and neoliberalism	Theories of sustainable development, Neoclassical growth theory, Postcolonial theories, Capability approach, Post-development theories	Additional resilience theories, post-growth theories
Goals	Geopolitical classification of the “Third World”; containment of communism, modernization of agriculture, rapid industrialization and technological progress, opening up of new markets	Aid for self-help, appropriate technologies, rural development, promotion of women	Increase in exports, balanced state budgets	Paradigm shift in development aid towards development cooperation, environmental and social compatibility of development, ensuring access to resources and infrastructure for all	
Consequences	Economic dependence and indebtedness increase, income disparities widen, poverty rises, ecological damage due to overuse and inappropriate technologies, undemocratic regimes are supported for geopolitical reasons	Selective progress in education and health, problems of debt and environmental damage	Living standards of the poorest population groups deteriorate severely, environmental exploitation, debt spiral, increase of dependency of the Global South on Global North countries	Focus shifts from purely economic development to human development; industrialized countries are obliged to implement sustainable development; emergence of a global development partnership	

Development theories

Modernization theory (1950s–1980s) emphasizes the transition from traditional societies to modern, industrialized societies as the basis for development. For proponents of modernization theory, the countries of the Global South are developing in the same direction as industrialized countries, but at a much slower pace. Modernization theory values economic growth, technological progress, and institutional adjustments. It is often associated with stage theories, which emerged in the 1950s and 1960s. Stage theories view development as a gradual process characterized by clearly defined stages or phases of development, with emphasis on economic growth and modernization. Prominent proponents include Walt Rostow, who identified five stages of development in his work, *The Stages of Economic Growth: A Non-Communist Manifesto* (i.e. traditional society, preconditions for take-off, take-off, drive to maturity, and mass consumption).

Dependency theory (1970s and 1980s) emphasizes the structural dependence and inequality between developed and developing countries. It argues that developing countries are trapped in an unjust global economic system and remain dependent on developed countries.

Neoclassical growth theory (as of the 1990s) focuses on the connection between capital accumulation, technological progress, and economic growth. It emphasizes free markets, trade, and investment as drivers of development.

Postcolonial theories (as of the 1990s) emphasize the historical and structural inequality between former colonial powers and colonies. They argue that development can only be achieved through a critical examination of the legacy of colonialism.

Sen's capability approach (as of the 1990s) emphasizes the importance of freedom, social justice, and human development. It emphasizes the necessity to improve the opportunities and capabilities of people to achieve development.

Post-development theories (as of the 1990s) criticize linear development models that impose Western ideas of progress, emphasizing instead local practices and values. Prominent thinkers in this field include Arturo Escobar, who critically reflects on the idea of development in his work *Encountering Development: The Making and Unmaking of the Third World*, and Vandana Shiva, whose *Decolonizing the North* calls for sustainable and fairer development policies in the Global South that reduce the influence of the Global North.

1.8 Sustainability models

1.8.1 Three dimensions of sustainability

The established three-sector model of sustainability, introduced in the 1987 Brundtland Report, seeks to balance environmental, economic, and social concerns. This model, which later also became known as the “triple bottom line” model, emphasized that economic development should be profitable while ensuring that environmental and social impacts remain neutral. On closer inspection, however, the model – initially depicted as a building with three pillars – has a major weakness. It was intended to suggest that if any one pillar was weak, the system as a whole would be unsustainable. However, it could be argued that removing one of the pillars – or even the two outer pillars – would not make the building collapse if the remaining pillar(s) were strong enough.

Despite its widespread use and overall success, the model struggles with transparency and equity. For example, while economic success is easily measured through quantitative indicators, it is more challenging to find similar metrics for social well-being, especially regarding its emotional aspects. Some critics argue that the model prioritizes conventional economic thinking, neglecting important social aspects. An overemphasis on human economic systems can also prevent us from seeing our dependence on non-human ecological systems.

1.8.2 Sustainability as an intersection

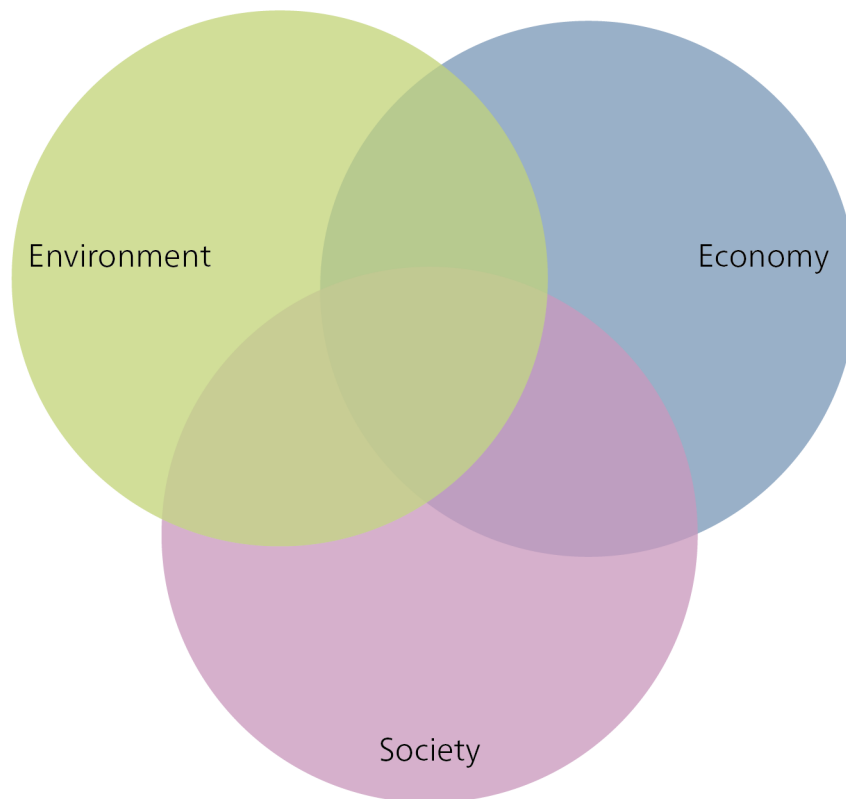


Figure 1.9: Sustainability as an intersection. Source: Own illustration.

A model displaying the three sectors as a Venn diagram (overlapping rings or circles, sometimes known as the intersection or triad model) offers an alternative to the separate pillars, emphasizing the interconnectedness of the three dimensions. This model illustrates that there can be a closer connection between two areas, and that the boundaries are fluid.

1.8.3 Nested sustainable development

Giddings et al. (2002) challenge the idea of separate spheres for the economy, society, and environment. They propose a “nested” model, where the economy is seen as part of society, which in turn is embedded in the environment. This model, they argue, ensures that economic decisions are always constrained by social and environmental factors. Global ecosystems form the basis for social systems (i.e. society). Embedded therein is the economy, which serves society’s needs. A potential drawback of this model, however, is that if the economy is the starting point for all sustainability considerations, there is a risk of prioritizing it over social and environmental considerations.

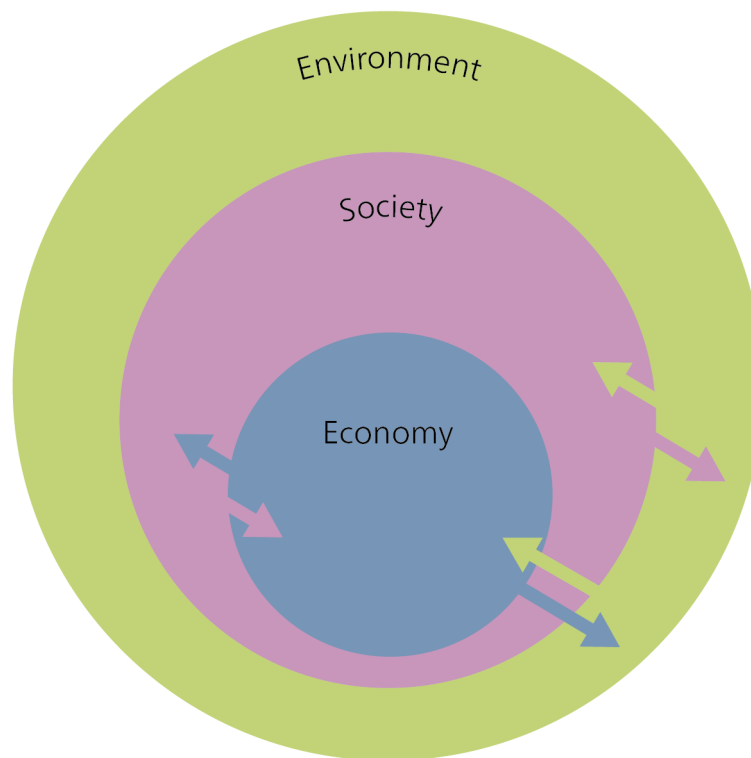


Figure 1.10: The nested model of sustainability. Source: Own illustration based on Giddings et al. (2002, 192).

1.8.4 A four-dimensional model of sustainability

One problem with the three-sector model of sustainability is that it is unclear what falls within the broad “social” sphere. It has therefore been suggested that a fourth dimension be added, to emphasize certain aspects that might otherwise remain hidden in the social sector. For example, Jon Hawkes, an Australian cultural commentator, has suggested adding “cultural vitality” as a fourth pillar of sustainability, in addition to environmental, economic, and social concerns.

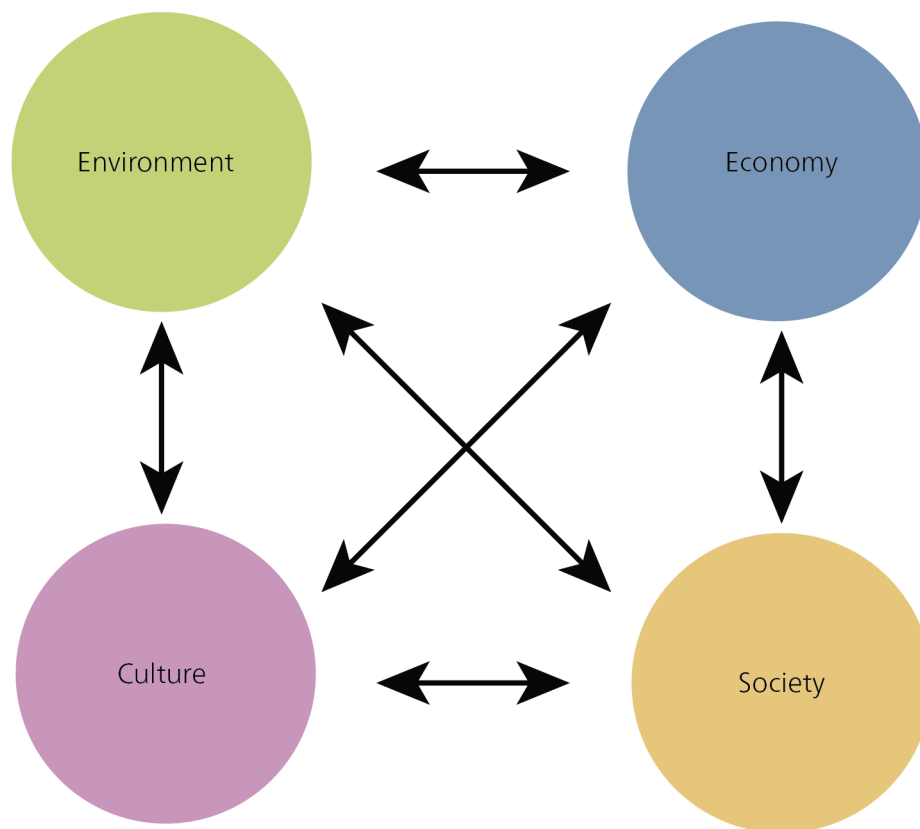


Figure 1.11: Four-dimensional model of sustainability. Source: Own illustration, adapted from Hawkes (2001).

Cultural sustainability is often seen as a part of social sustainability. Sometimes the term “sociocultural sustainability” is also used. In this context, “culture” refers to aspects such as equity, opportunities for participation, awareness of sustainability, and general operating and behavioural models (Murphy 2012). It’s clear that the social and cultural dimensions are closely linked. Cultural processes influence our social lives and how we view social sustainability, like the value we place on equality as a social goal. In turn, social structures and institutions influence cultural practices and judgements. Despite this strong connection, cultural and social sustainability can also be seen as separate dimensions or perspectives of sustainability.

Culture does not have its own Sustainable Development Goal (SDG). In the 2030 Agenda, culture is mentioned in relation to “civilization”, “diversity”, “interculturality”, “cultural heritage”, and “tourism”, under the following four SDGs: quality education (SDG 4), decent work and economic growth (SDG 8), sustainable cities and communities (SDG 11), and responsible consumption and production (SDG 12) (Duxbury et al. 2017).

The concept of cultural sustainability can be interpreted in various ways: as the fourth dimension of sustainability, as culture for sustainability, and as culture as sustainability. The interpretation of “culture as sustainability” requires a new way of thinking and a new paradigm in relation

to sustainability. In this view, culture itself is transformed towards sustainability. Cultural sustainability here means thinking about what needs to be preserved, what needs to change, and how we can implement these changes. This interpretation – viewing culture as sustainable development – defines culture in its broadest sense: as a comprehensive way of life and a continuously changing process. Our social order (such as capitalism and democracy), our values, and our ways of working are all cultural products. Culture thus encompasses all the other dimensions of sustainability, and transforming it becomes a key concern for sustainability.

The definition of cultural sustainability contains several important aspects: Firstly, it emphasizes the importance of intellectual growth and responsibility. This means acquiring knowledge, developing a better understanding of important issues, and participating in educational activities and events. It also means acting not only as individuals, but also as members of different communities. All problematic and unsustainable structures and operating models have been created and developed by humans. We humans, therefore, also have the opportunity to change them.

Cultural sustainability is required, for example, for a “Great Transformation” as advocated by Uwe Schneidewind (2018). In his work, Schneidewind emphasizes the need for comprehensive change in all dimensions of society, including culture, in order to achieve sustainable development. Recognizing cultural sustainability as a central paradigm of sustainability supports the vision of a comprehensive transformation that goes beyond technological and economic solutions and strives for a fundamental societal change of course.

1.9 Weak and strong sustainability

The academic debate on sustainability distinguishes between weak and strong sustainability (see e.g. Ott 2020). Neither concept can easily be dismissed, as both rely on assumptions that are difficult to refute. In practice, both concepts are considered partially true, with evidence both in favour and against. The key difference between weak and strong sustainability is the question of what we should sustain, and the degree to which what forms of capital can be substituted. This concerns, in particular, the following three forms of capital: natural capital, manufactured capital, and human capital.

Natural capital: the natural resources and ecosystems that are key to human well-being and sustainable development. These include land, water, air, biodiversity, and renewable energy. Natural capital is the basis for many economic activities, and it provides essential ecosystem services such as food, the purification of water and air, and cultural and aesthetic benefits.

Manufactured capital: the material resources and infrastructure used in production and economic activity. This includes buildings, machinery, technological equipment, means of transport, and physical infrastructure. Manufactured capital is closely linked to economic growth and productivity, and it plays an important role in economic development.

Human capital: the knowledge, skills, level of education, health, and creative abilities of people in a society. It encompasses individual and collective knowledge as well as the skills needed for economic productivity and social progress. Human capital is important for innovation, labour productivity, social participation, and the ability of a society to tackle challenges and to adapt.

1.9.1 Weak sustainability

Weak sustainability can be considered the “soft” or flexible option. Supporters of weak sustainability believe that an action is sustainable if it offers an advantage to the overall system, or at the very least, does not reduce its quality. Weak sustainability can therefore be seen as a basic requirement of sustainability, in that it aims to maintain a certain level of well-being and quality of life while ensuring equity. The concept of weak sustainability “assumes the extensive and, at least in principle, unlimited [...] substitutability of all types of capital” (Ott and Döring 2004, 41) and is thus based on the premise of neoclassical economics. Proponents of weak sustainability believe that natural capital can be substituted with other forms of capital. They emphasize the concept of “total capital”, arguing that the specific make-up of the capital inherited by future generations is irrelevant, as long as the overall benefits and well-being are sustained. This aligns with neoclassical utility theory, according to which it is irrelevant how utility is generated.

Weak sustainability supporters therefore believe that measures can be considered sustainable, even if they are carried out at the expense of natural capital – as long as this loss is compensated for by an increase in human or manufactured capital. This could justify the extraction of raw materials such as coal or excessive crude oil. And Carlowitz’s *Silvicultura oeconomica* forest, could, in principle, also be replaceable or degradable, provided that its natural and cultural functions can be fulfilled in other ways by future generations. This understanding of sustainability emphasizes the importance of technological progress for sustainable development.

Evaluating measures based on weak sustainability is challenging, due to the inherent complexities and numerous assumptions involved. This goes beyond fundamental questions about sustainability, such as “What is a good life?”, or “How do we measure the level of well-being?”. A key challenge lies in assigning a monetary valuation to different types of capital, especially those lacking a clear market price. To operationalize this, what factors should be taken into account? What is the value of natural beauty? What is the value of a rare bird or a whale? Buller’s *The Value of a Whale* (2022) impressively outlines the problems we face in trying to put a price on elements of nature.

Critics of the concept of weak sustainability point to several limitations. Firstly, they question the assumption that natural resources can be endlessly replaced by reproducible capital. Secondly, they doubt whether an increase in goods can truly compensate for the loss of environmental quality. Finally, concerns exist about our ability to develop new resources and the potential for exceeding critical threshold values. These criticisms have paved the way for the concept of “strong sustainability”.

1.9.2 Strong sustainability

In contrast to weak sustainability, strong sustainability emphasizes the preservation of natural capital. Proponents of this eco-centric theory are less optimistic about our ability to substitute natural resources with manufactured ones, believing these resource types to have limited interchangeability (Daly 1990; Ott 2001; Ott and Döring 2004). They argue that the individual elements of natural capital should be kept as constant as possible, to prevent, for example, species extinction.

Table 1.2: Weak and Strong Sustainability. Source: Derived from Eblinghaus and Stickler (1998); Michelsen et al. (2012); and Davies (2013).

Category	Very Weak Sustainability	Weak Sustainability	Strong Sustainability	Very Strong Sustainability
What should be preserved?	Total capital	Essential natural capital	Non-renewable natural capital	Nature has its own value
Why?	Maximization of economic profit and individual welfare (GDP)	+ Limitation of environmental damage and resource scarcity	Ensuring long-term living conditions for humans and nature	+ Respect for all forms of life and duties toward nature
How? (Policies)	Growth orientation through promotion of technology and innovation	Environmental regulations, resource efficiency, renewable energies	+ Conservation measures, ecological restoration, sustainable agriculture	–
Substitutability	In principle unlimited, natural resources can be replaced by technology and trade	Partly possible, but limited substitutability of ecosystem services	Limited, critical threshold for environmental changes	Low, recognition of unique values and relationships
Ethics	Anthropocentric, short-term benefits take priority	Anthropocentric with stronger consideration of environmental interests	Pathocentrism, recognition of intrinsic value of nature	Biocentrism, ecological integrity, ethical responsibility toward nature

While neoclassical economists are confident about substitution, proponents of strong sustainability emphasize the importance of prevention and anticipation, rather than aftercare and reaction. For example, reacting to the hole in the ozone layer by using sun cream, protective clothing, or medical aftercare fails to tackle the cause of the problem. Similarly, technological approaches such as geo-engineering are criticized for merely treating problems superficially. Geo-engineering refers to technical interventions in geochemical or biogeochemical cycles, for example to slow down climate change or ocean acidification.

Until now, reducing pollution has mainly focused on end-of-pipe technologies, such as flue gas cleaning systems on factory chimneys or catalytic converters in cars. But this also meant that we put off tackling long-term environmental problems, such as climate change or biodiversity loss, as long as their effects were not immediately apparent. This phenomenon is linked to the concept of externalization, in which environmental and social costs are shifted outwards, as described by Lessenich (2018). This means that pricing only takes into account the economic costs, partly because they are easier to quantify, while the social and environmental costs of providing goods and services are omitted or externalized, leading to distortions in market prices.

1.10 Sustainability strategies as principles of sustainable development

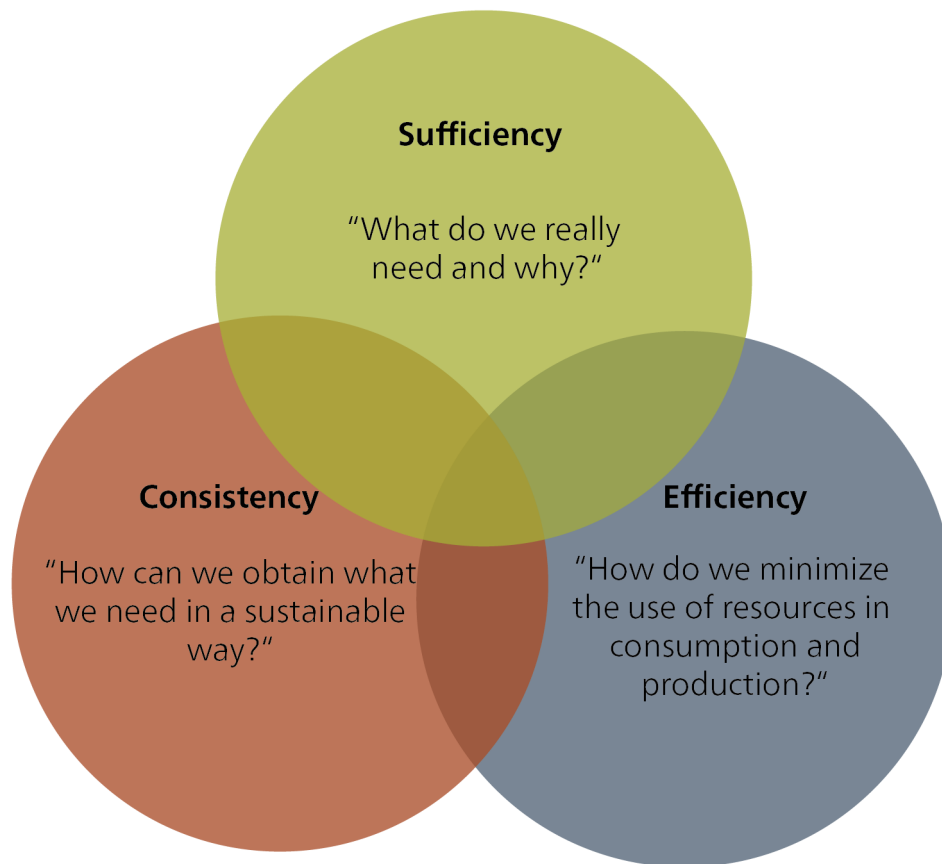


Figure 1.12: Sustainability strategies. Source: Own illustration.

1.10.1 Efficiency strategy

The efficiency strategy focuses on minimizing resource use (i.e. raw materials and energy) in manufacturing or service provision. This translates to reducing material consumption (material intensity), energy consumption (energy intensity), and emissions of harmful substances such as CO₂. Also known as “eco-efficiency”, this strategy is attractive to business and society, as it can reduce costs, resource use, and environmental impact. Implementing the efficiency strategy starts with improving production processes, primarily through technological advancements. Proponents believe it’s possible to double prosperity while halving natural resource use (von Weizsäcker et al. 1995). However, critics of the efficiency strategy are less optimistic, and caution against overestimating its impact on sustainable economic activity. They point to “rebound effects”, which can reduce or even wipe out the gains from efficiency increases (Paech 2012; Santarius 2014).

Putting Efficiency Strategies into Action: Examples

Energy-efficient lighting: Replacing conventional lightbulbs with energy-efficient LED bulbs lowers electricity use and helps reduce the demand for energy.

Fuel-efficient vehicles: The development of more fuel-efficient hybrid or electric vehicles can lead to lower fuel consumption and road transport emissions.

Efficient building technology: Installing smart HVAC (heating, ventilation, and air conditioning) systems in buildings helps to reduce energy consumption for heating and cooling.

Efficient water use: The use of water-saving fittings and systems in households and industrial companies reduces water consumption and minimizes waste.

1.10.2 Consistency strategy

While the efficiency strategy focuses on quantity (reducing resource use while achieving greater output), the consistency strategy (from the Latin *con* = together + *sistere* = to stop) strives to reconcile nature and technology by using resources repeatedly instead of only once. Instead of fossil fuel-based resources, products, and technologies, it seeks to use materials that are compatible with natural material cycles and processes. Also referred to as “eco-effectiveness”, this concept follows the principle of “cradle to cradle” rather than “cradle to grave”, and is based on the idea that intelligent systems generate only products, not waste. This can be achieved in two ways: Materials can either be biodegradable (e.g. a shampoo made with natural ingredients), or they can be designed with “technical nutrients” that remain in the technical cycle. This means that a disused product doesn’t end up in the rubbish, but instead enters the next cycle of use, for example through upcycling (e.g. reusing a computer casing or converting into, say, a shelving system). Like the efficiency strategy, the consistency strategy also starts by looking at how production processes can be optimized. Many believe that the consistency strategy has greater potential than the efficiency strategy, in terms of problem solving and achieving far-reaching changes.

However, critics believe the theory has limited applicability. For example, Gerolf Hanke and Benjamin Best (2013) argue that it cannot be applied equally to every type of good, and that implementing technologies to achieve it would require significant investment in production facilities and logistics. They also note that recycling processes themselves are associated with increased energy consumption, in accordance with the law of entropy:

“Every material economic process results in an increase in entropy, i.e. roughly simplified, that the elements on the material level are distributed more and more evenly, which ultimately only means that things and bodies wear out and a new concentration requires an ever-increasing amount of energy.” (Translated from Hanke and Best 2013, 11)

A special feature of the Earth’s ecosystem is solar energy, which provides a constant supply of energy from the outside and counteracts the increase in entropy on Earth. But we still need what is known as “grey energy” to manufacture energy generation systems, e.g. biofuel systems, solar cells, electric

car batteries, and wind turbines. Consistency strategies can incentivize industry to strive for a reduction in resource consumption and emissions.

Putting Consistency Strategies into Action: Examples

Renewable energy systems: The transition from fossil fuels to renewable energy sources such as solar energy, wind energy, and hydropower is an effort to align energy production with the natural rhythms and cycles of the environment.

Recyclable electronics: Designing electronic devices in a way that makes it easy to repair, upgrade and recycle them, to minimize resource consumption and environmental impact.

Sustainable construction: Constructing buildings using sustainable materials that have a low environmental impact and can be reused or recycled at the end of their useful life.

1.10.3 Sufficiency strategy

The concept of sufficiency, rooted in the Latin word *sufficere* (to suffice), challenges our assumptions about how much we need for a good life. It seeks to reduce resource and energy consumption by focusing on reducing the demand for resource-intensive goods and services. Sufficiency, which could also be described as frugality or adequacy, is critical of the constant creation of new needs through technology and advertising, especially given the limited supply of natural resources. Sufficiency urges us not to chase after every newly created need, and to fulfil our needs without consumption. While efficiency and consistency strategies focus on production, sufficiency focuses on consumption, albeit not exclusively: sufficiency can be practised to varying degrees and at different levels, from small changes in behaviour (sharing instead of buying) to significant changes in lifestyle (giving up air travel). Although it starts at an individual level, sufficiency can be applied at various levels, such as by companies (sufficiency-oriented product design) and governments (sufficiency policies). Sufficiency therefore seeks the right balance: How much do we need for a good life? And what do we not need?

Critics believe the sufficiency strategy has only limited potential and is unlikely to find broad sociocultural acceptance. However, proponents see it as crucial for sustainability policy, especially where efficiency and consistency strategies fall short. The concept of a post-growth economy draws on the principles of the sufficiency strategy.

Putting Sufficiency Strategies into Action: Examples

Sharing and communal use: Platforms and initiatives for sharing objects, tools, or vehicles make it possible to use resources more efficiently and reduce the number of products manufactured.

Plant-based diet: A predominantly plant-based diet is less resource-intensive than a meat-based one.

Reduction in working hours: Shortening the workweek can lead to a decrease in resource consumption, as producing fewer goods and services will save on energy and raw materials.

Buy local: Supporting local producers and markets helps to reduce transport routes and emissions.

1.10.4 The Rebound Effect: A flipside of efficiency



Figure 1.13: Rebound effect illustrated using the example of the automotive industry. Source: [@fietsprofessor](#) on LinkedIn.

“Any intelligent fool can make things bigger, more complex, and more violent. It takes a touch of genius – and a lot of courage to move in the opposite direction.” (Schumacher 1973)

Critics of the efficiency principle warn of the rebound effect, also known as the Jevons paradox (Jevons 1865/1965). This phenomenon occurs when efficiency improvements lead to lower prices for goods and services, thus increasing demand and negating the initial benefits of lower resource use.

Take this simple example: If passenger cars become cheaper due to improvements in efficiency, then people may choose a larger model when they next purchase a car. In addition, a fuel-efficient car is cheaper to run, as it requires less fuel per kilometre. This usually incentivizes people to drive more (either by making more frequent journeys or travelling longer distances) and reduce their use of public transport or their bicycle. As a result, efficiency gains that are technically possible are often not realized in practice, as the product in question is used more frequently or more intensively. Apart from the direct change in the use of the product (direct rebound), there may be additional changes in consumer behaviour that affect the environment. In this example, this could mean that the money saved on purchasing or using the car is spent on air travel instead (indirect rebound), which in turn cancels out some of the environmental benefits.

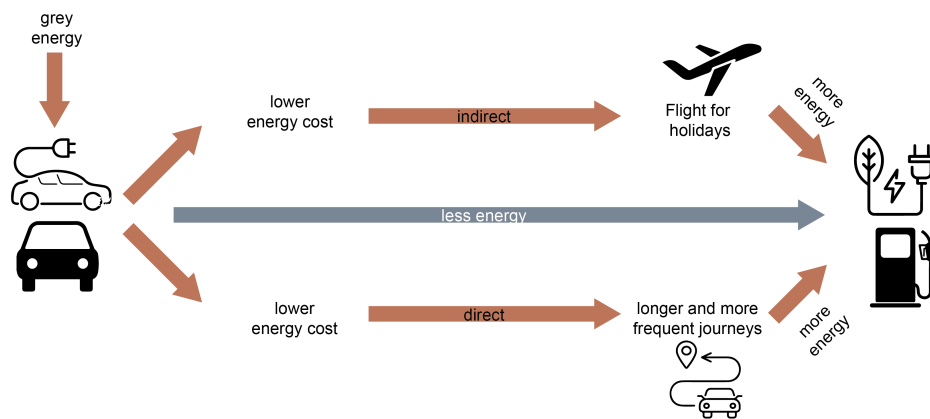


Figure 1.14: Indirect and direct rebound effects of energy-saving passenger cars. Source: Own illustration.

To understand the rebound effect, the literature focuses on the following three areas: 1) financial factors, 2) socio-psychological influences, and 3) regulatory effects, which we analyse in more detail below.

1. Financial factors are a major cause of the rebound effect. When efficiency leads to cost savings, financial resources are freed up. This can lead to people either consuming more of the same product (direct rebound effect) or investing in alternative goods (indirect rebound effect). The financial rebound effect can be based on several factors, including:
 - The income effect: when efficiency leads to real income gains for consumers, it can result in a direct or indirect rebound effect.
 - The reinvestment effect: when companies use cost savings to expand their production or invest in other products.
 - The market price effect: the macroeconomic effect in which demand in one sector stimulates demand in other sectors. For example, as cars become more efficient, demand for fuel may fall, making fuel cheaper and affecting demand in other energy-consuming sectors.

The extent and intensity of the financial rebound effect depend on factors such as price elasticity and consumer behaviour. While this happens at the individual level, it can cumulate into a macroeconomic market price effect with far-reaching consequences.

2. Socio-psychological influences include the “mental accounting” of individual consumers. For example, consumers who have purchased an efficient product may “allow themselves” to consume more. This is why increases in the efficiency of products that were previously considered harmful to the environment can, paradoxically, lead to an increase in consumption. This direct rebound effect is often referred to in social psychology as the “moral hazard effect”. If the decision to increase consumption is not made rationally, this is known as the “moral leaking effect”. For example, if you drive more after buying a

resource-efficient car, because the purchase alone is perceived as environmentally conscious. The indirect rebound effect can also be attributed to socio-psychological factors. The “moral licensing effect” states that the purchase of resource-efficient products can lead to an increase in the consumption of other products that may be harmful to the environment. For example, when the purchase of a fuel-efficient car is used to justify more frequent air travel.

3. Sometimes, government regulations or incentives to promote efficient technologies can have unintended consequences – this is what we mean by the “regulatory effect”. For example, if people are encouraged to purchase large electrical appliances devices due to government incentives for energy-efficient models. Such incentives may result in more frequent purchases of appliances, which – even if the newer model is more efficient than the old – partially cancels out the energy-savings effect. Furthermore, the introduction of efficiency technologies leads to new capacities and infrastructures, which can result in the creation of new markets. Wind turbines, for example, promote the development of new infrastructures and jobs, but at the same time require considerable resources. In addition, consumers who purchase a more efficient product may not necessarily discard their old product. Some end up using both products, representing an increase in overall consumption (Santarius 2014).

Rebound effects

The **direct rebound effect** occurs when improved energy efficiency makes it cheaper to consume a resource, leading people to consume more of it. For example, a more fuel-efficient car would be cheaper to run. This might encourage people to drive more, partially or completely cancelling out the energy savings from the efficiency increase.

The **indirect rebound effect** occurs when energy savings or resource efficiency gains in one area lead to the use of the freed-up resource in another area. This reduces the overall savings achieved. For example, if the money saved from more energy-efficient living is used instead for more frequent flying.

1.10.5 How big is the rebound effect?

It is difficult to know exactly how big the rebound effect is: empirical estimates vary widely, depending on the methods used and the effects taken into account. It is particularly challenging to clearly distinguish rebound effects from growth or structural changes. The type of product or service also influences the magnitude of the rebound effect. For example: If a commuter switches to a more efficient car, their costs decrease, but their available time remains fixed. Because a day contains only 24 hours, the commuter cannot increase their commuting time indefinitely. The rebound effect in this scenario is therefore relatively small. The situation is different for leisure activities, such as travelling by air. If costs fall due to more efficient aircraft or cheaper prices, there is a greater incentive to fly more often – for example, taking a second weekend trip instead of just one per year. As time is not such a tight constraint here, additional consumption can increase significantly – as can the rebound effect. Another important component is the degree of saturation with goods and services. Observations show that rebound effects tend to be lower in

high-income countries than in developing countries, where there is still a considerable need for additional consumption.

For example, the direct rebound effects associated with improved space heating efficiency can result in energy savings being 10-30% lower than what is technically achievable (e.g. Hediger et al. 2018). The rebound effects of transport vary even more. According to Andersson et al. (2019), the rebound effect related to private mobility might reduce potential savings by 7.5% in Denmark, 30% in Sweden, and 60% in Germany. In contrast, studies on lighting in private households have found very low rebound effects, of less than 10% (Sorrell 2007). Consequently, actual energy savings for these services are, on average, up to 25% lower than what is technically possible and predicted. Ultimately, the magnitude of the rebound depends on the specific context and can be mitigated through targeted policy measures.

This means that the actual energy savings for these services can be, on average, up to 25% less than what is technically possible and predicted. However, the exact magnitude of the rebound depends on the specific context and can be reduced by choosing appropriate measures.

Sometimes, albeit rarely, the savings may be overcompensated. This is known as the “backfire” effect. However, this is an exception, and the “backfire” effect is not a pure rebound effect, as it is linked to the effects of growth and structural change. A good example is digitalization, which can lead to short- and medium-term backfire effects (Peng et al. 2023).

1.11 Sustainability as a scientific discipline

What does “sustainable” mean? We could argue that sustainability means the existence and functioning – for as long as possible – of structures of various degrees of complexity. There are, in fact, many structures in the world that have existed for relatively long periods of time, such as atoms, certain ecosystems, life on Earth as a whole, or planetary systems and galaxies. But why are some structures relatively stable and able to exist “sustainably”, despite being exposed to regular disturbances that jeopardize their existence or function? Are there principles or mechanisms that could explain the continued existence of these diverse systems? And if so – could human societies learn from these principles?

As early as the mid-20th century, scientists began to recognize that fundamental principles connect all aspects of existence, shaping the universe’s order and dynamics. A key figure in this field was Karl Ludwig von Bertalanffy (1901–1972). His research began with his reflections on the “formation of form” in nature (1928), which he further developed in publications on the structure of life (1937), molecules and organisms (1940), and biology (1949). In 1950, he introduced a theory of open systems in physics and biology, a concept that significantly influenced the development of biophysics, fluid equilibrium, and modern developmental theories. His seminal work, *General System Theory*, was published in 1968.

Systems theory laid the foundation for a revolutionary world view that not only provided explanations for the development and function of so-called systems, but also integrated isolated knowledge from various disciplines. In 1975, in a significant contribution to systems theory, Ervin Laszlo distinguished between seven different system types (physico-chemical and biological systems, organ systems, socio-ecological and socio-cultural systems, and organizational and

technical systems). Today, systems theory provides the foundation for modelling complex systems and for cybernetics, which focuses on control and communication mechanisms within systems. The Club of Rome report used systems theory to analyse the growth of processes in the context of positive feedback.

1.11.1 Systemic effects: Escalation and feedback

In systems theory, the mechanisms of positive and negative feedback are used to explain the dynamics and stability of systems. Positive feedback reinforces changes in a system, while negative feedback tends to balance changes, returning the system to a stable state.

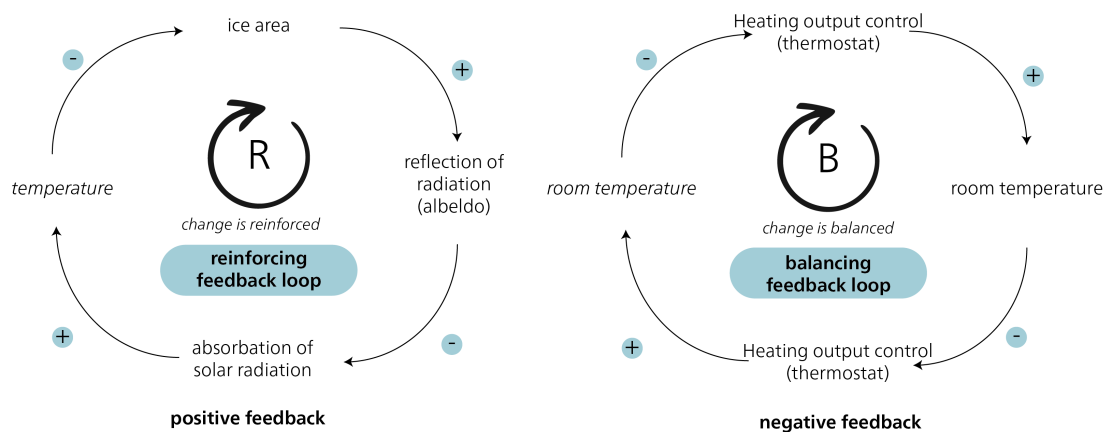


Figure 1.15: Positive/reinforcing and negative/balancing feedback loops. Source: Own illustration.

An example of a **positive feedback loop** is Arctic albedo amplification. Arctic ice reflects a significant portion of solar radiation back into space, limiting heat absorption by the ocean. However, as the ice melts, the darker seawater absorbs more heat, further warming the ocean and causing the remaining ice to melt more rapidly. This process is self-reinforcing, as reduced ice cover lowers the albedo, which in turn leads to greater solar absorption and further melting. This positive feedback has serious consequences for the climate system, as dwindling sea ice contributes to accelerated sea level rise and shifts in oceanic and atmospheric circulation patterns. The extent and speed of this occurrence took scientists by surprise (Stroeve et al. 2007).

An example of **negative feedback** is a thermostat that regulates the temperature in a room. If the room temperature rises above the set maximum, the thermostat recognizes this and activates the air conditioning to lower the temperature. As soon as the temperature reaches the set value, the thermostat switches off the air conditioning. This control circuit maintains the room temperature at a constant level.

Positive and negative feedback often work together and can cause complex behaviour in systems. Understanding these feedback mechanisms makes it possible to better understand and predict the interactions and behaviour of systems.

1.12 Sustainability science: origins and understanding

An emerging awareness – influenced not least by the publication, *The Limits to Growth* (Meadows et al. 1972) – led to increased research into global environmental problems. Three major international research programmes are particularly noteworthy here: 1) the International Geosphere-Biosphere Programme (IGBP) from 1987 to 2015; 2) the World Climate Research Programme (WCRP), founded in 1980; and 3) the International Research Programme on Biodiversity (DIVERSITAS), founded in 1991 and renamed in 2014 to Future Earth. Not only did these programmes initiate important basic research – they also provided an important impetus for greater consideration of the results in policy, as reflected, for example, in the UN's Brundtland Report and Agenda 21. Notably, these research programmes systematically brought together previously separate disciplines of the natural sciences for the first time, although they still largely neglected the interaction between society and the environment. As the interconnectedness of these topics began to gain traction, several research projects promoting their integration were launched, particularly in the US.

“A new field of sustainability science is emerging that seeks to understand the fundamental character of interactions between nature and society. Such an understanding must encompass the interaction of global processes with the ecological and social characteristics of particular places and sectors.” (Kates et al. 2001, 641)

The concept of sustainability science was officially introduced in 2001 at the “Challenges of a Changing Earth” congress in Amsterdam, by the International Council for Science (ICSU), the above-mentioned IGBP, the International Human Dimensions Programme on Global Environmental Change (IHDP), and the above-mentioned WCRP. The German-language term “Nachhaltigkeitswissenschaft” (sustainability science) can be traced back to a translation from English.

1.12.1 Understanding sustainability science

An oft used motto – “society has problems – universities have disciplines” – epitomizes the conventional structure of science into various disciplines. However, disciplinary science appears to be inadequately equipped to meet the challenges of sustainable development. Two main approaches to understanding sustainability science have emerged: on the one hand, it is seen as an independent discipline with its own theories and methods; on the other, as an amalgamation of different disciplines focusing on a common theme (Clark and Dickson 2003). While the first approach is not yet considered accurate, there is broad agreement that sustainability science should be seen as a platform that brings together science, practice, and visions – and incorporates contributions from the entire spectrum of natural, economic, and social sciences (Martens 2006, 38).

Sustainability science is distinct in two key ways. First, unlike traditional (“Mode 1”) research, it isn't driven by an original scientific programme. Instead, it is guided by the normative concept of sustainability, using this as a framework for scientific analysis. Second, as a distinct type of problem-oriented research, it differs from both basic and applied research. Clark (2007) introduced the term “use-inspired basic research”, which seeks to strike a balance by seeking new knowledge while also considering its potential applications.

Table 1.3: Mode-1 and Mode-2 science. Source: Based on Gibbons et al. (2010) and Martens (2006).

Dimension	Mode-1 science	Mode-2 science
Orientation	Academic	Academic and social
Disciplinarity	Mono-disciplinary	Trans- and interdisciplinary
Governance	Technocratic	Participative
Knowledge status	Certain	Uncertain
Approach	Predictive	Exploratory

1.12.2 Empirical-analytical vs normative

As sustainable development is a normative model, it is important to distinguish between descriptive (empirical-analytical) research and prescriptive (normative) research. Descriptive statements describe the current state (“is”), while normative statements describe the ideal state (“ought”). In everyday life, the distinction between “is” and “ought” is often not recognized, but it is crucial for science.

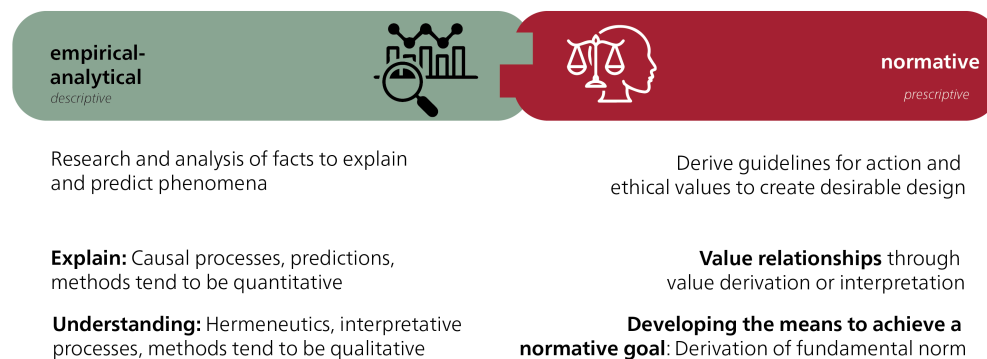


Figure 1.16: Distinction between empirical analytical and normative. Source: Own illustration.

It is important to note that “is” and “ought” statements have different qualities and focus areas, and they cannot simply be linked without further explanation. This understanding forms the basis of Hume’s law, which declares that normative statements about “what ought to be” cannot be derived from descriptive statements about “what is”. To give an extreme example: The fact that around 24,000 people starve to death every day does not allow any conclusions to be drawn as to whether this is good or bad. Without an evaluation, no normative statement can be made about whether this should be prevented. Hume pointed out that many scientists of his time often neglected this crucial distinction, and this is still the case today:

“In every system of morality, which I have hitherto met with, I have always remark’d, that the author proceeds for some time in the ordinary ways of reasoning, and establishes the being of a God, or makes observations concerning human affairs; when

of a sudden I am surpriz'd to find, that instead of the usual copulations of propositions, *is*, and *is not*, I meet with no proposition that is not connected with an *ought*, or an *ought not*. This change is imperceptible; but is however, of the last consequence. For as this *ought*, or *ought not*, expresses some new relation or affirmation, 'tis necessary that it shou'd be observ'd and explain'd; and at the same time that a reason should be given; for what seems altogether inconceivable, how this new relation can be a deduction from others, which are entirely different from it... [I] am persuaded, that a small attention wou'd subvert all the vulgar systems of morality, and let us see, that the distinction of vice and virtue is not founded merely on the relations of objects, nor is perceiv'd by reason." (Hume 1739, bk. 3, Part 1, Section 1)

The realization that *is* and *ought* statements differ does not mean that scientists cannot make *ought* statements. Rather, it means that due to the epistemological differences between the two categories, certain rules must apply when they are linked. Hume tried to establish the relationship between these two statement types through so-called "analytic bridge principles". Max Weber further formalized these considerations by replacing the bridge principles with the concept of "value relations". Weber's call for value freedom (or value neutrality) in empirical science stems from his observation that there is a clear distinction between factual questions ("*is*") and value questions ("*ought*") (Weber 1999, 245f, 509f). He argued that it is not the task of the empirical sciences to make value judgements. If scientists fail to make an analytical distinction between their personal values and their research goals, this could jeopardize the ability of science to produce verifiable and meaningful results (Weber 1999, 150f). In turn, it could lead to a blurring of the boundaries between scientific knowledge and knowledge from politics, religion, or everyday discussions.

Weber's call for value freedom in empirical science was often misunderstood (Weber 1999, 499f). It does not mean that research should do without values. (Non-scientific) values have a place in empirical research, particularly in the context of sustainable development, in the following ways:

- **Values can be an object of inquiry (research object):** Values play a key role in the field of environmental and sustainability psychology, for example. In discussions about lifestyles and subjective well-being, researchers study the personal values and beliefs of individuals.
- **Values influence the choice of research topic:** Values influence the selection of research objects and questions (epistemological interest), and thus the direction of research. This is particularly relevant in application-oriented and transdisciplinary research, where the needs and information of relevant social actors are integrated into the research process, for example through *joint problem framing* or *participatory vision development* (Schneider et al. 2019).
- **Value interpretations should be clearly stated:** Researchers can create value relationships, or make value interpretations (cf. Weber 1999, 522). A value interpretation, based on the hermeneutic method, concretizes the meaning of overarching value concepts and derives logical values and norms from them. For example, researchers can analyse the SDGs by examining the values underlying sustainable development. Other examples include norms derived from the concept of justice – such as the norm of human responsibility towards our environment, our social environment, and ourselves (see Michelsen and Adomßent 2014, 25; Schneider et al. 2019). It is important to make the original value explicit.

- **Developing the means to achieve a normative goal:** Sustainable development research often relies on value judgements to guide its pursuit of evidence-based solutions. This is comparable to political science, where normative theories aim to construct an ideal – i.e. just – form of government, based on the normative evaluation of what constitutes a just society. Science cannot prove the value of justice, but it can develop strategies and means to achieve this desired goal. This relationship between ends and means represents a logical connection between values and normative statements. The ultimate goal, or end, is determined by extra-scientific values. The researchers have to clearly state this relationship, and formulate their means accordingly. For example, an end–means connection could look like this: “To achieve goal X, Y is the most successful means under conditions b1, b2, and b3”. Sustainable development can be considered as an extra-scientifically given fundamental norm. For example, if research wants to make statements about how sustainable development can be achieved through solar energy, it could proceed as follows: “Switching to solar production could be a promising way of achieving sustainable development. This would organize production and distribution according to the needs of many people, minimize social inequalities, and create conditions for individual well-being.”

1.13 Types of knowledge for sustainable development

Sustainability research focuses on challenges that threaten the long-term safeguarding of social development conditions. In the context of SD as a challenge for society as a whole, the Forum for Climate and Global Change of the Swiss Academy of Sciences has described three different types of knowledge (ProClim 1997): systems, target, and transformation. These three knowledge types take place at three fundamental levels: 1) the analytical level, which aims to generate systems knowledge; 2) the normative level, which develops target and orientation knowledge; and 3) the operational level, which seeks to generate design or transformation knowledge. Figure 1.17 shows the three knowledge types in relation to empirical-analytical and normative research.

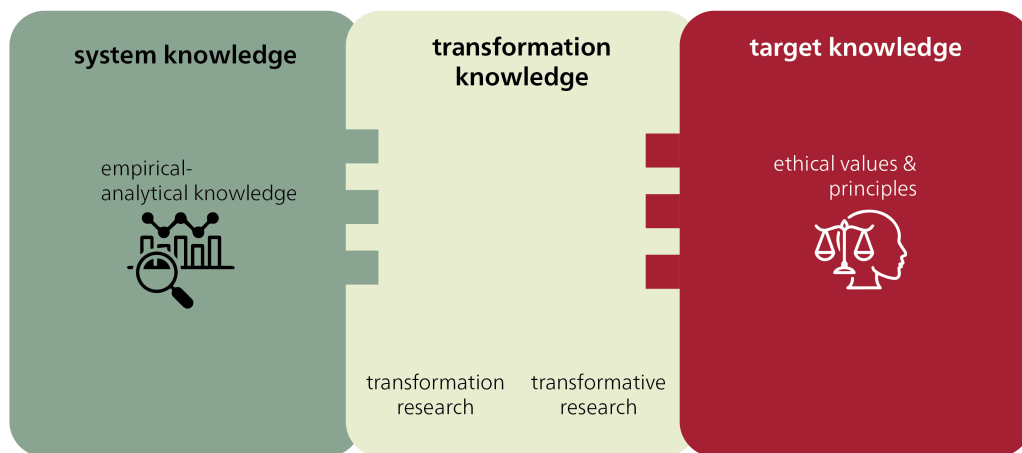


Figure 1.17: The three types of knowledge. Source: Own illustration based on ProClim (1997).

Systems knowledge refers to the understanding of the current state. In relation to sustainable development, research examines relevant structures and processes, describing and explaining their complexity and dynamics. It uses various methods, including qualitative and quantitative approaches, mixed methods, experiments, and systems dynamics approaches. It also uses forecasts, retrospectives, and scenario analyses. Most of the studies that generate systems knowledge fall within the field of empirical research.

Transformation knowledge is the knowledge of how we can get from the current state (“is”) to the desired state (“ought”). This requires an integration of both systems and target knowledge. In other words, research is looking for demonstrably effective solutions to the problems identified in the context of sustainable development. There are many methods to generate transformation knowledge.

Research on transformations, i.e. the change in social organization at various levels, can be empirical as well as normative. Research focusing on past and present transformations is empirical, while research focusing on concrete goals is normative. Accordingly, the German Advisory Council on Global Change (WBGU), distinguishes between “transformation research” and “transformative research”.

Another concept that is emphasized in the context of transformations is “transformative education”, which seeks to promote skills for a comprehensive and multi-layered understanding of change (Schneidewind 2013). As such, transformative education aims to strengthen society’s ability to reflect on issues, for example in terms of observation and in actively shaping transformation processes.

Target knowledge refers to the knowledge of what should be achieved and what should be avoided. Normative research is particularly important in this context. It is about interpreting and operationalizing the values of sustainable development and deriving further standards. Research

into the definition of threshold values is just as relevant as methods for assessments and visioning processes (cf. Swart et al. 2004; Wiek and Iwaniec 2014). Target knowledge gives us the answer to the question of what we should transform. CDE's study programmes are primarily guided by the following concepts of target knowledge for a sustainable economic and social system:

- The principles of sustainable development
- The 2030 Agenda and the SDGs (2015)
- The WBGU's Flagship Report "World in Transition: A Social Contract for Sustainability" (2011)
- The Global Sustainable Development Report GSDR (2019)

1.14 Sustainability Basics: The Quiz – Chapter 1

Which statement best defines "sustainable development" according to the Brundtland Report?

- ☐ Economic growth without environmental harm
- ☐ Development that meets present needs without compromising future generations
- ☐ Reducing CO₂ emissions by 50% before 2030
- ☐ The improvement of living standards through industrialization

The Brundtland Report (1987) defines sustainable development as development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

- False
- True
- False
- False

What is normativity?

- ☐ Refers to values and morally justified views about what should be done and how things should be
- ☐ Refers to the way things are right now
- ☐ Means that actions should be based on jointly agreed laws and norms

Normativity refers to values, norms, and morally justified views about what ought to be done and how things should be, rather than describing how things currently are.

- True
- False
- False

What is the "Great Acceleration"?

- ☐ The exponential growth in socioeconomic and Earth system trends starting in the mid-20th century
- ☐ A rapid increase in global resource use following World War I
- ☐ The rapid decline of industrialization during the 1960s

- ☐ The sudden reduction in biodiversity in the early 21st century

The “Great Acceleration” refers to the rapid and exponential increase in socioeconomic activities (such as population growth, industrialization, and resource use) and corresponding changes in Earth system processes that began around the mid-20th century.

- True
- False
- False
- False

What distinguishes “strong” from “weak” sustainability?

- ☐ Both concepts mean the same.
- ☐ Strong sustainability allows full substitution of natural capital.
- ☐ Strong sustainability requires preserving natural capital as non-substitutable.
- ☐ Weak sustainability restricts all technological innovation.

Strong sustainability assumes that certain ecological functions and natural capital stocks are critical and cannot be substituted by human-made or technological capital, whereas weak sustainability allows for substitution.

- False
- False
- True
- False

Which statements correctly describe the difference between strong and weak sustainability? (More than one statement may be correct.)

- ☐ Strong sustainability accepts unlimited substitution between natural and human-made capital.
- ☐ In weak sustainability, different forms of capital can substitute each other.
- ☐ Weak sustainability assumes that technological innovation can fully replace nature’s functions.
- ☐ In strong sustainability, natural capital is considered non-substitutable.

Weak sustainability is based on the assumption that different forms of capital (natural, human-made, social) can substitute each other and that technological progress can compensate for the loss of natural capital. Strong sustainability, in contrast, holds that natural capital is essential and cannot be substituted without risking irreversible damage.

- False
- True
- True
- True

Which of the following statements describe weak sustainability? (More than one statement may be correct.)

- ☐ Insufficient environmental protection.
- ☐ Giving priority to the needs of the current generation over the needs of future generations.
- ☐ The irreplaceability of human-made capital.
- ☐ The preservation of the total amount of all forms of capital from one generation to the next.
- ☐ The preservation of natural capital when it promotes or preserves human well-being.
- ☐ A company only fulfilling its statutory environmental and social responsibilities.
- ☐ Forms of capital being, in principle, mutually exchangeable.
- ☐ Increasing GDP.

Weak sustainability focuses on maintaining the overall stock of capital across generations, assuming that different forms of capital can substitute for one another. Natural capital is preserved primarily insofar as it contributes to human well-being, rather than being considered intrinsically irreplaceable.

- False
- False
- False
- True
- True
- False
- True
- False

Which of the following statements describe strong sustainability? (More than one statement may be correct.)

- ☐ The sixth wave of extinctions is considered to be the most important sustainability problem.
- ☐ An increase to the total amount of all forms of capital is called for.
- ☐ Environmental problems are resolved through innovative energy technologies.
- ☐ Preserving critical natural capital from one generation to the next is required.
- ☐ Nature is viewed as a form of capital that contributes to the well-being of people in an invaluable way.
- ☐ Natural capital and human-made capital complement each other and humans cannot survive without both.

Strong sustainability emphasizes the essential and non-substitutable role of natural capital for human survival and well-being. It requires the preservation of critical natural capital across generations and recognizes that nature provides indispensable contributions to human well-being that cannot be fully replaced by human-made capital or technological innovation.

- False
- False
- False
- True
- True
- True

What is the primary goal of the consistency strategy in sustainability?

- ☐ Reducing energy consumption in production
- ☐ Meeting consumer needs through environmentally sustainable methods
- ☐ Reducing social inequality
- ☐ Minimizing the use of raw materials in production

The consistency strategy aims to align economic activities and consumption patterns with natural systems by designing products, processes, and services in ways that are environmentally sustainable while still meeting human needs.

- False
- True
- False
- False

Which of the following statements about sustainability strategies are correct? (More than one statement may be correct.)

- ☐ Efficiency aims to use fewer resources per unit of output.
- ☐ Sufficiency encourages higher production levels.
- ☐ All three sustainability strategies have the same effect on consumption.
- ☐ Consistency means producing goods that can be recycled infinitely.

Efficiency focuses on reducing resource use per unit of output. Consistency aims at aligning production with natural cycles, for example through closed-loop or infinitely recyclable systems. Sufficiency questions levels of production and consumption, and the three strategies therefore differ in how they affect consumption.

- True
- False
- False
- True

Part II

Approaches to Sustainability

2 Approaches to Sustainable Development

The world is facing numerous challenges of sustainable development, including pressing environmental problems and social inequalities. Scientists, researchers, and activists are seeking innovative approaches to enable sustainable and equitable change. This chapter examines some of these approaches, which offer a complete rethink of current paradigms.

Approaches discussed in this chapter:

- Planetary boundaries framework
- Doughnut economics
- Approaches to a “great transformation”
- Green economy
- Post-growth approaches
- Implementing the 2030 Agenda

2.1 Debates about planetary boundaries

In 1798, British economist Thomas Robert Malthus published his influential essay, *An Essay on the Principle of Population*. Malthus’s core idea was that population growth would surpass Earth’s ability to produce food, sparking a debate on planetary carrying capacity that continues today. *The Limits to Growth* (1972), a report by the Club of Rome, built on Malthus’s ideas. It went beyond just food supply to consider a broader range of factors in calculating a planetary limit. These factors included resource availability, environmental pollution, and industrial output. The report remains relevant from an environmental point of view, as it challenges the assumptions of limitless growth, even though its specific predictions haven’t quite come true.

A significant development in sustainability science is the planetary boundaries concept, introduced in 2009. Unlike earlier theories focused solely on population limits, this framework uses Earth system science parameters. Researchers identified the Holocene epoch as a baseline, as this was a period of remarkable stability for human development. Significant deviations from this ideal state could push humanity towards uncertain “tipping points” – critical thresholds that can either interrupt previous progress, alter its course, or even accelerate it in unintended ways. An example might be the extinction of megafauna at the end of the last ice age, potentially linked to human arrival in the Americas. The planetary boundaries concept promotes the precautionary principle, urging action to minimize potential harm to both humans and the environment.

The concept, which puts forward nine planetary boundaries, was first introduced by Rockström et al. (2009). It was updated by Steffen et al. (2015). A recent update by Richardson et al. (2023) shows that six out of nine planetary boundaries have already been transgressed.

The planetary boundaries framework identifies nine critical Earth system processes. These processes regulate the planet's stability and include, for example, land system change and ocean acidification (see all planetary boundaries in Figure 2.8). Each boundary has an inner circle, within which it can operate safely ("safe operating space") and an outer circle, which represents increased uncertainty.

The latest update to the planetary boundaries framework paints a concerning picture. We're close to overstepping the safe operating space for ocean acidification, and regional atmospheric aerosol loading has already crossed its boundary. In a positive development, stratospheric ozone levels show some signs of recovery. However, the overall situation is alarming. The boundaries previously identified as transgressed (climate change, biosphere integrity (genetic diversity), land system change, and biogeochemical flows [N and P]) have all seen a worsening of their transgression since 2015.

The study added human appropriation of net primary production as a control variable for the functional component of biosphere integrity, arguing that this boundary has also been exceeded. In addition, the significant transgression of the planetary boundaries for phosphorus and nitrogen cycles, along with genetic biodiversity, raise the risk of fatal consequences.

Two of the nine planetary boundaries – biosphere integrity and climate change – are considered "core boundaries". These core systems encompass processes from many other subsystems and operate at a global scale. Reaching tipping points in these core systems could therefore push the entire Earth system into a new state.

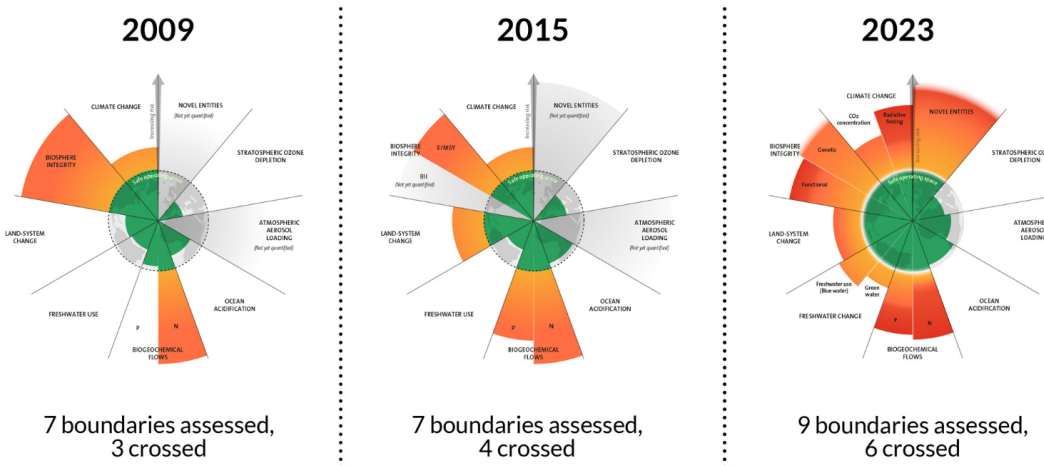


Figure 2.1: Development of control variables for all nine planetary boundaries. Source: Azote for Stockholm Resilience Centre, Stockholm University. Based on Rockström, Steffen, Noone, Persson, F. S. I. Chapin, et al. (2009), Steffen et al. (2015), Richardson et al. (2023). Licensed under CC BY-NC-ND 3.0.

2.1.1 Tipping points of the Earth's climate system

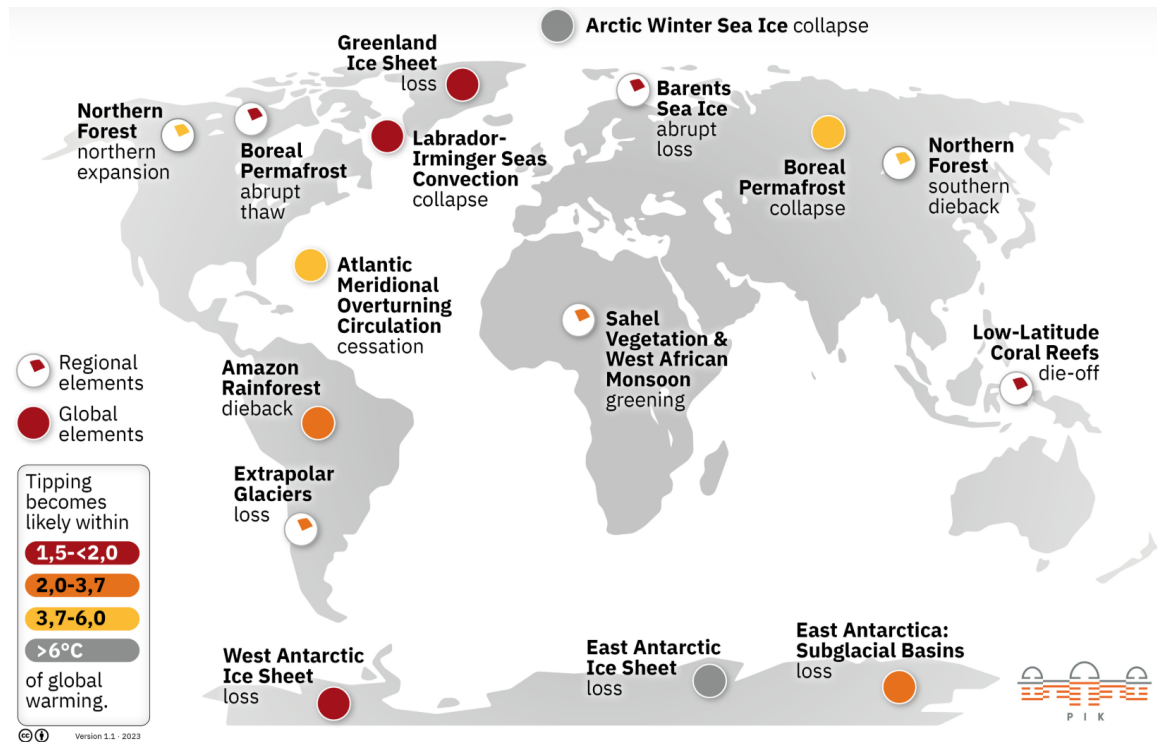


Figure 2.2: Spatial distribution of global and regional tipping elements. The colours indicate the temperature range in which tipping is likely to occur. Source: Designed at PIK, based on Armstrong McKay et al. (2022), expanded with interactions based on Lenton et al. (2019). Licensed under CC-BY.

Tipping points in the Earth's climate system are critical thresholds that, when crossed, can cause abrupt and often irreversible changes in the climate system. These tipping points can destabilize the climate and lead to accelerated climate change. Examples of tipping points include the melting of the Greenland ice sheet, the collapse of the Amazon rainforest, the thawing of permafrost soils, and changes in the Gulf Stream. If these tipping points are reached or exceeded, they can trigger self-reinforcing feedback effects that lead to further warming and an intensification of climate change.

The concept of tipping points emphasizes the urgency of limiting global warming and reducing greenhouse gas emissions. If we exceed the tipping points, it will become increasingly difficult to control climate change and minimize its effects.

2.1.2 Quantifying planetary boundaries

A recent development within the planetary boundaries framework is the concept of “safe and just Earth system boundaries (ESBs)” for the following domains: climate, the biosphere, water and nutrient cycles, and aerosols at global and subglobal scales (Rockström et al. 2023). The ESBs are based on modelling and literature review, and account for uncertainty through different levels

of likelihood. Staying within the ESBs protects stability and equity between species and future generations, although current generations, especially vulnerable groups, could still suffer harm. The authors therefore suggest stricter boundaries in some cases, and the addition of local standards to protect current generations and ecosystems. For example, they identify safe ESBs for warming (Rockström et al. 2023, fig. 1 and Table 1). These are based on reducing the probability of triggering climate tipping points, maintaining biosphere and cryosphere functions, and considering climate variability of the Holocene ($<0.5-1.0^{\circ}\text{C}$) and earlier interglacial periods ($<1.5-2^{\circ}\text{C}$).

The functions of the cryosphere include the preservation of permafrost in the northern high latitudes, the preservation of polar ice sheets and mountain glaciers, and the minimization of sea ice loss. The authors conclude that global warming of more than 1.0°C above pre-industrial levels, which has already been exceeded (IPCC 2021), could trigger tipping effects such as the collapse of the Greenland ice sheet or a localized abrupt thawing of the boreal permafrost with a moderate probability (Armstrong McKay et al. 2022). Global warming of one degree Celsius corresponds to the safe limit proposed in 1990 and the PB of 350 ppm CO_2 (Steffen et al. 2015). With a warming of more than 1.5°C or 2.0°C , the likelihood of triggering tipping points increases to high or very high.

2.1.3 Climate resilience

Resilience describes the ability of a system to withstand disruptions, “bounce back”, or recover from adversity. Originally used in psychology, resilience refers to the psychological robustness that an individual has actively acquired in dealing with challenges or stresses, particularly in childhood. In the context of ecosystems, resilience refers to the ability to absorb disturbances without a permanent systemic collapse, i.e. a collapse that would result in a different system regulated by new processes (Folke et al. 2010). More recently, the concept of resilience has been extended to social systems (see Section 2.2, *From planetary boundaries to the doughnut model*). Studies focus on which specific characteristics of a region need to be strengthened, to better prepare it for future crises and disasters related to climate change, terrorism, resource scarcity, or financial crises. The climate crisis, for example, requires both adaptation and mitigation measures. Resilience approaches offer a way of combining these two concepts rather than playing them off against each other.

Climate mitigation measures aim to reduce greenhouse gas emissions and curb climate change. Such measures include promoting renewable energies, improving energy efficiency, and expanding public transport. Resilience approaches emphasize the importance of climate mitigation, as limiting the rise in temperature will help to reduce the intensity and frequency of extreme weather events.

Climate adaptation measures aim to make societies and ecosystems more resilient to the current and expected effects of climate change. Adaptation measures include the development of early warning systems for extreme weather events, coastal protection against rising sea levels, the reduction of heat stress (e.g. through more urban green spaces), runoff or infiltration areas to reduce the damaging effects of heavy rainfall events, or the adaptation of agricultural practices to changing climatic conditions. Resilience approaches emphasize the need for climate adaptation to protect communities and ecosystems from the negative effects of climate change.

2.1.4 Conclusion

The concept of planetary boundaries tries to reduce complex ecological relationships to a small number of quantifiable limits. These specific limits and indicators for planetary boundaries are based on scientific findings that are not always clear or consistent. The boundaries are therefore contested by some scholars, who question the accuracy and reliability of the data and models used. Despite this criticism, the planetary boundaries framework makes a valuable contribution to the debate on sustainable development and raises awareness of the limited resources and resilience of our planet. To summarize:

- The planetary boundaries framework focuses on the ecological/biophysical limits of the Earth's resilience, and thus the environmental dimension of sustainable development.
- These limits to resilience – the planetary boundaries – focus on environmental factors that are considered fundamental to human survival.
- In normative terms, the framework aims to maintain the stable Earth system (“state of equilibrium”) of the Holocene, thereby mitigating threats to human survival.
- The planetary boundaries framework can be used to set concrete targets in the environmental dimension (e.g. at the global or national level).

Further readings

- Folke, Carl, Stephen R. Carpenter, Brian Walker, Marten Scheffer, Terry Chapin, and Johan Rockström. 2010. ‘Resilience Thinking: Integrating Resilience, Adaptability and Transformability’. *Ecology and Society* 15 (4). <https://www.jstor.org/stable/26268226>.
- Lenton, Timothy M., Johan Rockström, Owen Gaffney, et al. 2019. ‘Climate Tipping Points — Too Risky to Bet Against’. *Nature* 575 (7784): 7784. <https://doi.org/10.1038/d41586-019-03595-0>.
- Rockström, Johan, Joyeeta Gupta, Dahe Qin, et al. 2023. ‘Safe and Just Earth System Boundaries’. *Nature* 619 (7968): 102–11. <https://doi.org/10.1038/s41586-023-06083-8>.

2.2 From planetary boundaries to the doughnut model

Economist Kate Raworth's Doughnut Model is an innovative approach to sustainable development that recognizes social and environmental limits. This requires rejecting much of what has characterized 20th-century economics, as Raworth outlines in her 2017 book, *Doughnut Economics: Seven Ways to Think Like a 21st Century Economist*. Raworth depicts the ideal economy of the future in a simple image: a ring-shaped doughnut. The outer ring symbolizes an ecological ceiling that should not be crossed, as doing so would cause irreversible harm to the environment. The inner ring represents a social foundation covering people's basic needs, such as food, housing, and income. The challenge is to ensure that economic activities take place within this ring – the

doughnut – to ensure the well-being of both humanity and the environment (“safe and just space for humanity”).



Figure 2.3: The Doughnut of social and planetary boundaries. Source: <https://doughnuteconomics.org>.

The philosophy of the doughnut approach is based on three core principles: equitable distribution of wealth, regeneration of the resources used by the economy, and creation of wealth for all people. None of these principles should have to depend on economic growth, says Raworth (2017b). In other words, we don't need to pursue unlimited growth for the economy to flourish – instead, we should pursue development that is sustainable, balanced, and equitable. The transition to the doughnut model requires a fundamental change in the way we think and act. It's about moving from a growth paradigm to sustainable development, a development that takes social justice and environmental sustainability into equal account.

2.2.1 Conclusion

Despite the Doughnut Model's focus on the economy, critics say it doesn't discuss the underlying framework conditions (i.e. the structures, rules, and institutional organization of the economy) or how money is created and managed (i.e. the financial sector). The Doughnut Model focuses on what kind of economic activity is desirable (i.e. staying within the doughnut), but it doesn't explain how to get there. These points of criticism are being addressed by the Doughnut Economics Action Lab (DEAL, n.d.), which provides tools and strategies to implement the Doughnut Model in real-world settings.

- The Doughnut model builds on the concept of planetary boundaries, adding a social dimension and including goals and degrees of goal achievement.
- In normative terms, the model prescribes that
 - social goals should be achieved without overstepping the planetary boundaries. The planetary boundaries provide the biophysical framework within which the social goals should be achieved.
 - when setting concrete goals, measures etc. at sub-global levels, the global social goals must also be taken into account.

2.3 The debate about the Great Transformation

Concepts of the “great transformation” refer to fundamental changes in social, economic, political, and ecological systems that are necessary to create a sustainable and just society.

“Great transformation” was a term used by Karl Polanyi in his 1944 analysis that the shift to a free market in the 19th century brought about profound social, economic, and political changes that fundamentally transformed people's lives and relationship with nature. Polanyi argued that unbridled market dynamics caused social and environmental problems, and that society needed to develop mechanisms to regulate and balance these problems. Similarly, today's concepts of the great transformation emphasize the importance of regulation, redistribution, and the development of alternative economic models to create a sustainable and just society. While Polanyi stressed the need for social protection measures and the importance of integrating markets into social structures, current concepts of the great transformation additionally aim to fundamentally change production and consumption patterns to reduce environmental impact.

2.3.1 Flagship WBGU report: *World in Transition*

The flagship report by the German Advisory Council on Global Change (WBGU 2011), *World in Transition*, is an important publication that addresses the challenges and opportunities of a sustainable transformation of society. It was first published in 1994 and has since been updated several times. The report analyses global environmental changes, including climate change,

biodiversity loss, and resource consumption, and it proposes concrete policies and social change to promote sustainable development.

World in Transition is significant to debates on the great transformation, as it argues for a far-reaching transformation of business, policy, and society. This vision extends beyond mere adaptation, demanding fundamental shifts in the structures and patterns of economic activity and daily life. The report underscores the urgency of driving forward the transition to a climate-friendly and environmentally sound economy and way of living. It emphasizes the critical role of a comprehensive sustainability policy. Finally, it identifies innovation, technology, education, institutional reforms, and international cooperation as key levers for achieving sustainable development.

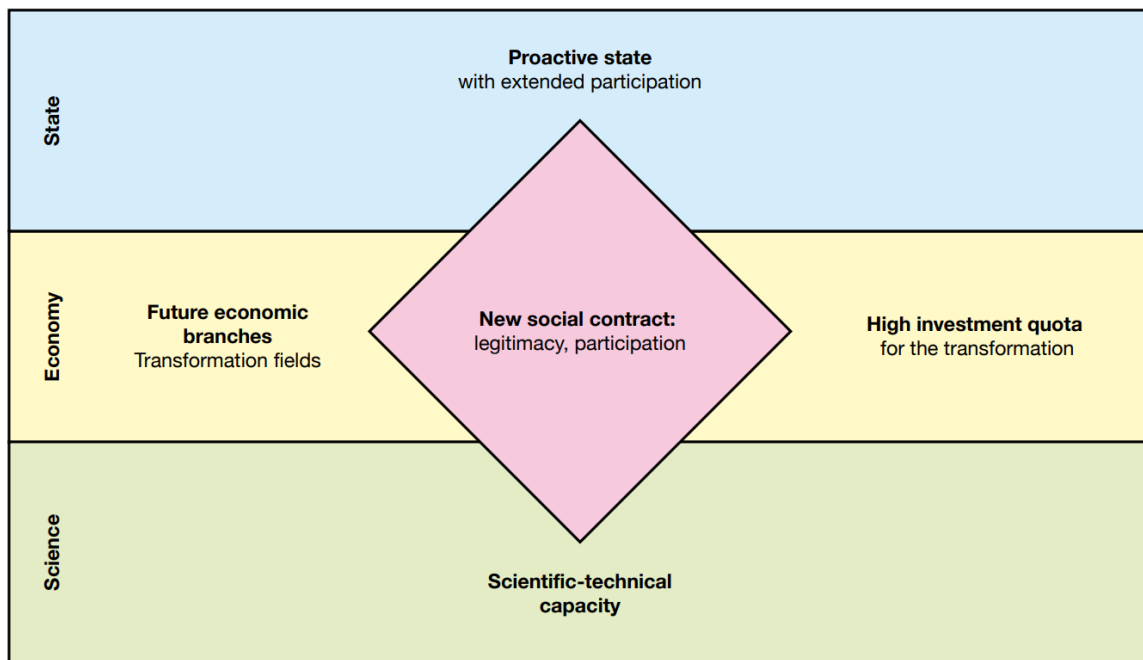


Figure 2.4: New social contract. Source: WBGU (2011, 275).

In Chapter 8, *World in Transition* distinguishes between two concepts: “transformation research and education” and “transformative research and education”. These concepts are key to promoting fundamental change towards sustainability.

Transformation research analyses processes of change, particularly in the context of sustainable development. It seeks to understand the dynamics and complexity of transformations, and to identify possible courses of action for a sustainable future. Transformation research offers an inter- and transdisciplinary perspective, and it involves different actors and stakeholders in the research process.

Transformation education is education that aims to impart the knowledge, skills, and values needed to achieve a sustainable transformation of society. It includes formal and informal educational measures that enable people to actively participate in transformation processes and to promote sustainable thinking and action.

Transformative research not only generates knowledge but also goes a step further by actively striving for change towards sustainability. It thus aims to influence social practice and develop solutions and innovations for sustainable development. Transformative research works closely with partners from practice and strives to translate knowledge into action.

Transformative education fosters a shift in mindsets, values, and behaviour towards sustainability. It goes beyond imparting knowledge, by also promoting critical awareness, empathy, and action for sustainable changes in society.

The WBGU believes that the state should take on tasks that are not adequately performed by individuals or the private sector. It recognizes that the state plays a decisive role in creating framework conditions and shaping political measures to promote sustainable development. The state can drive innovative solutions, steer investments, establish regulations, and implement policies that enable a sustainable transformation.

“The important thing for Government is not to do things which individuals are doing already, and to do them a little bit better or a little worse; **but to do those things which at present are not done at all.”** John M. Keynes, *The End of Laissez-Faire*, 1926

However, the WBGU also emphasizes that effective government intervention should take place in collaboration with other actors and stakeholders, including civil society, the private sector, and academia. Jointly shaping a sustainable future is about creating partnerships and involving a variety of expertise.

2.3.2 Schneidewind’s “art of the future”

The German economist and politician, Uwe Schneidewind, discusses the concept of the great transformation in his 2018 book, *Die grosse Transformation: Eine Einführung in die Kunst des gesellschaftlichen Wandels*, which translates as *The Great Transformation: An Introduction to the Art of Social Change*. Schneidewind (2018) points out that a great transformation isn’t dictated by the unstoppable development dynamics of modern societies or by a technocratic blueprint for an ecologically just society. Instead, he sees it as a process that should be actively shaped by many actors. As such, it’s important to have a clearly defined normative compass and to develop the ability to navigate complex social, cultural, economic, and technological processes. He describes this ability as the “art of the future” – the skill of making desirable futures possible – and thus builds a bridge to Harald Welzer’s “FuturZwei” approach. Welzer, a psychologist and sociologist, emphasizes the importance of imagination, creativity, and engagement in bringing about transformative change and developing alternative visions of the future.

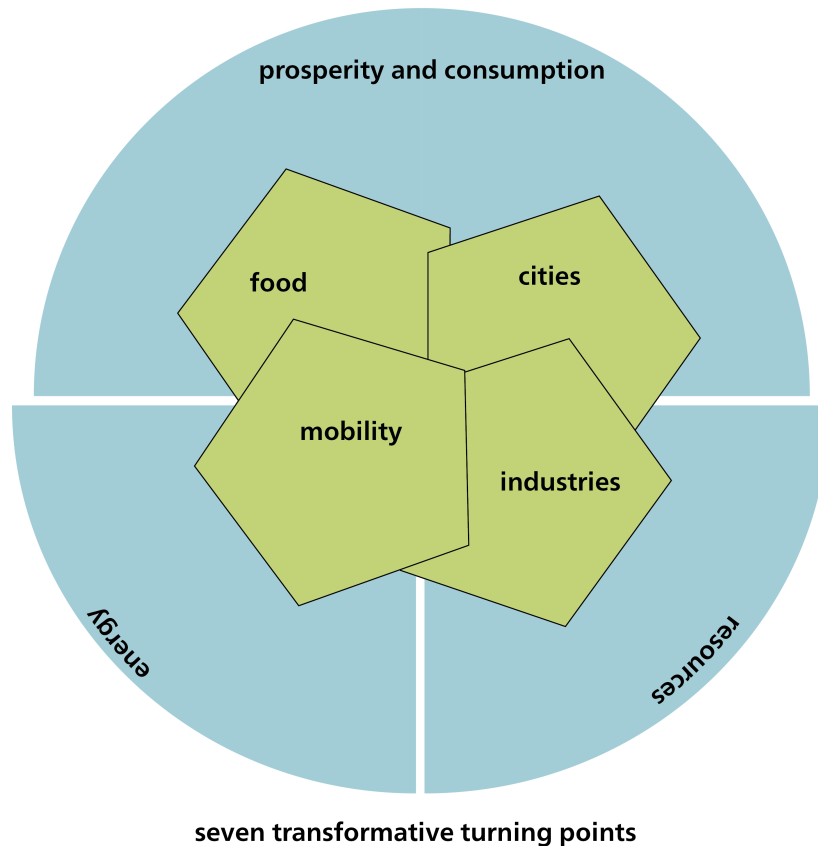


Figure 2.5: Understanding the interplay of the seven turning points of the great transformation. Source: Own illustration based on Schneidewind (2018).

For Schneidewind (2018), the great transformation has seven transformative turning points. The four dimensions of the “art of the future” (technological, economic, cultural, institutional) can be found in all seven turning points. Resource and energy transformations are fundamental. However, according to the concept of “double decoupling”, they are inconceivable without a comprehensive transformation in ideas surrounding prosperity and consumption. Envisaging the desired transformation is easier in more specific sectors, such as mobility, food, cities, and key energy- and resource-intensive industries.

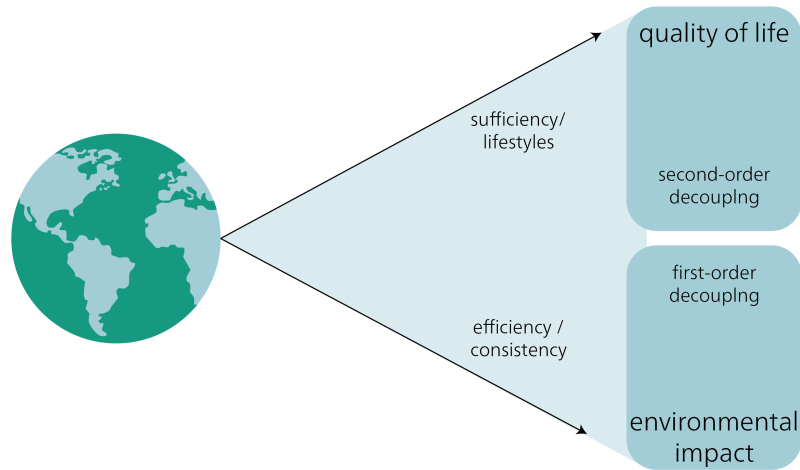


Figure 2.6: Double decoupling. Linking technology and prosperity models. Source: Own illustration based on Schneidewind (2018).

“Decoupling” refers to the separation of economic activity from environmental impact. A key concept in the sustainability debate, **double decoupling** states that sustainable development can only be achieved through increases in technological efficiency in combination with new models of prosperity and consumption. It aims to promote a more comprehensive and systemic understanding of innovation, to include both technological and social innovations.

Double decoupling refers to the following two types of decoupling:

- **First-order decoupling:** This focuses on increasing resource efficiency and consistency within the traditional economic growth model, mainly through technological advancements, and
- **Second-order decoupling:** This focuses on “sufficiency”, decoupling quality of life and a “good life” from the traditional economic growth model, as measured by GDP.

2.3.3 Global Sustainable Development Report

In September 2019, the first Global Sustainable Development Report (GSDR) was published by a group of 15 independent scientists appointed by the UN Secretary-General. Intended for publication every four years, the aim of the GSDR is to synthesize existing knowledge and identify pathways to achieving the Sustainable Development Goals (SDGs). The latest report (2023) highlights the significant gap between current progress and achieving the SDGs. Building on the 2019 report, it introduces capacity building as a new lever to accelerate progress.

TRANSFORMATIONS TO THE SDGS: ENTRY POINTS AND LEVERS

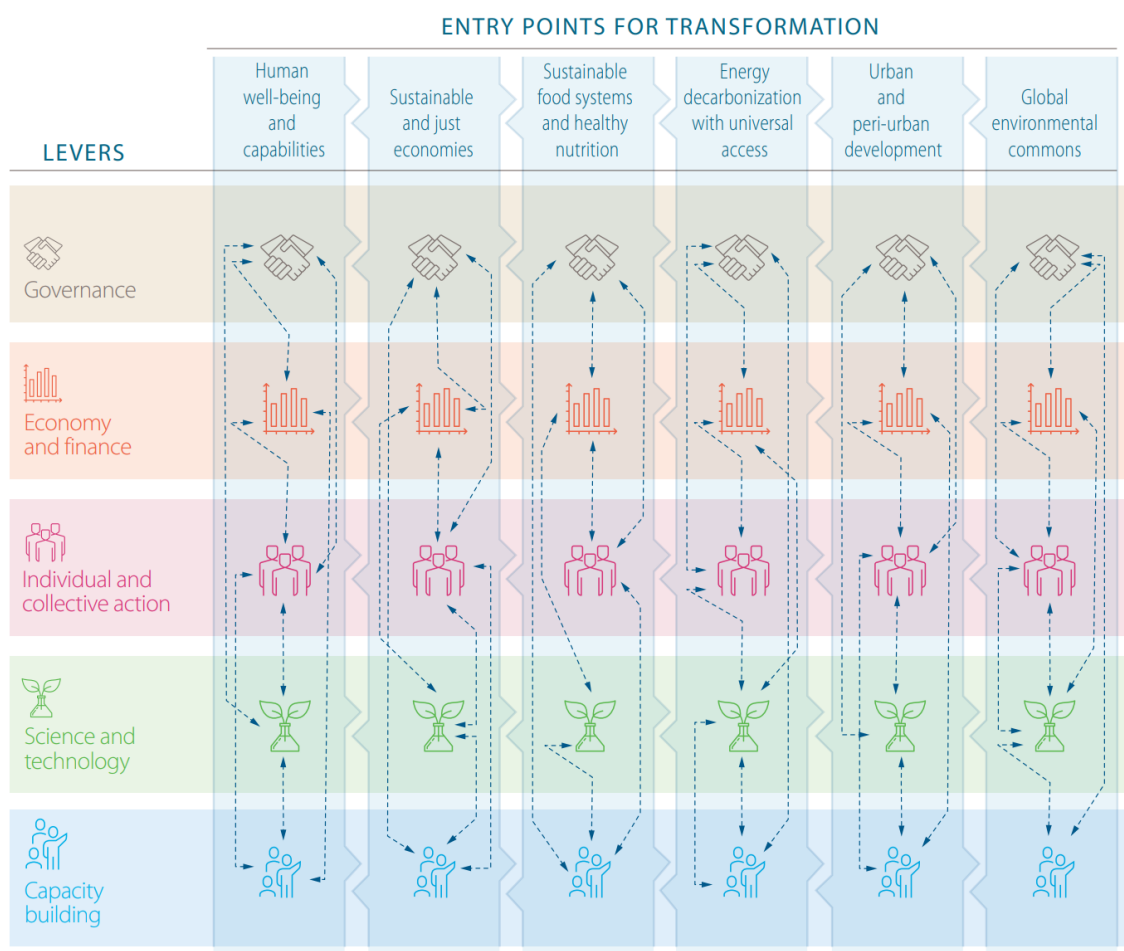


Figure 2.7: Levers and entry points for achieving the SDGs. Source: Independent Group of Scientists appointed by the Secretary-General (2023).

The GSDR proposes six key areas, or “entry points”, with the highest potential to drive the large-scale and swift transformations needed. To initiate these transformations, active and multifaceted collaboration is crucial among actors from diverse fields: governance, business and finance, individual and collective action, and science and technology.

2.3.4 Conclusion

“Great transformation” concepts

- consider it necessary and possible to steer social development towards sustainability.
- are holistic concepts for managing social development. They propose entry points at various areas of society and at several levels of action.

- propose key areas with a major leverage effect (e.g. the WBGU proposes an energy transition, urban transition, land use transition, and transformative education and research).

Further reading

- Polanyi, Karl. 1944. *The Great Transformation: The Political and Economic Origins of Our Time*. Beacon Press.

2.4 Green economy

In a pioneering use of the term “green economy”, David Pearce and Edward Barbier launched their groundbreaking *Blueprint for a Green Economy* series in 1989. The series was the first to present economic frameworks and programmatic approaches for achieving the dual goals of economic prosperity and ecological well-being (e.g. Pearce et al. 1989, 2000; Barbier and Markandya 2013). The core principle of the green economy challenges the traditional view that economic growth and environmental well-being are inherently at odds. It proposes a paradigm shift away from bans and restrictions, instead favouring economic incentives and strategies that promote environmental sustainability. This approach fosters models of economic and technological development that align with both ecological and social goals.

In the run-up to Rio+20, the 2012 United Nations Conference on Sustainable Development, the green economy concept was finally developed into a guiding principle that shaped the debate (UNEP 2011; Bär et al. 2011). “Green economy” was one of two key themes of Rio +20; the other was the “institutional framework” for sustainable development. Accordingly, numerous documents on the green economy were published in 2011 and 2012, and a seminal definition of the term was published by the organizing agency of Rio+20, the United Nations Environment Programme (UNEP), in the report, *Towards a Green Economy: Pathways to Sustainable Development and Poverty Eradication* (UNEP 2011). The OECD also contributed its own response to the financial crisis with its resolution on “Green Growth” (OECD 2009).

“UNEP defines a green economy as one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities. In its simplest expression, a green economy can be thought of as one which is low carbon, resource efficient and socially inclusive. In a green economy, growth in income and employment should be driven by public and private investments that reduce carbon emissions and pollution, enhance energy and resource efficiency, and prevent the loss of biodiversity and ecosystem services.” (UNEP 2011, 2)

While Rio+20 cemented the concept of a green economy, the need for a fundamental global change in thinking was already widely accepted. It had become clear that while environmental protection measures cost money, they also have a positive economic impact by creating jobs (see e.g. UBA 2008) and sparking innovation in technological efficiency advancements (cf. e.g. von Weizsäcker et al. 1995). The green economy concept systematizes such findings and calls for programmes to overcome the apparent contradictions between economy and ecology, growth and resource conservation, and prosperity and environmental protection. The concept thus marks a significant

paradigm shift. Adhering to planetary boundaries or ecological guidelines in a green economy does not necessarily mean forgoing economic growth and technological progress (Rockström, Steffen, Noone, Persson, F. S. Chapin, et al. 2009; Sachverständigenrat für Umweltfragen (SRU) 2012). Instead, the idea is that by harmonizing these apparently contradictory aims, we can achieve them even more efficiently. The EU Green Deal exemplifies this concept. The EU aims to be the first climate-neutral continent by 2050, aiming for a modern, resource-efficient, and competitive economy with net-zero greenhouse gas emissions. This plan seeks to decouple growth from resource use while ensuring a just transition that leaves no one behind.

2.4.1 Conclusion

The core thesis of the green economy is that environmental and economic goals are not contradictory. Instead, they can be reconciled through appropriate economic incentives and strategies. The green economy aims to foster both economic growth and environmental sustainability through public and private investment in low-emission, resource-efficient, and socially equitable economic systems.

Critics, however, have raised concerns about the effectiveness of green economy measures. They argue that the proposed technologies and incentives are insufficient to bring about the far-reaching systemic changes needed. An overemphasis on economic growth and technological solutions could potentially neglect essential structural and behavioural changes. And they warn that the global transition to a green economy could leave behind disadvantaged communities and developing countries, if their specific needs are not addressed.

2.5 Post-growth societies

The post-growth debate emerged from concerns raised in the 1970s, after publication of the influential Meadows Report, *The Limits to Growth* (1972). As previously mentioned, this report highlighted the Earth's finite capacity to sustain humanity in the face of unrestrained economic growth.

The post-growth debate advocates qualitative growth or even zero growth, and criticizes the effects of the “modern” economy and lifestyles (e.g. Daly 1999). It argues that the compulsion for constant growth is making us exceed ecological limits and leading to negative social and ecological consequences. “Ecological economics” is an important concept in this respect, as it aims to develop alternative models and approaches for evaluating economic growth.

The dilemma in this debate is that most approaches to a sustainable economy assume that a growth-independent economy should not be profit-driven. However, capitalist economies are existentially dependent on growth (e.g. Binswanger 2019; Oberholzer 2021). This dilemma is key to the question of how to organize a successful transition from the current – unsustainable – system to a sustainable economic and social system. One main approach is to reduce dependencies on growth and promote alternative models that focus on sufficiency, without neglecting strategies that focus on efficiency and consistency. This requires changes in production and consumption patterns, in social norms, and in the policy framework. The post-growth debate emphasizes

the need for a comprehensive transformation that encompasses the environmental, social, and economic dimensions – and aims to achieve a balance between human needs and planetary boundaries.

The concept of “sufficiency” is an integral part of the post-growth debate (Schneidewind and Zahrnt 2013). Sufficiency aims to reduce overconsumption and promote alternative lifestyles, consumption habits, and production patterns. To promote the widespread adoption of sufficiency, a legal and institutional framework that incentivizes and facilitates sufficiency-oriented practices is necessary.

The four pathways to a politics of sufficiency

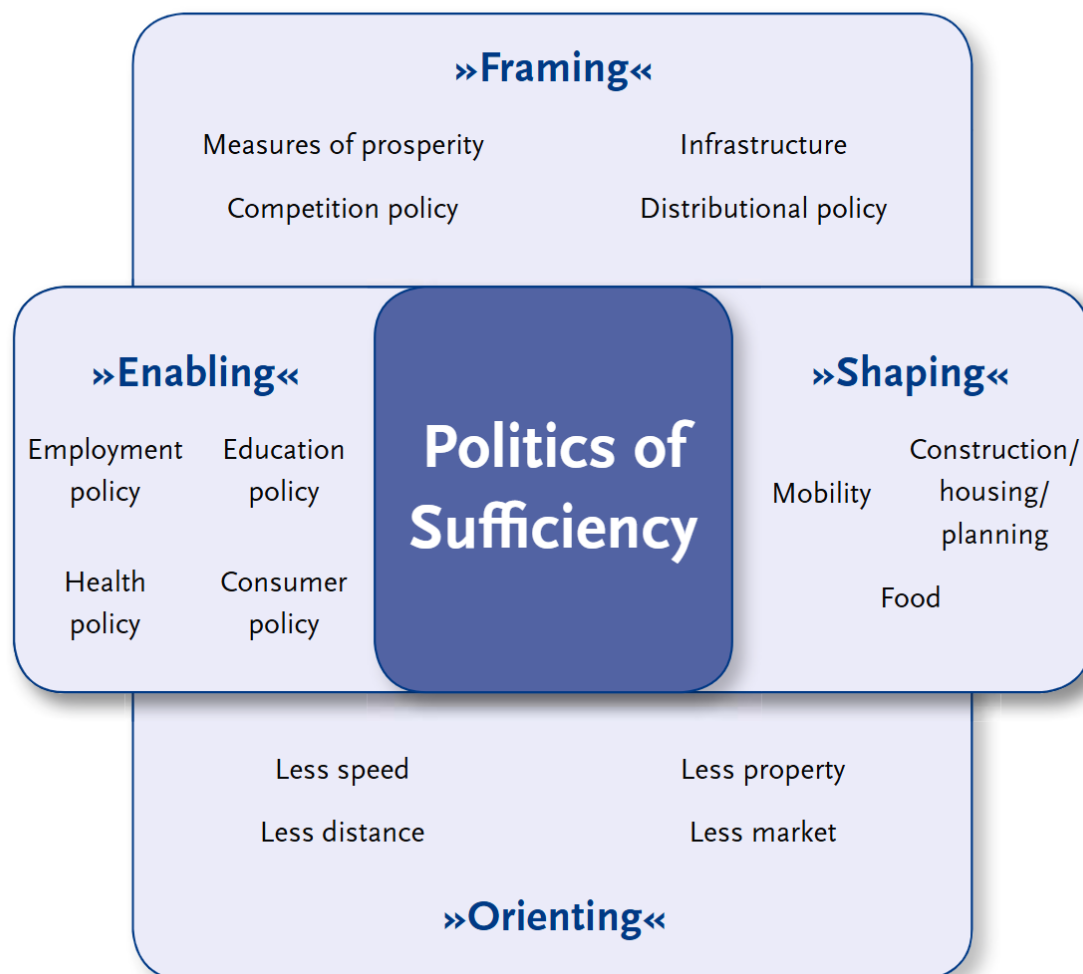


Figure 2.8: The four pathways to sufficiency. Source: Schneidewind and Zahrnt (2014).

An overall policy to promote sufficiency can actively shape our choices, through attractive sufficiency-oriented offers and services. It can also foster awareness and provide guidance for

adopting sufficiency-oriented lifestyles and practices. This comprehensive approach aims to shift consumption towards what is necessary and meaningful, ultimately reducing excessive resource use.

2.5.1 Conclusion

Post-growth debates analyse and criticize modern society's dependency on economic growth, and the negative environmental and social effects of this growth. Rather than focus solely on technological progress and market forces, theories of a post-growth society advocate for changes in society, structures, and institutional frameworks. Overall, post-growth debates emphasize the need for sustainable approaches to achieve a comprehensive transformation of the economy and society. Transformation towards a post-growth society requires a reorientation of values, structures, and institutional frameworks. The challenge is to find ways of shaping the transition to an economy that is independent of growth, while at the same time ensuring social justice and ecological sustainability.

Critics of post-growth debates argue that a rejection of economic growth could have a negative impact on prosperity and social progress. They fear that an economy independent of growth could lead to stagnating innovation, fewer jobs, and falling living standards. Constructively addressing the challenges of growth and developing viable alternatives are therefore important aspects of enabling sustainable and equitable change.

2.6 Indicators for sustainable development: From principles to practice

A key concern of sustainable development is the ability to systematically make visible not only the progress achieved – but also the setbacks experienced along the way – at the global, national, and local levels. Indicators play a crucial role in this effort: they help capture complex socio-ecological realities, enable comparability, and support evidence-based policymaking. Precisely because sustainability is a normative objective, particular attention must be paid to the selection, design, and use of indicators.

Importantly, indicators are not purely technical measurement tools. They inevitably reflect specific values, world views, and political goals. Their significance is not solely based on “objective data”, but also on decisions about what should be measured, how, and for what purpose. For instance, Gross Domestic Product (GDP) has long served as the predominant benchmark for economic success – despite its failure to take into account environmental damage, social inequality, unpaid care work, and ecosystem services.

Sustainability science and environmental policy distinguish between different types of indicators, each serving a specific function (de Vries 2024):

- **State indicators** describe the condition of a system or resource – such as CO₂ concentration in the atmosphere, the proportion of protected forest area, or average life expectancy. These are primarily used to monitor trends and system states.

- **Causal indicators** capture underlying drivers or influencing factors that lead to changes within a system. These include, for example, fossil fuel consumption or socioeconomic drivers like urbanization rates or consumption patterns.
- **Input or intervention indicators** refer to institutional or policy measures that are introduced in response to problems. Examples include subsidies for renewable energy, legal regulations, or public awareness campaigns. They track the presence and nature of interventions.
- **Performance or output indicators** measure the results or impacts of these interventions in relation to predefined goals. These may include emission reductions, the share of recycled materials, or progress in social justice.

This classification illustrates that individual indicators often lack significance in isolation. It is the interaction of indicators – e.g., within impact models or causal frameworks – that enables a robust assessment of sustainability trends and policy responses.

Over the past decades, a wide range of alternative indicator systems has been developed to provide a more comprehensive picture of societal development beyond GDP. These approaches aim to integrate economic, social, and environmental dimensions while making distributional and non-market contributions visible.

A significant milestone was the development of the **Human Development Index (HDI)** by the United Nations Development Programme (UNDP) in the 1990s. The HDI combines three dimensions: life expectancy at birth (as a proxy for health), average and expected years of schooling (education), and per capita income adjusted for purchasing power (material living standard). The HDI sought to broaden the narrow monetary focus of GDP by offering a capability-based understanding of development as “the expansion of people’s choices” (Sen 1999). Today, it remains a standard reference in development economics (see UNDP Human Development Reports since 1990).

A more advanced approach is the **Genuine Progress Indicator (GPI)**, which starts with personal consumption (as GDP does) but adjusts for external costs such as environmental degradation, resource use, crime, and income inequality, to reveal trade-offs between costs and benefits of economic growth. It also adds value to unpaid services like domestic work and volunteering. First proposed by Daly and Cobb (1989) as Sustainable Economic Welfare (ISEW), the GPI is well established in ecological economics (see Talberth et al. 2007). Unlike GDP, which typically increases over time, the GPI has stagnated or declined in many high-income countries since the 1980s – indicating a possible decoupling between growth and well-being.

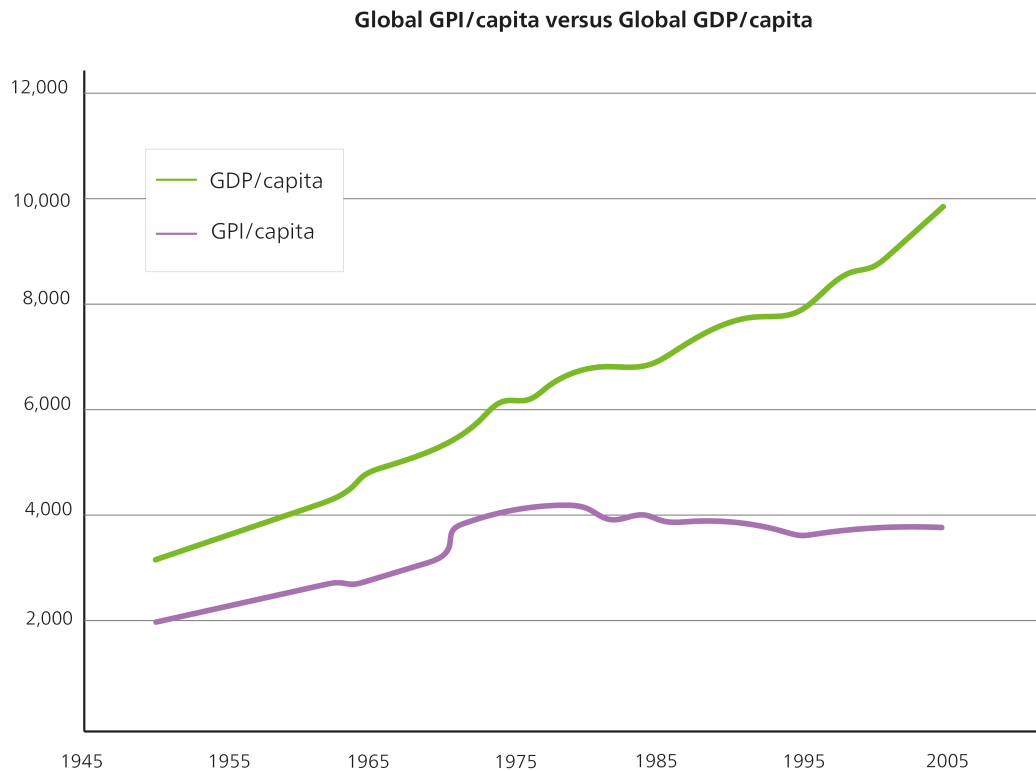
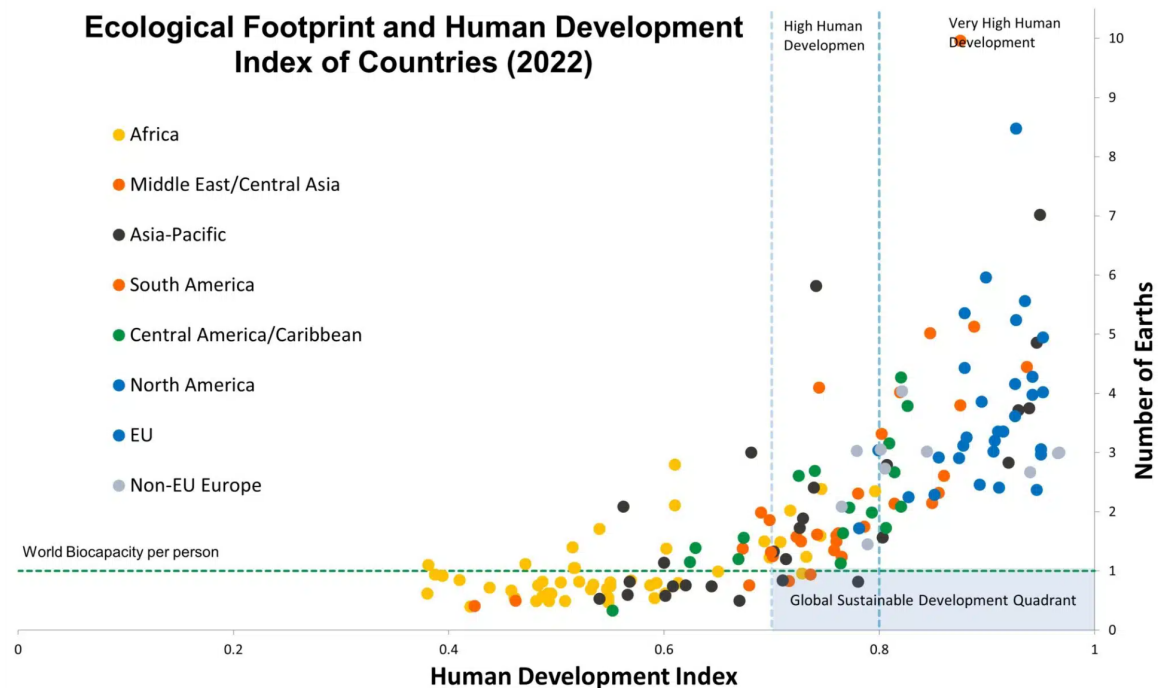


Figure 2.9: Comparison of global Gross Domestic Product (GDP) and Genuine Progress Indicator (GPI). Source: Own illustration based on Kubiszewski et al. (2017).

Another commonly used measure is the **Ecological Footprint**, which translates human resource use into global hectares. It represents the biologically productive area needed to meet the resource demand of an individual, region, or country – including land for food production, settlement, and carbon absorption. Developed by Wackernagel and Rees (1996) and promoted by the Global Footprint Network, it is especially prevalent in environmental education. The Ecological Footprint illustrates whether a population is living within its ecological means – i.e. whether it exceeds its fair share of global biocapacity.



Source: Ecological Footprint: York University Ecological Footprint Initiative & Global Footprint Network. National Footprint and Biocapacity Accounts, 2025 Edition. Human Development Index: Human Development Report, 2025, UNDP.

Figure 2.10: Correlation between Ecological Footprint and Human Development Index. Source: <https://www.footprintnetwork.org>.

The **Happy Planet Index (HPI)** offers a more subjective approach, combining life satisfaction (survey-based), life expectancy, and ecological footprint to calculate “wellbeing per unit of environmental input”. Developed by the New Economics Foundation, the HPI shows that high life satisfaction does not necessarily require high resource consumption. Middle-income countries like Costa Rica often score higher than industrialized nations with unsustainable consumption patterns (NEF 2006).

In the German-speaking world, the **National Welfare Index (NWI)** has been introduced as a GPI-based tool adapted to national data availability. It includes dimensions such as income distribution, environmental burden, and health costs. Initially developed by the Institute for Ecological Economy Research (IÖW) and the Environmental Policy Research Centre (FFU) at FU Berlin, the NWI has been tested in several German federal states (Diefenbacher and Zieschank 2011).

Another particularly relevant measure is the **Multidimensional Poverty Index (MPI)**, developed by the Oxford Poverty and Human Development Initiative (OPHI) in collaboration with the UNDP. It captures poverty across three core dimensions: health, education, and standard of living. Indicators include child mortality, school attendance, access to electricity, and housing conditions. Unlike income-based measures, the MPI provides a nuanced understanding of deprivation across overlapping dimensions. Bader et al. (2017) demonstrate the value of the MPI in a longitudinal study on poverty in the Lao PDR, showing how rapid economic growth reduced income poverty while simultaneously exacerbating disparities in education, infrastructure, and health access.

The Ecological Footprint – Explanation and Critique

The **Ecological Footprint** measures how much biologically productive land and sea area (in global hectares) is required to sustain the resource use of an individual, population, or country over time. It includes land required for CO₂ absorption, food production, infrastructure, and other human needs. Developed in the 1990s by Mathis Wackernagel and William Rees, the concept has become one of the most widely recognized tools for environmental communication.

The Ecological Footprint is often used in public awareness campaigns, notably in connection with the annually published *Earth Overshoot Day* – the symbolic date when humanity's resource consumption exceeds Earth's capacity to regenerate those resources in the same year.

The main limitations of the Ecological Footprint indicator, based on Global Footprint Network (2020):

- It doesn't take into account non-ecological aspects of sustainability, such as social well-being. Having a footprint smaller than the biosphere is a necessary minimum condition for a sustainable society, but it is not the only condition. In addition, in terms of resources, even if the ecological footprint is within biocapacity, poor management can still lead to depletion. A footprint smaller than biocapacity is necessary for making quality improvements replicable and scalable.
- It doesn't track the amount of non-renewable resource stocks available, such as oil, natural gas, coal, or metal deposits. The footprint associated with these materials is based on the regenerative capacity used or compromised by their extraction and, in the case of fossil fuels, the area required to assimilate the wastes they generate.
- It doesn't directly take into account activities that are inherently unsustainable, such as the release of heavy metals, radioactive materials, and persistent synthetic compounds (e.g. chlordane, polychlorinated biphenyls (PCBs), chlorofluorocarbons (CFCs), polyvinyl chloride (PVC), dioxins, etc.). These are activities that need to be phased out independently of their quantity, as there is no biocapacity budget for using them. Where these substances cause a loss of biocapacity, however, their influence can be seen.
- It doesn't directly measure ecological degradation, such as increased soil salinity from irrigation, which could affect future bioproductivity. However, if degradation leads to reductions in bioproductivity, then this loss will be captured when measuring biocapacity in the future. Moreover, by looking at only the aggregate figure, "under-exploitation" in one area (e.g. forests) can hide over-exploitation in another (e.g. fisheries).
- It doesn't identify where and in what way the capacity of ecosystems is vulnerable or resilient. The footprint is merely an outcome measure documenting how much of the biosphere is being used compared with how productive it is.

Despite these limitations, the Ecological Footprint remains a useful entry point for engaging broader audiences in discussions on global ecological justice and planetary responsibility – especially in educational and communication settings.

2.7 Between simplification and complexity: Interactions between SDGs

A key challenge in designing sustainability indicators lies in balancing simplification and complexity. Indicators are intended to support decision-making – and they must account for a wide array of interactions, uncertainties, and contextual dependencies.

This is particularly evident in the 2030 Agenda for Sustainable Development. While the Agenda comprises 17 distinct goals and numerous targets, these goals and targets do not exist in isolation. Instead, they are intricately interconnected: progress in one area may reinforce, neutralize, or hinder progress in another. Consequently, indicators that illuminate these relationships are essential for coherent policy design.

Breu et al. (2020) present a methodology based on the interactions framework developed by Nilsson et al. (2016), which systematically classifies the relationships between SDGs on a seven-point scale, from strongly negative (–3) to strongly positive (+3). They applied this approach to Switzerland's national sustainability framework, revealing areas where policy actions toward one SDG could support or conflict with actions favouring other SDGs. For example, measures to reduce poverty (SDG 1) could conflict with biodiversity goals (SDG 15) if they are not designed to be environmentally sustainable.

2.8 Sustainability Basics: The Quiz – Chapter 2

Which two planetary boundaries are considered “core boundaries”?

- ☐ Ozone depletion and land use
- ☐ Ocean acidification and nitrogen cycles
- ☐ Freshwater and aerosols
- ☐ Climate change and biosphere integrity

Within the planetary boundaries framework, climate change and biosphere integrity are identified as core boundaries because they play a fundamental role in regulating the state of the Earth system and strongly influence the other planetary boundaries.

- False
- False
- False
- True

Which of the following is NOT one of the nine planetary boundaries identified by Rockström et al.?

- ☐ Climate change
- ☐ Ocean acidification
- ☐ Global financial stability
- ☐ Biosphere integrity

The planetary boundaries framework identifies biophysical Earth system processes that regulate the stability and resilience of the planet. Climate change, ocean acidification, and biosphere integrity are part of this framework, whereas global financial stability is not a planetary boundary.

- False
- False
- True
- False

Which of the following tipping points does NOT belong to the Earth's climate system?

- ☐ Greenland Ice Sheet melting
- ☐ Gulf Stream collapse
- ☐ Amazon rainforest collapse
- ☐ Ocean plastic pollution

Tipping points in the Earth's climate system refer to large-scale components of the climate that can undergo abrupt and potentially irreversible change. The Greenland Ice Sheet, the Gulf Stream (AMOC), and the Amazon rainforest are all commonly discussed climate tipping elements, whereas ocean plastic pollution is an environmental problem but not a climate-system tipping point.

- False
- False
- False
- True

What term refers to the decoupling of economic growth from environmental degradation while maintaining prosperity?

- ☐ Resilience
- ☐ Planetary boundaries
- ☐ Green economy
- ☐ Double decoupling

The concept of a green economy refers to an economic model that aims to achieve economic growth and human well-being while reducing environmental risks and ecological scarcities, thereby decoupling prosperity from environmental degradation.

- False
- False
- True
- False

What does the inner circle of the Doughnut Model represent?

- ☐ The ecological ceiling
- ☐ The safe and just space for humanity
- ☐ The social foundation

☐ The planetary limits

In the Doughnut Model, the inner circle represents the social foundation, which defines the minimum social standards necessary for a just and equitable life for all people.

- False
- False
- True
- False

Which of the following statements describe post-growth approaches? (More than one statement may be correct.)

- ☐ Emphasis on GDP increase
- ☐ Critique of growth dependency
- ☐ Focus on sufficiency and well-being
- ☐ Advocacy for structural change

Post-growth approaches critically question the dependence of modern societies on continuous economic growth. They emphasize sufficiency, well-being, and quality of life rather than GDP growth, and they call for structural changes in economic and social systems to enable sustainable and just futures.

- False
- True
- True
- True

Part III

Transforming (Un-)sustainable Systems: Key Areas and Strategic Approaches

3 Environmental Sustainability

The economy is integral not only to society, but also to nature. Humans are living beings and therefore reliant on material resources to fulfil their needs. They also require intact ecosystems and a climate that is suitable for human habitation. When we speak of “environmental sustainability”, we generally mean the protection of the diversity and functioning of natural ecosystems and the services they provide for future generations. This is an anthropocentric definition, as it focuses on human activities and how these seek to ensure the long-term sustainability of nature for human benefit. This perspective is also reflected in our everyday language and our dualistic separation of “human” and “environment”, a separation that goes back to the 19th century and persists to this day. For example, we separate science into “natural science” and “social science”, and our analyses often conceptualize environmental damage as “externalities”. Another understanding of environmental sustainability emphasizes the importance of maintaining the self-regulation of the Earth’s climate system. This view highlights the interaction and feedback between various components of the Earth’s climate system.

3.1 The normative dimension of environmental sustainability

Although our understanding of how ecosystems work is based on thoroughly researched empirical information (and therefore constitutes “systems knowledge”), environmental sustainability remains a normative concept. This means that it is based not only on scientific findings, but also on an ethical evaluation (Figure 3.1). The various interpretations of environmental sustainability offer different answers to the question of what should be preserved, and why. Each approach to environmental sustainability therefore expresses the desired state of the ecological environment, both now and in the future – i.e. what aspects of nature should be preserved or should survive from the present to the future. For example, when we talk about ecosystem services, the intention is often to preserve ecosystems in a way that continues to support and maintain our own welfare (in this sense, it is an anthropocentric perspective).

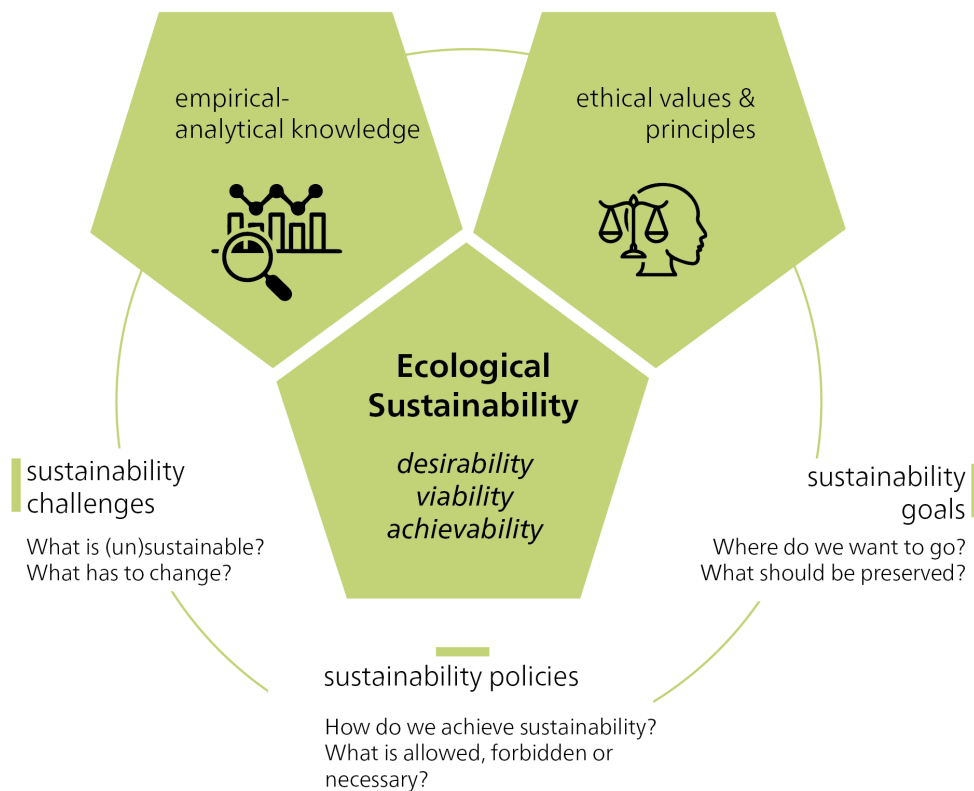


Figure 3.1: The normative dimension of environmental sustainability. Source: Own illustration.

Ethical judgments and justifications shape the selection of which ecological properties and functions we preserve for future generations, and which we deem intrinsically valuable in nature. What constitutes a desirable state for ecosystems, and why? What aspects of the ecological environment deserve protection, and what is the aim of this protection? Is it necessary to keep ecosystems as pristine as possible, and how do we define “naturalness”? To what extent should we use ecosystems for human purposes without restriction, and are there areas that we should leave for organisms to use, with minimum human intervention? Interpretations of environmental sustainability define the goals being pursued and for whom, why, and how we are pursuing them (Figure 3.1).

Specific examples of ethical assessments in the context of environmental sustainability

Wildlife conservation: Should we intervene in natural wildlife populations to protect endangered species and maintain the balance of ecosystems? Or should we leave nature as untouched as possible, even if this means losing some species?

Biotechnological interventions: Is it morally acceptable to use biotechnological methods, such as genetic engineering, to alter the ecological characteristics of organisms and potentially influence ecosystems?

Conservation of endangered species: Should we focus on protecting and preserving endangered species to maintain biodiversity, or should we focus on preserving broadly available species that are more important for the human diet or the economy?

Protecting ecosystems: Should we establish protected areas to preserve threatened ecosystems and species, even if this restricts local communities in their economic activities? How can we strike a balance between conservation and sustainable use?

3.2 Ecosystems, ecosystem management, and ecosystem services

An ecosystem is a group of living organisms and their physical surroundings that interact with each other. Ecosystems can range from small, like a flowerpot, to large, like an ocean. When similar ecosystems are found in a larger region with the same climate, they are called biomes. Energy and material flows are important for the functioning of an ecosystem. Some of these flows, such as the carbon cycle, take place at the global level, while others are more localized. Most ecosystems obtain their energy from the sun and can influence the Earth's climate through their interactions (Britannica 2025).

Ecosystem services are the benefits that humans derive from ecosystems such as mangrove forests, oceans, or wetlands. The services provided by ecosystems include the provision of clean water, the prevention of flooding, the promotion of crop growth, and the provision of places for leisure activities. The Millenium Ecosystem Assessment, published in 2005, divides ecosystem services into four categories (cf. green Box in Figure 3.2): **provisioning services**, **regulating services**, **cultural services**, and **supporting services** (Millenium Ecosystem Assessment 2005). Note that the supporting services, which are primarily fundamental biophysical processes, enable and guarantee the other services in the first place.

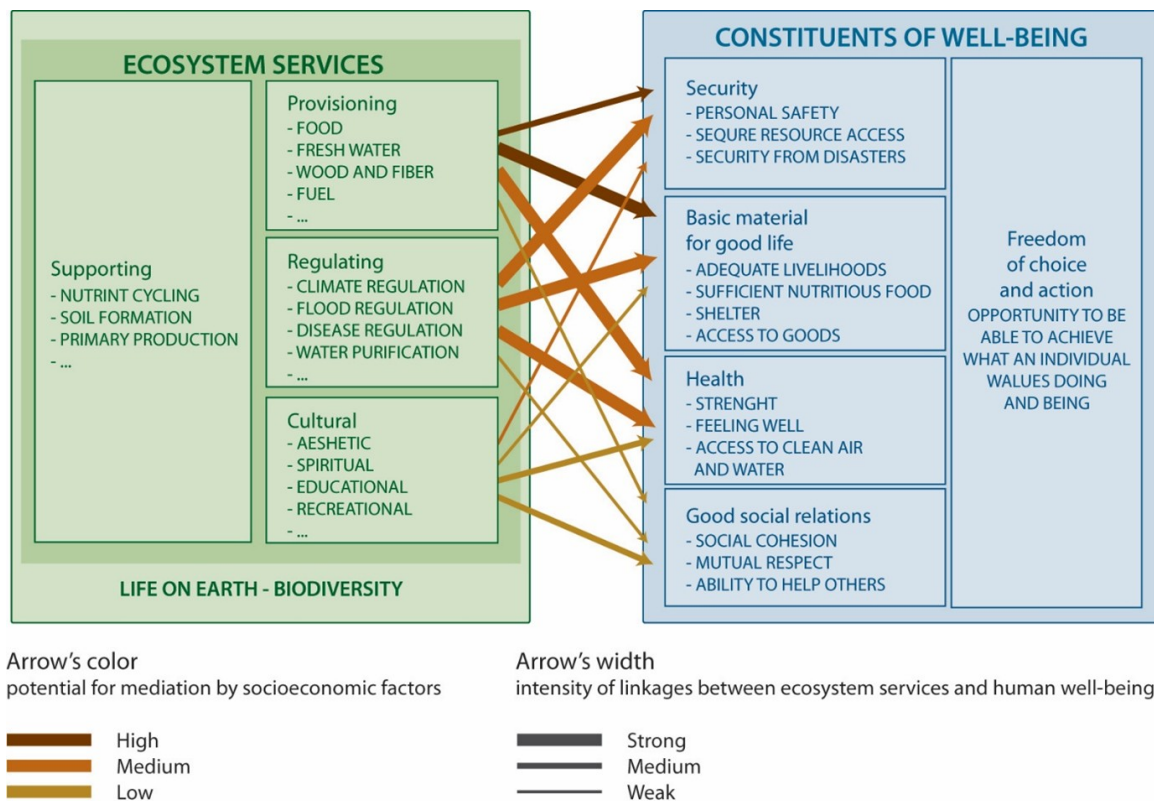


Figure 3.2: Millennium Ecosystem Assessment. Source: Millennium Ecosystem Assessment (2005). Licensed under CC.

According to the Millennium Ecosystem Assessment (2005), these ecosystem services can improve our quality of life and our economy (cf. blue Box in Figure 3.2). While this way of looking at ecosystems views humans as separate from the environment, it emphasizes the relevance of ecosystem services for human well-being. Ultimately, the concept of ecosystem services seeks to place greater consideration on the value of nature in policy and economic decisions. However, the ecosystem services concept also plays a role in landscape and spatial planning. For example, it's used to analyse how spatially effective (legal) regulations impact ecosystem services, and consequently, human well-being (cf. Albert et al. 2016).

Case study: Ecosystem services in Madagascar

This landscape on the eastern coast of Madagascar offers a case study in ecosystem services. Ecosystem services in this area include firewood (which is used by local inhabitants for cooking) and rice (which is a staple food). Describing agricultural products such as these as ecosystem services illustrates that they only become “services” through human demand. Ecosystem services are not given in nature per se: instead, they represent a perspective through which humans view and describe the environment. Other ecosystem services in this landscape include the water cycle, which is vital for local inhabitants, as they rely on rivers for drinking water. Water is also used for agricultural irrigation and other purposes. Climate regulation is another significant service, important both locally and globally.

This landscape clearly illustrates that the demand for and importance of ecosystem services vary among individuals or groups. Such diverse claims on ecosystem services from the same land can lead to **conflicts of interest**. Demand for agricultural products or firewood, for example, can conflict with climate-regulating and biodiversity-promoting (international) demands on the rainforest.

Humans have had an influence on many ecosystem services, both directly and indirectly. While these impacts have often been negative, some have been positive. For example, replacing natural ecosystems with cultivated agroecosystems has increased the value of ecosystem services for humans. Take, for example, dry meadows and pastures in Switzerland. Only through centuries of forest clearing and extensive agricultural use in alpine locations could the ecosystem services these areas provided (in this case, fodder production) be used and increased. At the same time, these anthropogenically influenced sites harboured rare meadow plants and pollinators, thus fostering a high level of biodiversity. These areas are also vital for tourism, as they significantly shape the alpine landscape. Consequently, these extensively used areas generally offer greater overall ecosystem service multifunctionality than intensively farmed grasslands (Richter et al. 2024).

Many of the negative effects on ecosystems and their services result directly from human land use and overfishing. In addition, indirect factors such as climate change and invasive non-native species significantly impair the stability and functionality of ecosystems. Two research projects with significant CDE involvement – Woody Weeds and Woody Weeds+ – have demonstrated that invasive woody plants such as *Prosopis juliflora* have a major impact on biodiversity and local livelihoods in East Africa. Originally introduced to combat desertification and timber extraction, this species displaces native vegetation and is not suitable as animal feed without processing ([CDE project website](#)).

3.2.1 The monetary valuation of ecosystem management

The term “nature’s services” first appeared in the publication “How much are Nature’s Services Worth?” (Westman 1977). A synonymous term, “ecosystem services”, was coined a few years later (Ehrlich and Ehrlich 1981). The concept of ecosystem services and the thinking behind it are closely linked to economic philosophy and practice (Gómez-Baggethun et al. 2010). Around 20 years after the introduction of “nature’s services”, Robert Costanza and colleagues estimated that the **global value of 17 ecosystem services was worth 33 trillion US dollars per year** (Costanza et al. 1997). A

recalculation in 2011 showed that this value had already fallen by between 4.3 trillion and 20.2 trillion US dollars per year (Costanza et al. 2014). It is predicted that by 2050, the value of ecosystem services will either drop by up to USD 51 trillion per year, or rise by up to USD 30 trillion per year, depending on land use scenario (Kubiszewski et al. 2017).

Calculating the **monetary value of ecosystem services** typically involves three steps and requires high-quality and differentiated data. 1) The first step is to measure the change in ecosystem services after an intervention, thus placing the focus not on the status quo, but on the difference. 2) The second step is to evaluate the impact of this change on people's socioeconomic well-being. 3) The third and final step is to place a monetary value on the resulting change in productivity or prosperity. There are **many different methods for calculating the monetary value of ecosystem services** (Figure 3.3). One such method is “green accounting”, which uses existing market values, if available, or exchange values as an approximation (e.g. comparing avalanche barriers and protection forests). Another method is cost–benefit analysis, which typically distinguishes between stated preferences and revealed preferences.

One such method is “green accounting”, which uses existing market values, if available, or exchange values as an approximation (e.g. comparing avalanche barriers and protection forests). Another method is cost–benefit analysis, which can be divided into two types: stated preferences and revealed preferences.

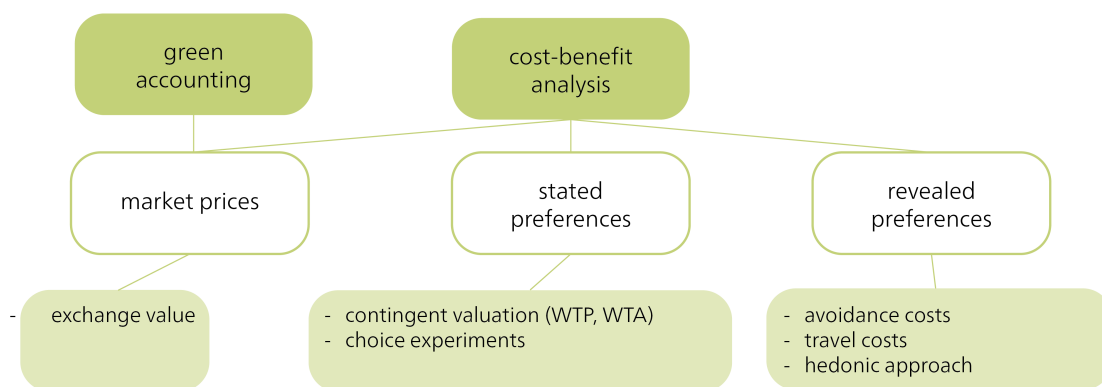


Figure 3.3: The monetary value of ecosystem services. Source: Own illustration.

Stated preferences involve people stating (e.g. in a survey) the extent to which they would be willing to pay for certain ecosystem services, or which changes they would be willing to accept (contingent valuation). Another variation of this method involves choosing between different (usually sub-optimal) options, each representing a trade-off based on changes in ecosystem services (which are then analysed econometrically). In contrast, **revealed preferences** are based on actual behaviour. For example, they involve analysing avoidance costs (e.g. how much people pay for noise barriers) or calculating travel costs (which assumes that the time and travel costs incurred for visiting a destination reflect the value of the place, and thus, in a broader sense, an ecosystem service). Other methods include using hedonic approaches to determine the extent to which favourable or unfavourable environmental influences, and thus their quality, can affect the value of a property.

Each of these methods has its advantages and disadvantages. They are criticized mainly for their one-sided assessments and failure to account for social inequalities, power relations, and cultural differences. Consequently, these methods are limited in their ability to promote fair and environmentally sustainable decisions.

The principle behind **Payments for Ecosystem Services (PES)** involves compensating land users for maintaining a component of a specific ecosystem service. Such compensation may, for example, be paid by organizations or companies for which these specific ecosystem services are important. A well-known example of PES is the “Pagos por Servicios Ambientales” (PSA) programme in Costa Rica. This national programme, which has been operational since 1997, was set up to reduce deforestation and promote reforestation by paying landowners to maintain, reforest, or sustainably manage forest areas. The payments aim to promote the provision of ecosystem services such as carbon sequestration, water conservation, biodiversity conservation, and scenic beauty. The payments in this project come from various sources, including a tax on fuels and international funding from organizations interested in carbon offsetting. Studies show that the programme has significantly reduced the rate of deforestation in the country and promoted the conservation of biodiversity, while at the same time increasing the income of landowners (Porrás et al. 2013).

3.2.2 Criticism of ecosystem services, and further and alternative concepts

While the concept of ecosystem services has been recognized and applied in the scientific and policy debate, it has also been widely criticized (see e.g. Kosoy and Corbera 2010; Norgaard 2010). One argument is that by focusing on ecosystem services, we reduce the complex and diverse values of ecosystems to their monetary benefits for humans (cf. Schröter et al. 2014). This **economic reduction of nature** neglects the non-quantifiable but nevertheless important aspects of nature, such as spiritual or intrinsic values (Fairhead et al. 2012). The commodification of nature through ecosystem services could ultimately also lead to an exploitative relationship between humans and nature (cf. Schröter et al. 2014) or **an exacerbation of social inequalities between powerful actors and marginalized groups** (Dawson et al. 2021), as the concept does not explicitly consider justice (Loos et al. 2023, 478). In some cases, it is even assumed that the concept of ecosystem services could conflict with – and undermine – biodiversity conservation (cf. Schröter et al. 2014).

“The flurry of enthusiasm for optimizing the economy by including ecosystem services has blinded us to the more important question of how we are going to make the substantial institutional changes to significantly reduce human pressure on ecosystems, especially by the rich, and to adapt to and work effectively with the rapid ecosystem changes being driven by existing and foreseeable climate dynamics.” (Norgaard 2010, 1220)

In response to this criticism, the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES), (see Section 3.4, *Biodiversity Loss*) has further developed the concept of ecosystem services and developed another concept, that of **Nature’s Contribution to People** (NCP). NCPs are defined as “all the positive contributions, or benefits, *and* occasionally negative contributions, losses or detriments, that people obtain from nature.” (Pascual et al. 2017, 9). The concept may resemble ecosystem services, but it goes further and recognizes that nature and its contributions to a good life can be perceived and viewed in different ways, depending on the cultural and

institutional context. The concept seeks to include different world views, such as Indigenous knowledge systems, thus taking into account both intrinsic (i.e. non-anthropocentric) and relational values of nature (Díaz et al. 2018). Overall, this approach also includes context-specific perspectives, and it gives greater consideration to the justice dimensions than the concept of ecosystem services does (Loos et al. 2023).

However, such **plural valuations** are also subject to criticism and pose various challenges. For example, Jacobs et al. (2023) point out that current approaches to integrating multiple values and perspectives into ecosystem assessments risk merely promoting pseudo-participation, while existing **structures of power and discrimination** remain unchanged. Plural valuations aim for inclusivity and democracy, but they can inadvertently lead to conflict, power imbalances, and unclear outcomes if not carefully designed. Scientists have also criticized the above-mentioned NCP concept, which is often seen as an inadequate development of **utilitarian environmentalism**. This is a strongly Western and anthropocentric perspective that maintains a dualistic separation of humans and nature (cf. Muradian and Gómez-Baggethun 2021). Muradian and Gómez-Baggethun (2021), propose that “in order to induce transformative change in human-nature relations we need a shift from a morality of utility to a morality of care, a reallocation of property rights, and the extension of the community of justice to non-human entities.” A notable example of this transformation is the 2017 recognition of New Zealand’s Whanganui River as a legal person (see Charpleix 2018).

3.3 Planetary boundaries

The current era is called the Anthropocene because human activities are exerting an increasingly profound impact on the Earth and its systems. Although this term has not (yet) been recognized as an official geologic or geochronological epoch by geological bodies (Zhong 2024), it is used to describe the enormous human impact on the environment. There is no uniform definition of the origins of the Anthropocene; possible markers include human impact on greenhouse gas concentrations through rice cultivation, the global spread of radioactivity from nuclear testing, or the appearance of plastic particles in geological sediments. The Industrial Revolution is generally considered the start of the Anthropocene.

A quantitative definition of environmental sustainability was proposed by Rockström, Steffen, Noone, Persson, F. S. I. Chapin, et al. (2009) and Steffen et al. (2015), who identified nine planetary boundaries: climate change, biosphere integrity, land-system change, freshwater change, biogeochemical flows, ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, and novel entities. These planetary boundaries serve as a scientific reference point and show the limits within which our global socioeconomic systems must operate to ensure the well-being of humanity.

This chapter focuses on three of these boundaries, described here as **climate change**, **land use change**, and **biodiversity loss**. These three crises are mutually reinforcing and form a complex web of interactions.

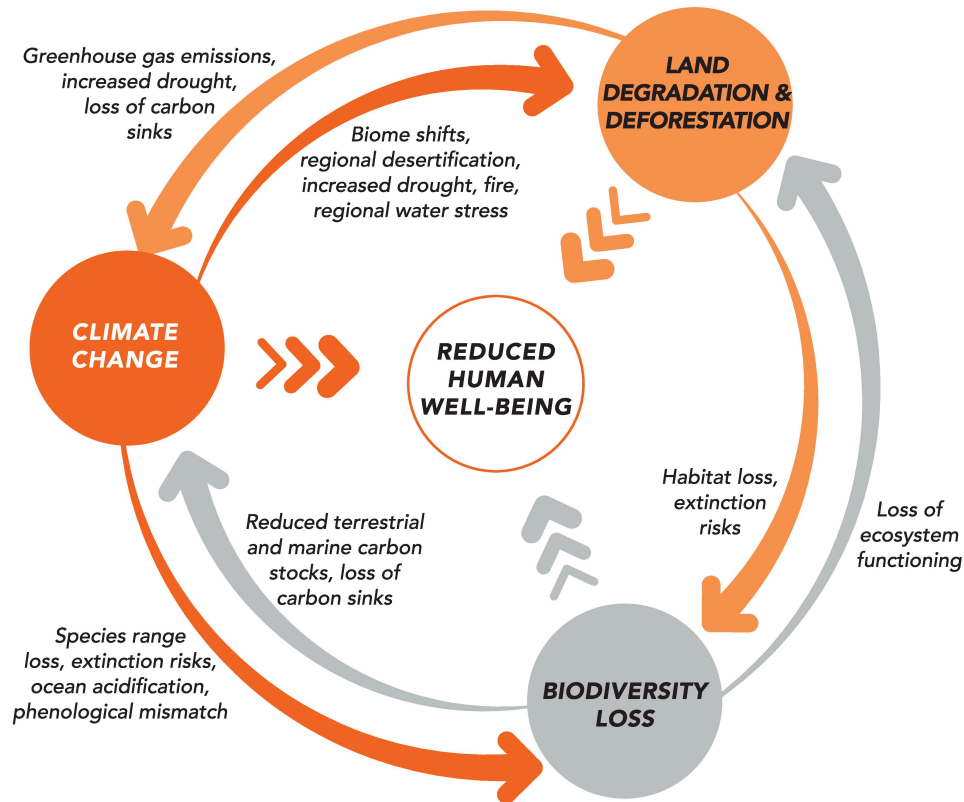


Figure 3.4: Interactions between biodiversity, climate change and land use. Source: UNEP (2021). Licensed under CC-BY.

3.3.1 The climate crisis (climate change)

Climate change – or, to use a term more illustrative of the urgency, the climate crisis – is one of the most pressing issues of our time. It receives much media attention and is exemplary of a “wicked” problem (Hulme 2009, 9). It is the focus of the Fridays for Future movement that was initiated by Greta Thunberg, and it is a topic of discussion at least, on the (global) political stage. While local conditions and natural fluctuations contribute to some uncertainties in climate models and statistics, the scientific evidence clearly shows that rising global temperatures have an enormous impact on environmental sustainability and human well-being. This is why climate change is one of the nine planetary boundaries.

The term **climate** describes the average state of the atmosphere (temperature, precipitation, humidity, etc.) over a period of at least 30 years at a specific location. By contrast, **weather** describes the short-term state (i.e. from seconds to weeks) and includes weather phenomena such as thunderstorms or heat waves (NCCS, n.d.). The **climate system**, in turn, encompasses not only the atmosphere but also the hydrosphere (oceans, lakes, rivers, groundwater), the cryosphere (glaciers, snow cover, sea and lake ice, permafrost), the biosphere (living organisms), and the lithosphere (soils or the outermost layer of the earth). In particular, it refers to the physical, biological, and chemical interactions and interdependencies between these five spheres. In addition, the climate system is influenced by radiation from space (see Figure 3.5).

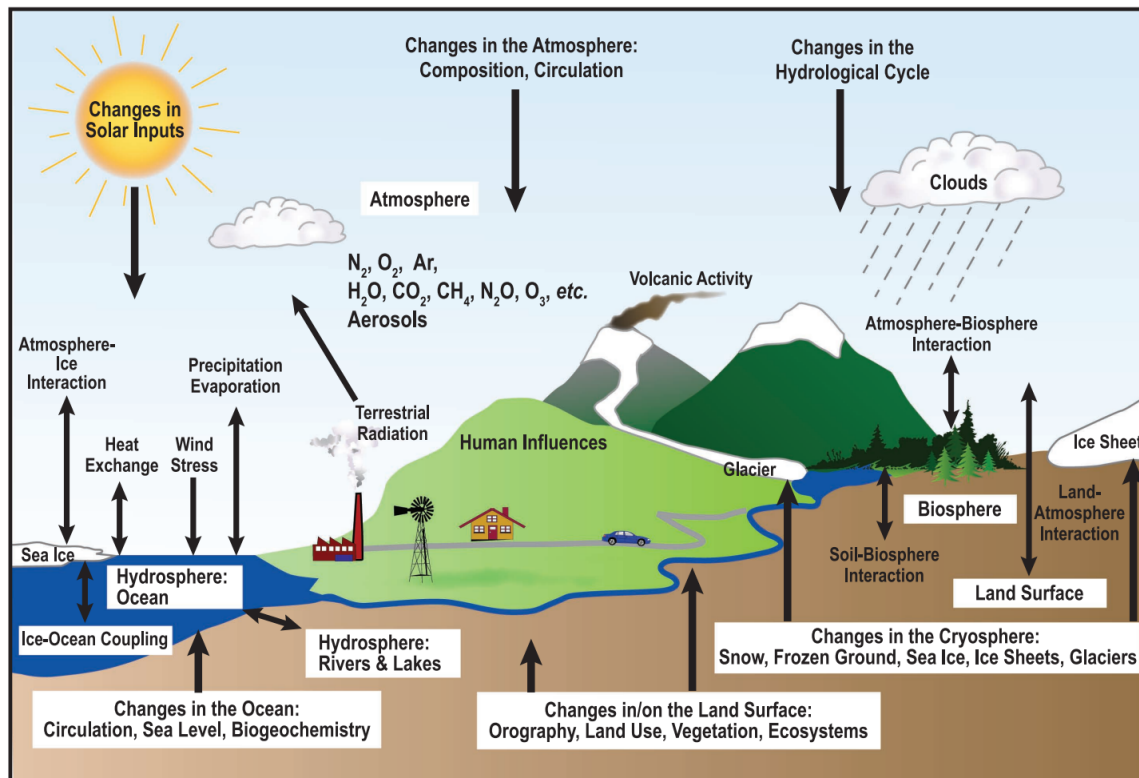


Figure 3.5: The components of the climate system, their processes, and interactions. This includes the atmosphere, biosphere (life), cryosphere (ice), hydrosphere (water), land, and human influences, as well as the interactions between these components (shown with two-way arrows). The parts of the system that can change are labeled as “Changes”. Source: IPCC AR4 FAQ (https://archive.ipcc.ch/publications_and_data/ar4/wg1/en/faq-1-2.html).

Material and energy flows between the various components of the climate system, such as the carbon cycle, can act as **feedback mechanisms**. We have already learnt about such feedback mechanisms in the chapter on systemic effects (see Chapter 2, *Approaches to Sustainable Development*). These can either be positive, if the feedback reinforces the original change, or negative, if the feedback attempts to reverse the original change. Here are examples of a positive and negative feedback effect in the climate system:

- **Positive feedback:** When the Earth’s surface temperature increases, so does water evaporation, creating a positive feedback loop. Water vapour is the most significant **greenhouse gas** in the atmosphere. As temperatures rise, more water vapour enters the atmosphere, increasing the reflection of heat back to the Earth’s surface and leading to further evaporation. This cycle accelerates warming and atmospheric water vapour accumulation. On Venus, this feedback mechanism intensified the greenhouse effect to the point where all liquid surface water evaporated.
- **Negative feedback:** Increased atmospheric CO_2 can enhance photosynthesis, a process known as **CO_2 fertilization** and an example of a negative feedback mechanism. Although an increase in carbon dioxide can cause plants to bind more carbon in biomass than before, it is

far from sufficient to compensate for the increasing CO₂ content in the atmosphere, which is mainly caused by the burning of fossil fuels and changes in land use. The ability of plants to take up additional CO₂ is limited by various factors such as nutrient availability, water resources, and temperature conditions. Therefore, this feedback mechanism is insufficient in compensating for the increase in atmospheric CO₂ concentrations caused by human activities.

Positive feedback mechanisms or the failure of negative feedback can lead to significant **thresholds** being exceeded in the climate system. These thresholds are referred to as **tipping points of the climate system** and are closely linked to the concept of planetary boundaries (see Section 2.3, *The debate about the Great Transformation*). When these thresholds are exceeded, abrupt and often irreversible changes can occur in the climate system, destabilizing the climate and leading to accelerated climate change. The concept of tipping points emphasizes the urgency of limiting global warming and reducing greenhouse gas emissions. Exceeding these thresholds will make it increasingly difficult to control climate change or minimize its effects. Ultimately, the Earth risks becoming a large greenhouse (“Hothouse Earth” scenario) (see Figure 3.6).

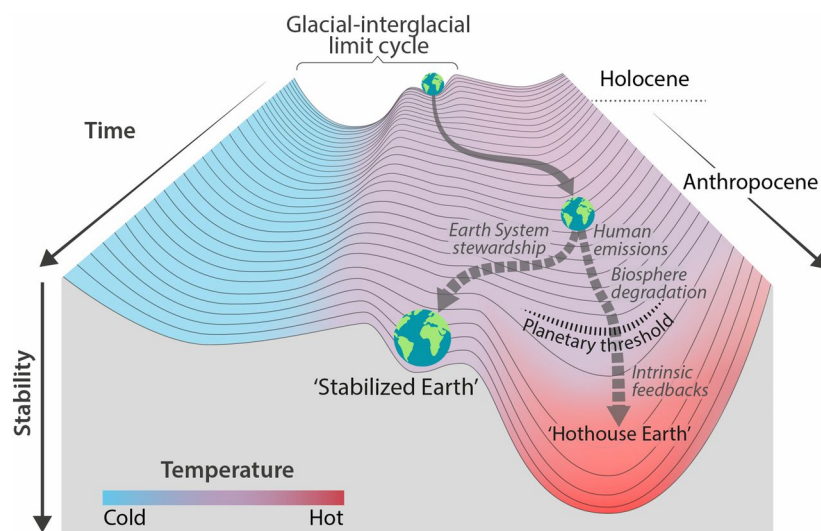


Figure 3.6: Stability landscape showing the pathway of the Earth System out of the Holocene and thus, out of the glacial–interglacial limit cycle to its present position in the hotter Anthropocene. Source: Steffen et al. (2018). Licensed under CC BY-NC-ND 4.0.

A total of 16 tipping points are assumed to exist (Figure 3.7), including the Siberian permafrost soils or the West Antarctic Ice Sheet. These tipping points have different threshold values, meaning that some will be reached earlier and with less global warming than others. If these tipping points are reached or exceeded, they can trigger self-reinforcing feedback effects that lead to further warming, initiate further tipping points, and intensify climate change. Some of these tipping points are already in a critical state, as explained impressively by Johan Rockström, one of the founders of the Planetary Boundaries concept (Rockström 2024). In the Amazon, for example, some areas now emit more CO₂ than they absorb – becoming carbon *sources* rather than sinks – due to forest fires, climate-change-induced water stress, and deforestation (see Section 3.3.2, *Land use change*) (Gatti et al. 2021; Harris et al. 2021). Consequently, the Amazon is already in transition and

losing resilience (Flores et al. 2024). Another worrying and very real tipping point is the Atlantic Meridional Overturning Circulation, or AMOC (Rahmstorf 2024a, 2024b). The AMOC is nearing its tipping point (van Westen et al. 2024), with projections suggesting it could collapse as early as the middle of this century (Ditlevsen and Ditlevsen 2023). A collapse would lead to increased European heat waves and sea-level rise along the north-eastern US coast (Rahmstorf 2024a).

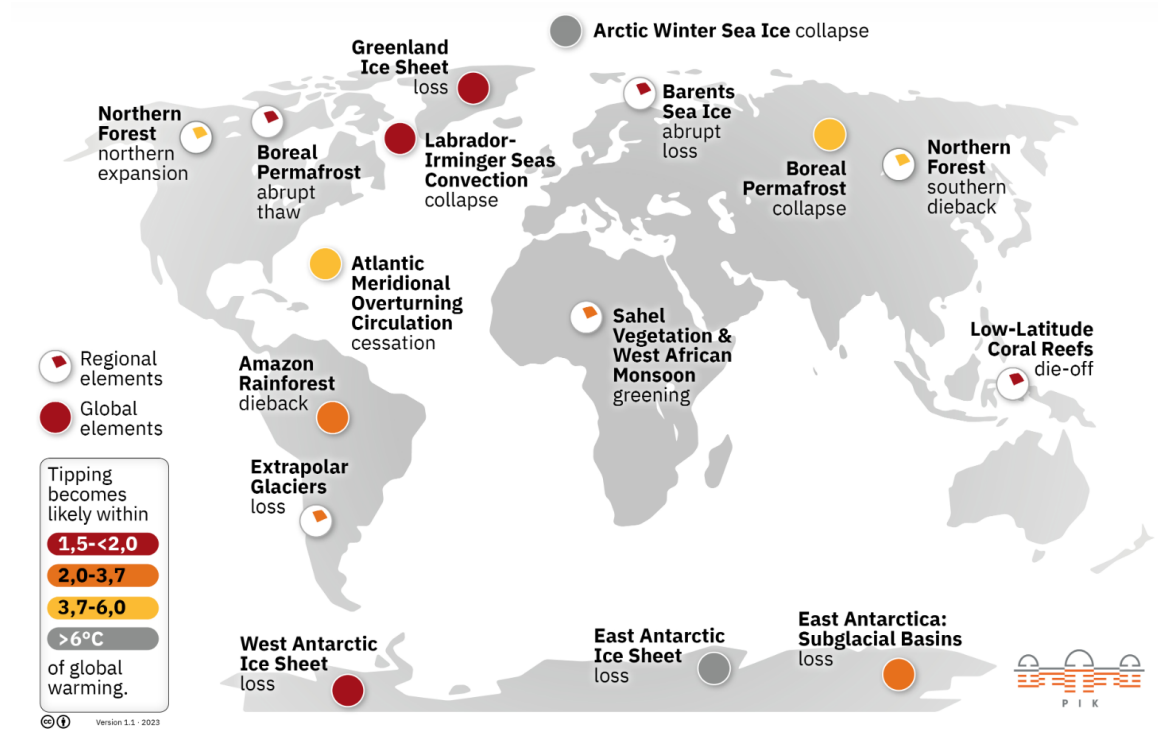


Figure 3.7: Spatial distribution of global and regional tipping elements. The colours indicate the temperature range in which tipping is likely to occur. Source: Designed at PIK, based on Armstrong McKay et al. (2022), expanded with interactions based on Lenton et al. (2019). Licensed under CC-BY.

But what is causing the Earth to warm and climate change to occur? In addition to natural causes of climate change such as geological, long-term causes (Earth's axial rotation, Milanković cycles, volcanic eruptions), the primary driver of global warming is the **greenhouse gas effect**. The greenhouse gas effect is a process where short-wave solar radiation enters the Earth's atmosphere and is re-emitted from the Earth's surface as long-wave infrared radiation. Greenhouse gases – such as carbon dioxide, methane, and water vapour – absorb some of this infrared radiation and emit it in all directions, causing some of the heat to return to the Earth's surface. Without this effect, the Earth would be too cold to support life. Today, however, there are too many greenhouse gases in the atmosphere due to anthropogenic emissions, causing the earth to warm up.

The largest share of human CO₂ emissions, around ¾ of the global total, originates from energy production, i.e. primarily electricity and heat generation (Figure 3.8). This is largely due to the combustion of coal, gas, and oil, with emissions from industry (comprising 24.2% of the global total), transport (16.2%), buildings (17.5%), and agriculture (18.4%). The main drivers of agricultural emissions are livestock farming, deforestation for agricultural land, and the use of nitrogen

fertilizers on agricultural soils.

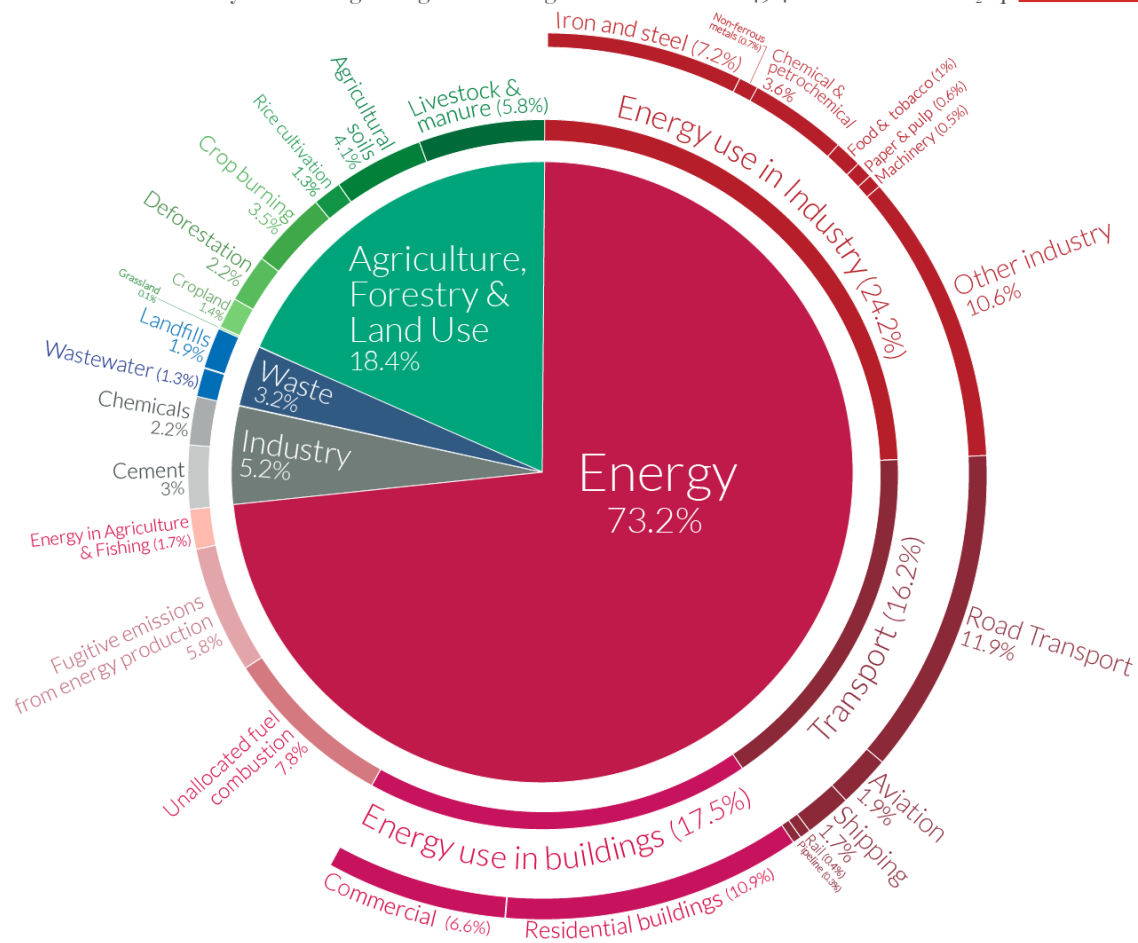
Within the transport sector, air traffic accounts for 1.9% of global emissions (see Figure 3.8). In Switzerland, however, flights account for about 11% of national emissions. These flight emissions are largely attributable to international air traffic (FOEN 2025), although the calculations exclude Basel-Mulhouse Airport, which is located on French soil. Despite this evidence, Switzerland has not yet introduced a CO₂ tax on kerosene (the main component of aviation fuel), although there is one on fossil fuels such as heating oil and natural gas. It should however be noted that climate impact is not solely determined by greenhouse gas emissions (Allen et al. 2018; Lee et al. 2021; Neu 2021). The warming potential of air travel can thus be significantly higher than indicated by greenhouse gas emissions alone, as non-CO₂ effects from flights – such as water vapour, nitrogen oxides (NO_x), and soot – also affect the climate (Lee et al. 2021).

Globally, air travel emissions have more than doubled since 1990, driven primarily by international flights. Despite a sharp drop in the number of flights in 2020 due to the pandemic, flight emissions have rebounded and are projected to surpass 2019's record highs in the coming years.

Global greenhouse gas emissions by sector

Our World
in Data

This is shown for the year 2016 – global greenhouse gas emissions were 49.4 billion tonnes CO₂eq.



OurWorldinData.org – Research and data to make progress against the world's largest problems.

Source: Climate Watch, the World Resources Institute (2020).

Licensed under CC-BY by the author Hannah Ritchie (2020).

Figure 3.8: Global greenhouse gas emissions by sector. Source: <https://ourworldindata.org/ghg-emissions-by-sector>.

Recent calculations and climate reports indicate that the Earth's temperature has risen by around 1.2°C in the last 10 years (2014-2023) compared to pre-industrial levels (1850-1900) (UN 2024). In addition, 2023 was the warmest year on record, with a global surface temperature of 1.45°C above the pre-industrial baseline. This may not sound like much, but it represents a substantial heat increase, given the oceans' vast size and heat storage capacity (Ritchie 2020).

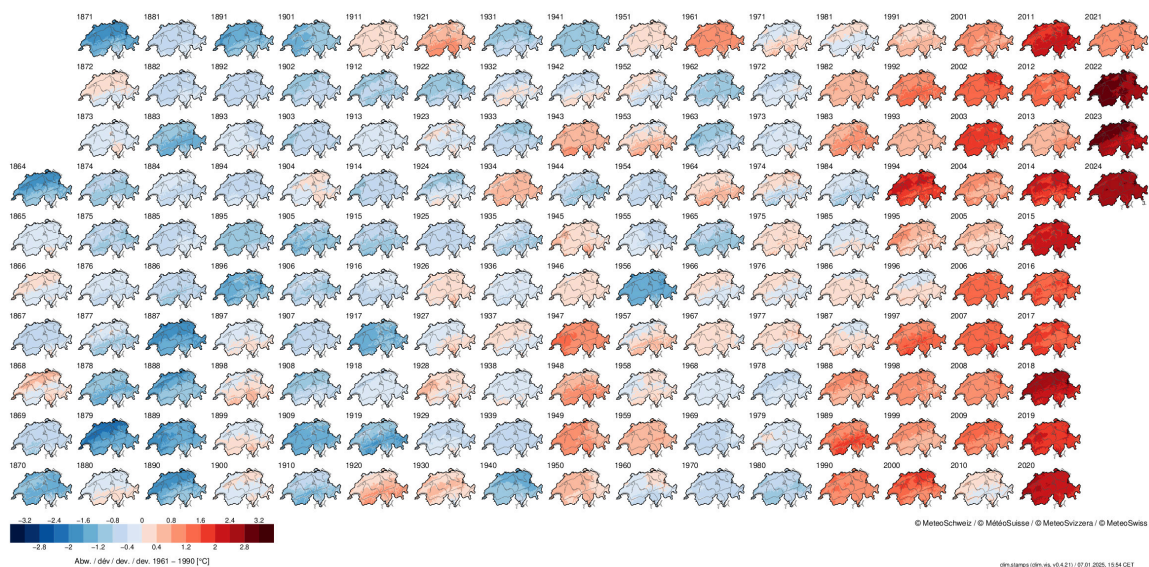


Figure 3.9: Switzerland's changing climate from 1864 to 2024. Source: <https://www.nccs.admin.ch/nccs/en/home/climate-change-and-impacts/climate-basics/what-is-climate-.html>.

The **Paris Climate Agreement**, adopted in December 2015, was the first international treaty where industrialized and emerging countries jointly pledged to curb greenhouse gas emissions. Its core objective is to limit global temperature rise to a maximum of 2°C, with efforts to pursue a 1.5°C limit, relative to pre-industrial times. In 2018, the Intergovernmental Panel on Climate Change (IPCC) published a Special Report containing some startling facts: Most notably, that we only have few years left – until 2030 – to avoid exceeding 2°C. The next major decision in international efforts that built on the Paris Climate Agreement was the 2022 climate conference, COP 27. At COP 27, signatories agreed to establish the Loss and Damage Fund, a result of decades of prior discussion, designed to help developing countries to receive financial support after extreme weather events.

The Intergovernmental Panel on Climate Change (IPCC)

The **Intergovernmental Panel on Climate Change (IPCC)**, established in 1988 by the United Nations and the World Meteorological Organization, synthesizes scientific research on climate change for policymakers worldwide. Its periodic IPCC reports provide a comprehensive assessment of the current state of knowledge on climate change, including the physical foundations, impacts, risks, and adaptation strategies. These reports are based on thousands of studies and are recognized as the most authoritative source of information on climate change. The IPCC reports serve as the basis for international negotiations and national climate policies, including the Paris Agreement of 2015.

c) The extent to which current and future generations will experience a hotter and different world depends on choices now and in the near term

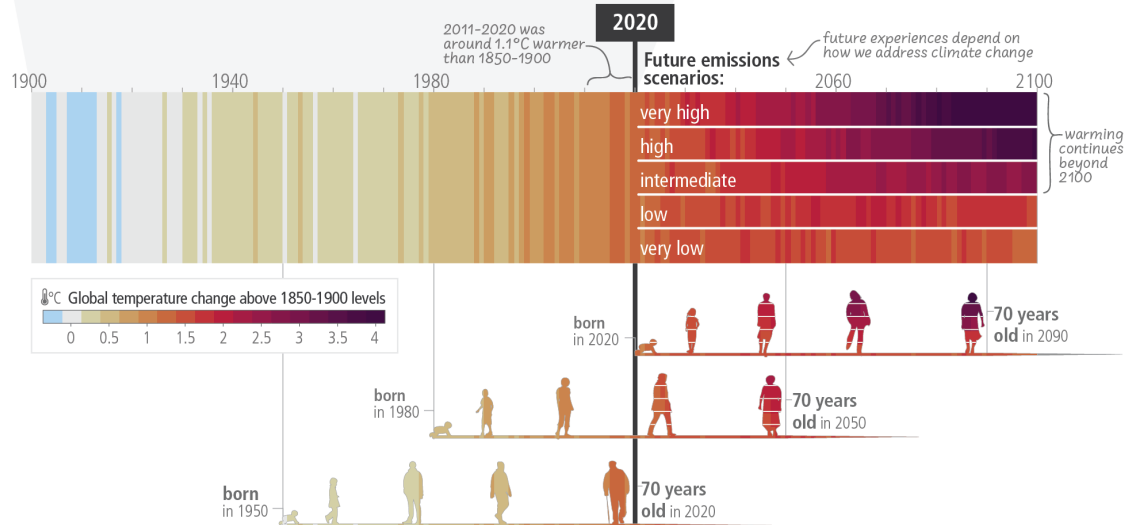


Figure 3.10: Excerpt from AR6 Synthesis Report Figure SPM.1 (c): Observed (1900–2020) and projected (2021–2100) changes in global surface temperature (relative to 1850–1900), which are linked to changes in climate conditions and impacts, illustrate how the climate has already changed and will change along the lifespan of three representative generations (born in 1950, 1980 and 2020). Source: IPCC (2023).

But what will it take to prevent the temperature from rising by more than 1.5 degrees? A whole lot. Anthropogenic CO₂ emissions must fall by 45% between 2010 and 2030, and net CO₂ emissions must fall to zero by 2050. While we are adopting greener technologies for marginal activities like car purchases or building efficiency, we still risk neglecting the serious threat posed by the carbon that is already in the atmosphere. It's important to distinguish between ongoing emissions (flow) and accumulated atmospheric carbon (stock). Flow plays a subordinate role for the planet – it's the stock that drives global warming. Climate change is therefore primarily a stock problem, rather than a flow problem. This is a challenging concept, not only for laypeople. Indeed, Sterman and Sweeney (2007) observed that most adults had difficulty in grasping the stock aspect of climate change.

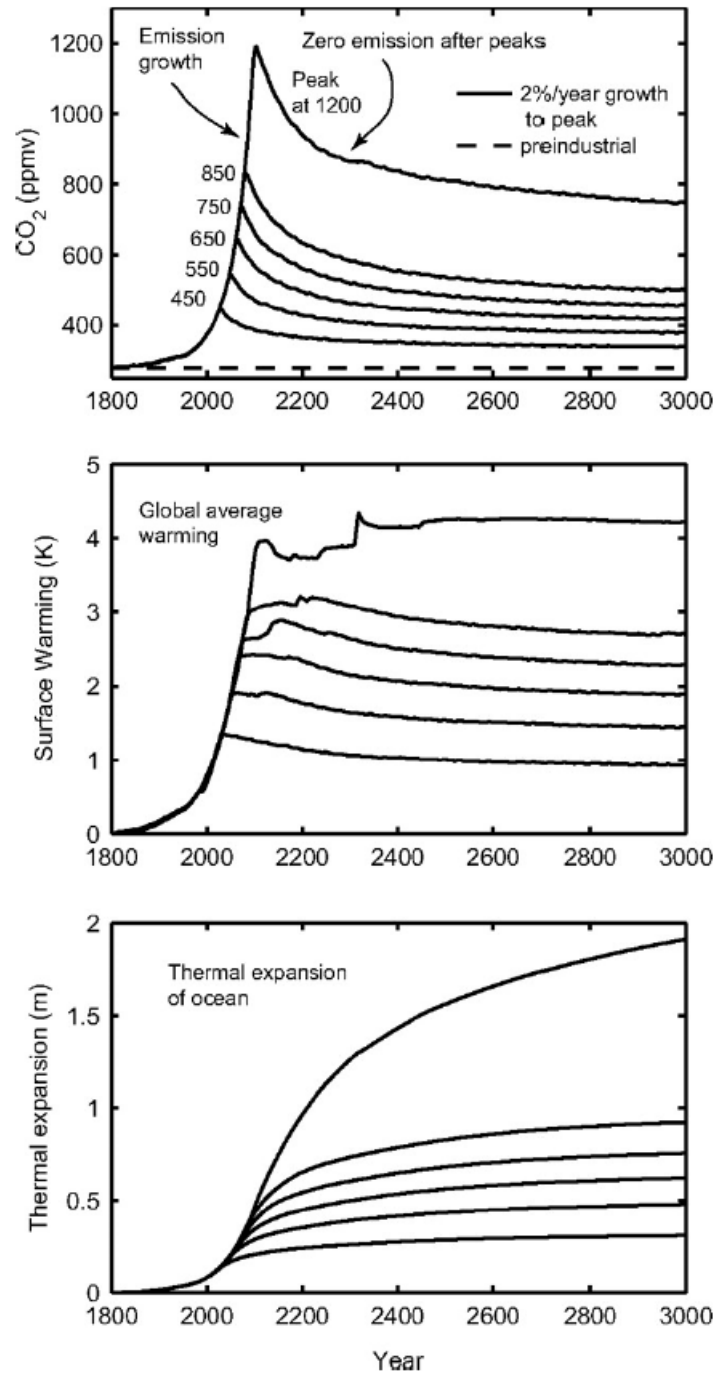


Figure 3.11: Carbon dioxide and global mean climate system changes (relative to preindustrial conditions in 1765). Source: Solomon et al. (2009).

An effective mental model is the bathtub analogy: if the inflow of water through the tap (i.e. our emissions, or flow) exceeds the outflow of water through the drain (carbon absorption by rainforests, oceans, etc.), the water level (the atmospheric carbon stock) increases. The past five

years, and twenty of the last twenty-two, have been the warmest on record. This warming trend is a direct result of the rising water level in that “bathtub”.

Impacts: Climate change is expected to have a range of direct and indirect impacts, some of which are already occurring. These include sea level rise, shrinking glaciers, threats to water and food security, an increase in extreme weather events, adverse health effects related to temperature, heatwaves, droughts and excessive precipitation, wildfires, and an increase in geomorphological activities and thus natural hazards due to permafrost thaw.

3.3.2 Land use change

Land use change, like deforestation for agriculture or urban expansion, has a significant impact on the **environmental dimension of sustainability**. Land use change threatens biodiversity and disrupts ecosystem functions. It can also increase greenhouse gas emissions, alter water cycles, or deplete valuable agricultural land. This can result in a reduction of ecosystem services (see Section 3.2, *Ecosystems, ecosystem management, and ecosystem services*), ultimately restricting human well-being. Land use change is one of the nine **planetary boundaries** that has already been exceeded, according to Richardson et al. (2023).

Some key definitions are needed to understand land use change:

- **Land cover** refers to the characteristics of the land surface and its immediate subsurface, including biota, topography, and human structures (Lambin et al. 2000, 322).
- **Land use** is the way in which humans use land cover (Lambin et al. 2000, 322).
- **Land use change** represents the change from one land use to another (Lambin et al. 2000, 322).
- **Land degradation** means “reduction or loss, in arid, semi-arid and dry sub-humid areas, of the biological or economic productivity and complexity of rainfed cropland, irrigated cropland, or range, pasture, forest and woodlands resulting from land uses” (UN 1994, 4).
- **Land systems** “represent the terrestrial component of the Earth system and encompass all processes and activities related to the human use of land, including socioeconomic, technological and organizational investments and arrangements, as well as the benefits gained from land and the unintended social and ecological outcomes of societal activities” (Verburg et al. 2013, 433).
- Land can also be understood as a **socio-ecological system** (Chapin et al. 2006) in which the social and ecological subsystems interact and depend on each other. This system is characterized by mutual feedback, with humans considered integral components of nature (Verburg et al. 2015).

Various **drivers**, i.e. reasons or causes, are responsible for the occurrence of land use change. These factors vary depending on the type of land use. For example, the main drivers of global deforestation, a form of land use change, are commercial agriculture, slash-and-burn agriculture, commercial forestry, urbanization, and forest fires (Curtis et al. 2018). Deforestation has already led to an alarming decline in biodiversity (see Section 3.4, *Biodiversity loss*) and the disruption of

carbon cycles (see Section 3.2, *Ecosystems, ecosystem management, and ecosystem services*). For example, the Agriculture, Forestry, and Other Land Use (AFOLU) sector is a major contributor to **global anthropogenic greenhouse gas emissions**, accounting for approximately 13% of CO_2 , 44% of CH_4 , and 81% of N_2O emissions (IPCC 2019). These interactions between climate change and land use/land use change show that land systems and climate systems should be considered systemically, rather than in isolation (Richardson et al. 2023). Figure 3.12 is a schematic representation of these interactions between land and greenhouse gases.

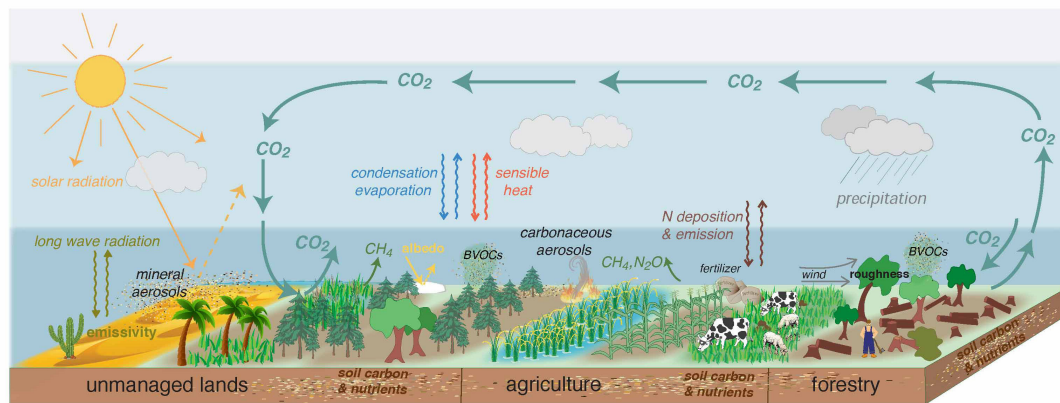


Figure 3.12: The structure and functioning of managed and unmanaged ecosystems that affect the local, regional, and global climate. Source: IPCC (2019).

Beyond their impact on climate-relevant greenhouse gas flows, land use and land use change also disrupt other **biogeochemical material cycles**, such as nitrate, phosphorus, or sulphur cycles, linking them to another already-exceeded planetary boundary, that of biogeochemical flows (Richardson et al. 2023). A classic example of this is **eutrophication**, a process that occurs when agricultural fertilizer, such as phosphorus, enters nutrient-poor waters and leads to excessive growth of algae, known as algal blooms. As these blooms decompose, they deplete the oxygen level, ultimately restricting or even preventing aquatic life (Galloway et al. 2008; Smith et al. 2006).

Land use change in the tropics: the case of Madagascar

In the tropics, the forms of agriculture best adapted to local conditions (i.e. without fertilizers or other inputs) are agroforestry and slash-and-burn agriculture. **Agroforestry** is a permanent land use system in which annual crops are planted alongside trees, or where perennial crops (e.g. manioc, vanilla, bananas, and lychee) are planted together. By contrast, **slash-and-burn agriculture**, also known as **shifting agriculture**, refers to a land use system where a different area is cleared each year and a crop such as maize is grown, while the previously cultivated fields are left fallow to allow nutrients to rebuild.

As mentioned above, slash-and-burn agriculture, such as that practised in Madagascar, is one of the main causes of global deforestation, which is why governments and NGOs have been fighting slash-and-burn agriculture since colonial times. One challenge with slash-and-burn agriculture is that when fallow periods become too short due to a lack of land, farmers expand into unused forest, clearing it to maintain the cycle of slash-and-burn. This deforestation results in substantial biomass that needs to be burnt. In Madagascar, for example, more than 6,000 burnt areas were recorded within one month from mid-September to mid-October 2023.

This is why deforestation in Madagascar is blamed on the local population. But slash-and-burn agriculture is a necessary survival strategy for local inhabitants, who have few alternatives and rely on it to secure land for future generations. This conflict of objectives is specifically a **land use conflict**.

Land use conflicts in Switzerland: the Solar Express

Another example of a land use conflict can be observed in Switzerland, in a conflict of interest between a decentralized renewable energy project and landscape services (see Kienast et al. 2017). In 2022, the Swiss government launched a “solar offensive” in the form of a project called “Solar Express”. The aim was to promote alpine photovoltaic systems through high subsidies that would cover up to 60% of investment costs. In the summer of 2024, the groundbreaking ceremony for the first large-scale plant, covering an area of 300,000 m², took place in Sedrun, in Switzerland’s easternmost canton of Graubünden. However, such large-scale solar plants clash with other land uses, particularly affecting agricultural land, nature conservation areas, and tourist regions. Solar installations alter landscapes and could affect flora and fauna, leading to fears among farmers and nature conservation organizations about the loss of valuable farmland and biodiversity areas. At the same time, supporters of renewable energies are calling for the rapid implementation of these projects, to reduce Switzerland’s dependency on fossil fuels and protect the climate. This highlights the tension between the need for renewable energy on the one hand, and the desire to preserve the landscape and the environment on the other.

Deforestation in tropical regions is a form of land use change that is influenced not only by local needs, but also by international decisions and demands (Curtis et al. 2018). This suggests that preventing deforestation must be addressed at both local and international levels (DeFries et al. 2010). However, distant factors influence not just deforestation, but land use change in general (Meyfroidt et al. 2013). In science, such phenomena and processes are referred to as **telecoupling**

(Liu et al. 2013). Telecoupling describes the socioeconomic and ecological interactions between distant human and natural systems, analysing how actions in one place (the sending system) can affect another place (the receiving system). For example, new EU regulations require that products entering the EU market from the end of 2024 are deforestation-free ([Regulation on Deforestation-free products – European Commission](#)). These regulations are likely to have significant ecological (and social) effects in the countries of cultivation (Coenen et al. 2023).

Land rights and land tenure

Land use and land use change are closely linked to issues of **land rights** and **land tenure**, as these factors largely determine who is authorized to use land, and how it is used. Secure land tenure and clearly defined land rights are crucial for sustainable land use practices, as they incentivize long-term investment in land management and promote the protection of resources. Insecure or unclear land rights, on the other hand, can lead to land use changes that harm the environment, such as uncontrolled deforestation. These issues become particularly explosive in the context of **land grabbing**, in which large areas of land are acquired by foreign (or even domestic) investors, often without adequate consultation with or compensation for the local population. Land grabbing can lead to conflicts that undermine existing land rights and change land use in a way that jeopardizes environmental and social sustainability.

This chapter has shown that land, land use, and land use change are a multifaceted topic. Underscoring the importance of this issue, Meyfroidt et al. (2022) summarized ten facts about land systems for sustainability (Figure 3.13, left), grounded in empirical research. These facts, in turn, lead to ten associated challenges (Meyfroidt et al. 2022). For example, Fact 2 (“Land as complex systems”) leads to the challenge that the consequences of interventions in land systems are difficult to predict and understand, which makes it difficult to develop effective measures or policies regarding land use. Several issues discussed in this chapter align with these facts, such as telecoupling and Fact 5, “Distant connections”.

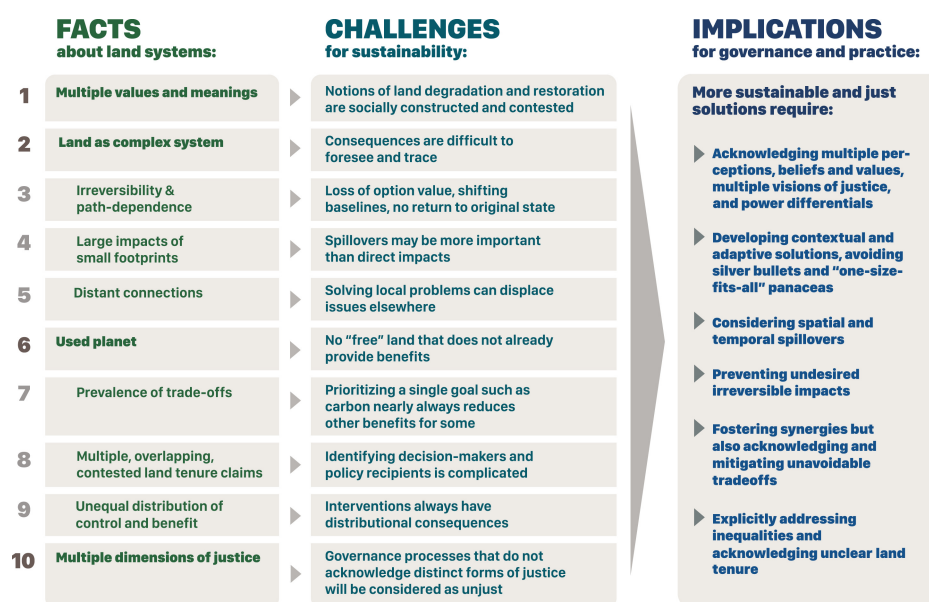


Figure 3.13: Ten empirical realities (facts) about land systems that have strong, general support. Challenges summarize the issues that arise from each fact when trying to manage and govern land systems for sustainability. Implications summarize how governance of land systems for sustainability could be improved by acknowledging these facts and challenges. The 10 facts are structured through four higher-level facts (1, 2, 6, and 10) and several lower-level facts that derive from these higher-level ones (3, 4, and 5 deriving from 2; 7, 8, and 9 deriving from 6), yet they all express specific realities that imply distinct challenges. Source: Meyfroidt et al. (2022). Licensed under CC BY 4.0.

Example: Land investments in Laos

Another example of land use change and its far-reaching implications is commercial land investment in Laos. The Lao People's Democratic Republic is a mountainous, landlocked country with a high poverty rate and a few small urban centres. Situated between China, Vietnam, Thailand, Cambodia, and Myanmar, Laos emerged from its communist isolation in the 1990s. The country is rich in natural resources such as land, forests, and minerals. To promote national economic growth, the Government of Laos is pursuing a strategy of **turning land into capital**, i.e. converting “unused” land into productive land in the hope of achieving rural development through spillover effects. This strategy is mainly implemented through commercial land investments, where the government grants land concessions to domestic and foreign investors.

But to what extent does this policy actually contribute to sustainable development? To find out, scientists from the Centre for Development and Environment (CDE) worked with the Government of Laos to take stock of land concessions and examine the quality of the investments. A national digital information system was developed, enabling Laos to manage its land concessions effectively. The system also revealed that not all investors fulfil their promises, calling into question the sustainability of this land use strategy.

The changes in land use towards banana plantations, other tree plantations such as rubber plantations, mines, and hydropower projects have led, among other things, to a loss of forest areas, with around a third of all concessions being granted within national forest areas. This has also led to a loss of important ecosystem services and a reduction in biodiversity. Beyond the ecological impact on the local environment, the land investments have also had socioeconomic consequences. Despite increased financial income for some, the land investments have negatively affected the well-being of the local population in many areas, reducing food security and access to land and natural resources vital to subsistence-oriented livelihoods (Nanhthavong et al. 2021; Hett et al. 2020)

<https://youtu.be/J6QpmuWYbfE?si=mkIRErt6Khp6Vpca>

3.4 Biodiversity loss

Biodiversity is defined as the variety of life in all its forms, ranging from genes to species to ecosystems (UN 1992a). It is an essential foundation for the well-being of humankind and thus a substantial component of **environmental sustainability**, forming the basis for numerous ecosystem services (Haines-Young and Potschin-Young 2010) or Nature's Contribution to People (Antunes et al. 2024). Biodiversity is also closely linked to several of the **planetary boundaries**, especially land-system change and biosphere integrity (Steffen et al. 2015). It is clear that an unchecked loss of biodiversity could trigger tipping points in ecosystems, disrupting the Earth's balance and severely impairing the ability of our planet to support human life. The protection and restoration of biodiversity is therefore considered key to a sustainable future.

Yet **biodiversity** is currently in an alarming state (see box on *Biodiversity monitoring*). Worldwide, species diversity has been declining for several decades (Butchart et al. 2010; Díaz and Malhi 2022), a downward trend that is predicted to continue (Díaz and Malhi 2022). As a result, there is already

discussion of a sixth mass extinction (Ceballos et al. 2020). To counteract this development, the **Convention on Biological Diversity** (CBD) was ratified at the 1992 Rio Earth Summit (UN 1992a), with Swiss participation. This legally binding agreement means that if Switzerland sends delegates to subsequent conferences, any new provisions must be legally implemented.

Biodiversity monitoring

Biodiversity monitoring is an important tool for tracking changes in biodiversity. The International Union for Conservation of Nature (IUCN)'s Red List of Threatened Animal and Plant Species, maintained since 1964, plays a particularly important role in this. It categorizes the status of a species into one of nine categories, from “not endangered” to “extinct”. Figure 3.14, for example, shows the percentage of species in each category at the global level. The same categories are also used in Switzerland. Such databases, among other benefits, help improve policymaking and nature conservation planning, and facilitate more appropriate allocation of (financial) resources in research and society. Despite the extensive and established monitoring of species, cautious studies assume that only between 1.5% and 18% of terrestrial vertebrate species on our planet have been identified and described (cf. Moura and Jetz 2021).

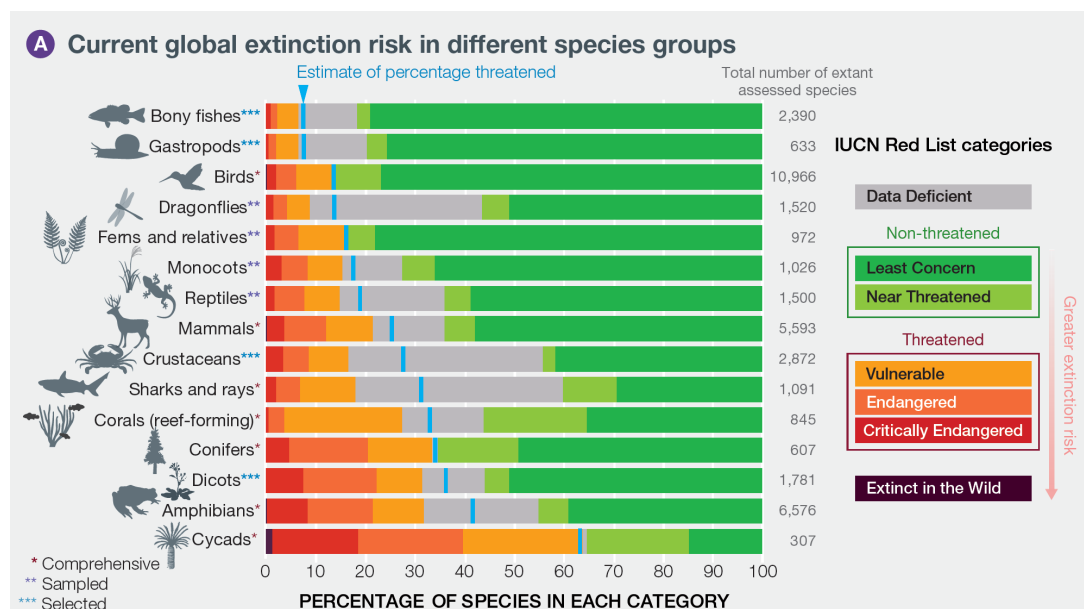


Figure 3.14: Current global extinction risk in different species groups. Source: IPBES (2019), Figure SPM3.

Despite these international efforts, global biodiversity remains under unprecedented pressure. This is due to direct and indirect **drivers** (Figure 3.15) that have significantly accelerated over the past 50 years (IPBES 2019, 25). For example, human activities such as **land use change** have drastically altered and reduced natural habitats, leading to a significant loss of species and genetic diversity (Pereira et al. 2010). Species loss is also closely linked to anthropogenic **climate change** (see Section 3.3.1, *The climate crisis*), as rising temperatures are making or will make many areas and regions uninhabitable for certain species (WWF 2022, 16). This decline jeopardizes not only the stability of ecosystems, but ultimately also the long-term sustainability of human societies

(Cardinale et al. 2012).

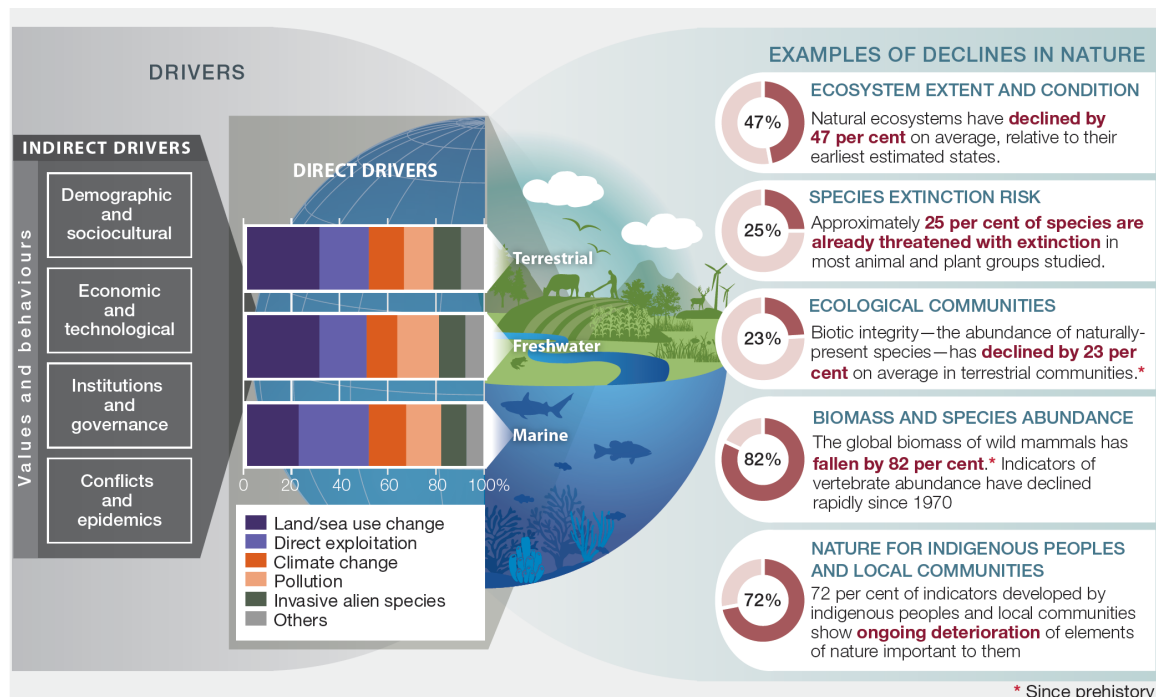


Figure 3.15: Examples of global declines in nature, emphasizing declines in biodiversity, that have been and are being caused by direct and indirect drivers of change. Source: IPBES (2019), Figure SPM2.

The latest major international agreement, the Kunming-Montreal Global Biodiversity Framework (UNEP-CBD 2022), sets new goals and targets (Figure 3.16) aimed at making global biodiversity conservation more focused. A key target is Target 3, which specifies that by 2030, at least 30% of all terrestrial and marine areas, particularly those of high ecological importance, must be conserved through a system of protected areas. Studies have shown that well-managed protected areas can effectively conserve biodiversity. However, as of 2024, only 16.32% of terrestrial areas and 8.35% of marine areas were protected (Protected Planet 2025). More radical approaches even call for 50% protected areas (the Half Earth Project).



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categorized into three types: intrinsic (“nature for nature”), instrumental (“nature for society”), and relational (“nature as culture”) (Durán et al. 2023) (Figure 3.17). The concept of ecosystem services, for example, falls under **nature for society**, as it corresponds to a utilitarian view of nature in which society derives the greatest possible benefit from nature (without harming it). By contrast, a nature park that prioritizes strict nature protection or the rewilding of areas, for example, can be classified as embodying intrinsic values, because human intervention and thus its impact on nature is minimized. This **nature for nature** narrative also extends to concepts like smart cities, for example, as these seek to limit human activity to urban areas as efficiently as possible. The **nature as culture** vision, for its part, portrays a world where values such as reciprocity and harmony define human relationships with nature at every level. Here, biological diversity and cultural diversity are jointly preserved and managed within closely linked biocultural systems. These systems are supported by local, self-determined governance structures that respect Indigenous sovereignty and local identities. Economic exchange prioritizes the social value of goods over their monetary value. This vision includes, for example, local and decentralized food production as well as communities that operate in a resilient, biocultural network with deep spiritual connections to nature.

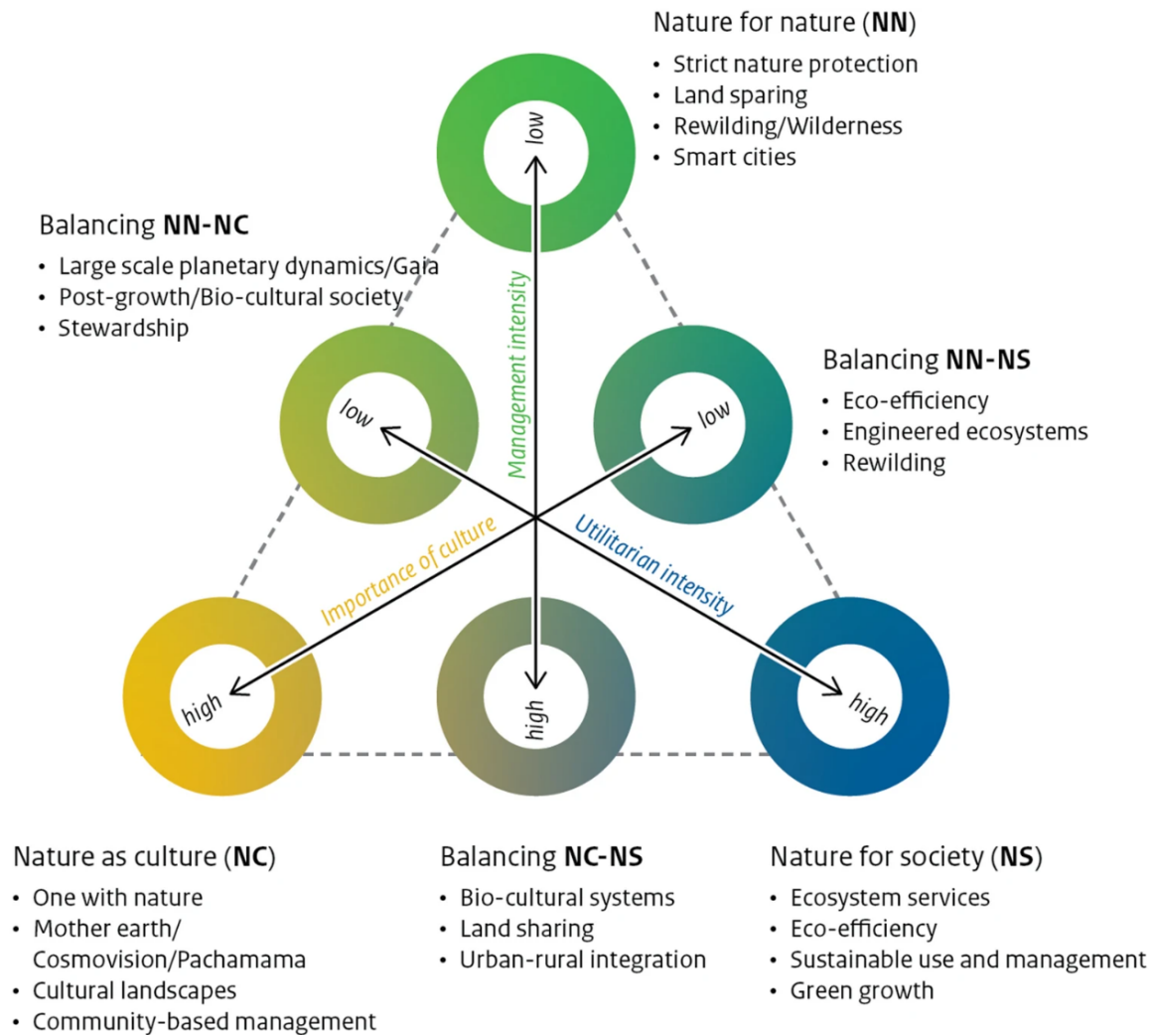


Figure 3.17: The Nature Futures Framework (NFF) and the three main perspectives on nature conservation. Source: Durán et al. (2023), based on Pereira et al. (2020). Creative Commons CC BY 4.0.

The diverse values associated with biodiversity and nature – and to what extent they should be protected – therefore require a normative assessment. They are based not only on empirical systems knowledge but also on ethical principles, as explained at the start of Chapter 3.

As we have seen, and as with other sustainability challenges and dimensions, there is no established consensus on how to best implement and achieve sustainability in the environmental dimension. In this dimension, too, sustainable development remains a social negotiation and decision-making process. Given the diversity of different perspectives and interests of various stakeholders, preferences regarding approaches and measures also vary. To select suitable measures and harmonize the various levels of responsibility (Figure 3.18), we can, for example, refer back to Hans Jonas's (1979) ethics of responsibility, which we learned about at the start of the book. Integrating these various areas of responsibility can foster a holistic and sustainable ethic that appropriately considers the needs and interests of all aspects of life and nature.

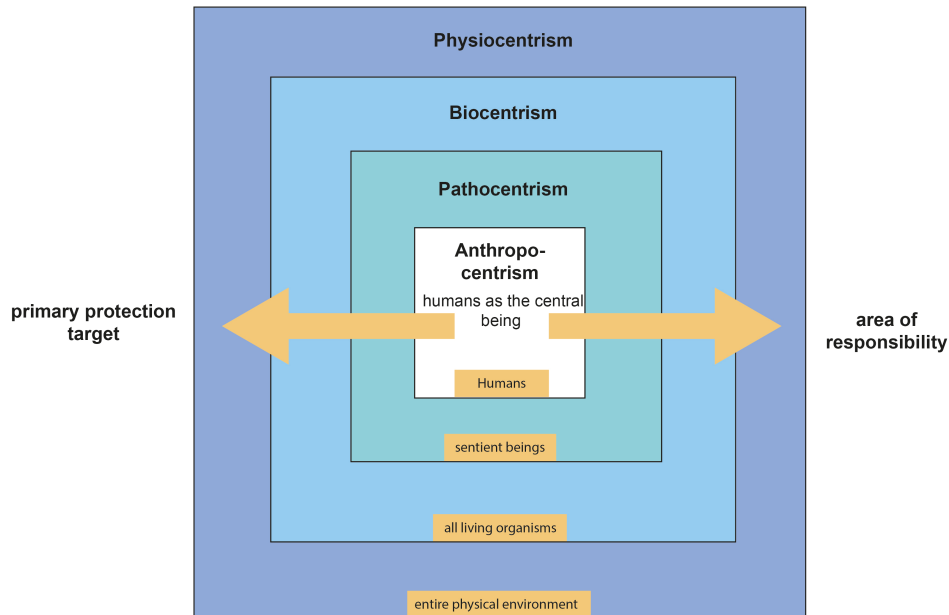


Figure 3.18: Different ethical approaches. Source: Own illustration based on Carnau (2011, 143.).

3.5 Strategic approaches

These days, various strategies are being pursued or at least considered to achieve a transformation in the environmental dimension of sustainable development (see Figure 3.19 for a non-exhaustive list). Some of these strategies were previously discussed in this textbook. While the approaches mentioned do not necessarily have to be transformative per se, they can certainly have an impact on the environmental dimension of sustainability if applied appropriately. The approaches presented in this section are categorized according to the “Levers for Transformation” of the Global Sustainable Development Report (GSDR). The 2019 GSDR defines four levers: governance, economy and finance, individual and collective action, and science and technology. The 2023 GSDR added a fifth lever, capacity building, as this is essential for a transformative process both as a lever itself and as a supporting factor for the other levers (Independent Group of Scientists appointed by the Secretary-General 2023).

TRANSFORMATIONS TO THE SDGs: ENTRY POINTS AND LEVERS

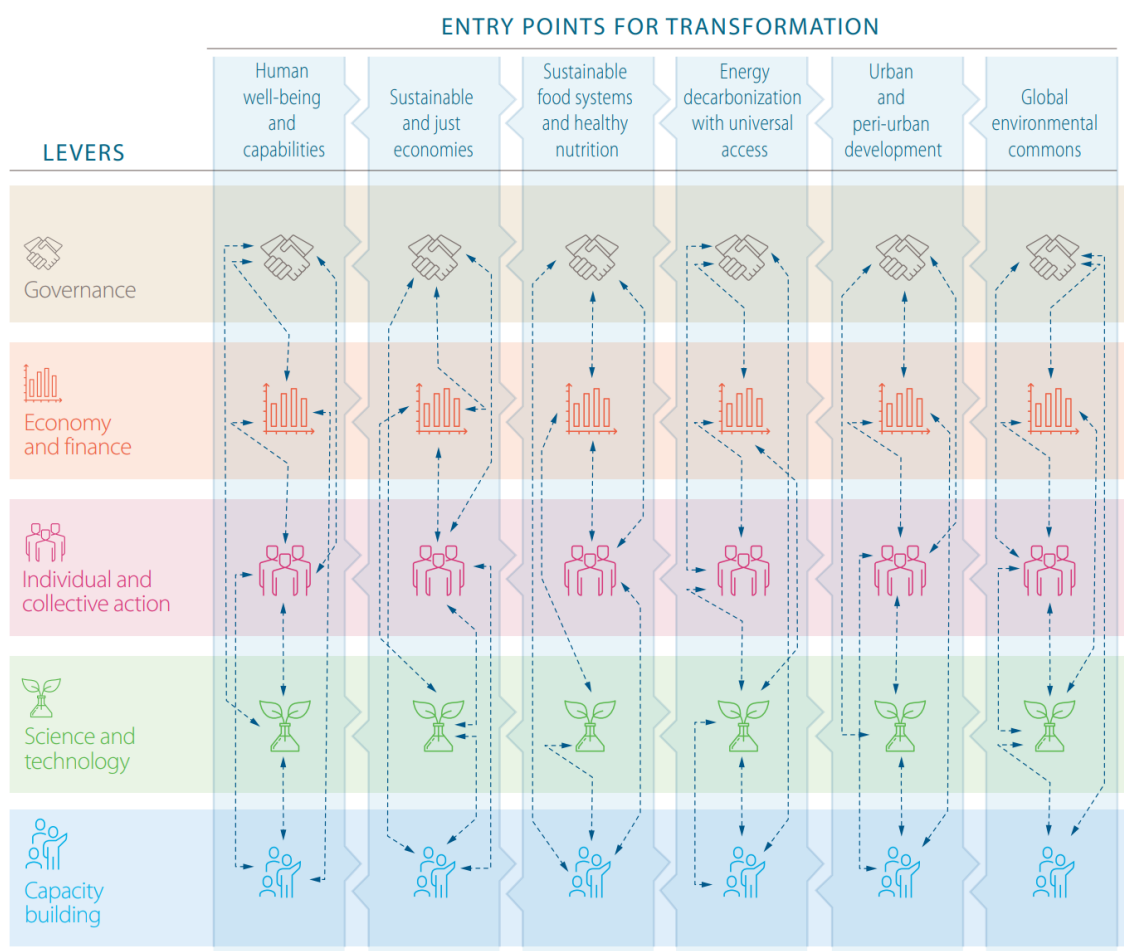


Figure 3.19: Entry points for transformation to the Sustainable Development Goals (SDGs), as proposed by the Global Sustainable Development Report 2023. Source: Independent Group of Scientists appointed by the Secretary-General (2023).

For each lever, we present selected strategies related to climate, land use, or biodiversity (our three topics of focus in this Chapter):

The **governance** lever involves institutions and spaces that determine the direction of development by setting goals, coordinating measures, creating regulations, and facilitating funding at national and regional level. Good governance should also promote synergies, take into account conflicts of interest, and strengthen cooperation between a wide range of stakeholders. Governance thus plays a key role in the development and implementation of environmental sustainability strategies, as it can establish the policy framework for sustainable practices through regulatory mechanisms, incentive structures, and laws. Governance levers include:

- Creation of protected areas to preserve biodiversity

- Subsidies for sustainable practices, such as direct payments for biodiversity-friendly farming methods

The **economy and finance** lever primarily refers to the significant public and private investments required to achieve the SDGs and transformation (globally, between USD 1.4 and 2.5 trillion per year). This lever is thus a key enabler of environmental sustainability strategies and policies, for example through the funding of sustainable technologies. Examples include:

- Payments for Ecosystem Services (PES)
- REDD+ programmes to reduce deforestation and promote sustainable land use

The **science and technology** lever refers to social and technological innovation as well as cost-effective and scalable technologies. This includes investing in research and development, fostering greater international cooperation, and enabling access to proven technologies (especially for low-income countries), where science can help explain complex interrelationships and develop evidence-based solutions. Essential levers for sustainability transformations include:

- Carbon sequencing technologies
- Nature-based solutions that use natural processes to address climate change and environmental problems
- Innovative land use systems, such as agroforestry

Individual and collective action

Social change often begins in people's hearts and minds through local organization and mobilization before translating into laws and economic policy. When a critical mass adopts new practices or norms, this can be supported by education, information campaigns, financial incentives, and legislation, and extended to the whole of society. Examples of sustainability strategies and policies in the field of individual and collective action include:

- Sustainable consumption campaigns that motivate consumers to choose environmentally friendly products
- Promotion of solidarity-based farming models (SoLaWi) that strengthen regional and organic food production

Capacity building

Capacity building to support the transformation to achieve the SDGs is multifaceted and depends on the goal, the required transformations, and the specific country or regional context. Among other things, competences are required in the areas of strategic orientation, innovation, coordination, learning, and resilience. Environmental sustainability strategies and policies related to the capacity building lever include:

- Establishment of advice centres or digital platforms to enable continuous learning and the exchange of best practices, e.g. in the field of agroforestry
- Supporting local initiatives to improve knowledge transfer

Note: It is important to be aware that approaches aren't always synergistic; often, an approach addressing an environmental dimension of sustainability involves trade-offs with other approaches or with the social or economic dimensions of sustainability.

Further reading

- Guyer, Madeleine, Myriam Steinemann, and Nina Saalismaa. 2022. *Biodiversity and Sustainable Development*. Climate Change & Environment Nexus Brief. Swiss Agency for Development and Cooperation.

3.6 Sustainability Basics: The Quiz – Chapter 3

Which of the following statements about the concept of ecosystem services are correct? (More than one statement may be correct.)

- ☐ Four different types of ecosystem services can be distinguished: regulating services, provisioning services, cultural services, and supporting services.
- ☐ The concept of ecosystem services represents the direct and indirect benefits that humans obtain from ecosystems.
- ☐ The term “ecosystem services” is a scientific term that describes the congregations of nature-oriented religions (so-called nature religions), such as animism, during which nature as a whole is worshipped.
- ☐ Three different types of ecosystem services can be distinguished: services of avoidance, improvement, and shifting.
- ☐ The concept of ecosystem services represents the accumulation of biomass within an ecosystem within a year, measured in kilojoules.
- ☐ The concept of ecosystem services represents a measure for biodiversity and therefore also a measure for an ecosystem's resilience.
- ☐ The concept of ecosystem services represents all measures taken by humans in order to preserve ecosystems.
- ☐ Unlike the concept of Sustainable Development, the concept of ecosystem services is not anthropocentric, but ecocentric.

The ecosystem services concept refers to the direct and indirect benefits that humans derive from ecosystems and is therefore explicitly anthropocentric. A commonly used classification distinguishes four types of ecosystem services: provisioning, regulating, cultural, and supporting services. The other statements confuse ecosystem services with biodiversity metrics, ecological productivity measures, conservation actions, alternative classifications, or unrelated cultural or religious concepts.

- True
- True
- False
- False
- False

- False
- False
- False

What are the four main categories of ecosystem services?

- ☐ Economic, social, environmental, cultural
- ☐ Material, spiritual, industrial, social
- ☐ Provisioning, regulating, cultural, supporting

Ecosystem services are commonly classified into four main categories: provisioning services (e.g. food and raw materials), regulating services (e.g. climate regulation), cultural services (e.g. recreational and spiritual benefits), and supporting services (e.g. nutrient cycling and soil formation).

- False
- False
- True

Which of the following statements apply to the concept of ecosystem services, and which do not? (More than one statement may be correct.)

- ☐ Ecosystem services can be divided into the following categories: supporting, provisioning, regulating, and cultural.
- ☐ Ecosystem services represent the direct and indirect contributions of ecosystems to human well-being.
- ☐ Ecosystem services represent a measure of biodiversity and, consequently, the resilience of an ecosystem.
- ☐ Ecosystem services represent the annual accumulation of biomass within an ecosystem measured in kilojoules.

The ecosystem services concept refers to the direct and indirect contributions of ecosystems to human well-being and is commonly structured into four categories: provisioning, regulating, cultural, and supporting services. It does not serve as a direct measure of biodiversity or ecosystem resilience, nor does it describe biophysical productivity metrics such as annual biomass accumulation.

- True
- True
- False
- False

Which of the following are types of feedback mechanisms in the climate system? (More than one statement may be correct.)

- ☐ Positive feedback
- ☐ Neutral feedback
- ☐ Negative feedback

Feedback mechanisms in the climate system describe processes that either amplify (positive feedback) or dampen (negative feedback) changes in the climate system. Neutral feedback is not commonly used as a category in climate science.

- True
- False
- True

Which of the following are direct drivers of biodiversity loss? (More than one statement may be correct.)

- ☐ Economic policy
- ☐ Climate change
- ☐ Pollution
- ☐ Land use change

Direct drivers of biodiversity loss are factors that have an immediate impact on ecosystems and species. These include land use change (e.g. habitat destruction), pollution, and climate change. Economic policy, by contrast, is considered an indirect driver, as it influences biodiversity loss through socioeconomic pathways rather than through direct ecological pressure.

- False
- True
- True
- True

Which of the following statements correctly describe land use change and related concepts? (More than one statement may be correct.)

- ☐ Land degradation refers to the reduction or loss of biological or economic productivity of land due to land use, particularly in dryland areas.
- ☐ Land use describes how humans utilize land cover, for example for agriculture, settlement, or infrastructure.
- ☐ Land use change can negatively affect biodiversity, ecosystem functions, and ecosystem services, thereby restricting human well-being.
- ☐ Land use change refers to changes in land cover only, independent of human activities.
- ☐ Land cover refers to the physical and biological characteristics of the land surface, including vegetation, topography, and human-made structures.
- ☐ Land systems only describe ecological processes and exclude social, economic, and technological factors.

Land cover describes the biophysical characteristics of the land surface, while land use refers to how humans use that land. Land use change involves shifts from one land use to another and is closely linked to human activities. Such changes can threaten biodiversity, disrupt ecosystem functions, increase greenhouse gas emissions, and reduce ecosystem services, thereby affecting human well-being. Land degradation captures the loss of land productivity due to land use, and land systems explicitly integrate social, economic, technological, and ecological dimensions.

- True
- True
- True
- False
- True
- False

Which characteristics distinguish “Nature’s Contributions to People (NCP)” from “ecosystem services”? (More than one statement may be correct.)

- ☐ Integrates Indigenous perspectives
- ☐ Based solely on monetary valuation
- ☐ Includes intrinsic and relational values
- ☐ Focuses only on provisioning functions

The concept of Nature’s Contributions to People (NCP), introduced by IPBES, broadens the ecosystem services framework by explicitly including intrinsic and relational values of nature and by integrating Indigenous and local knowledge systems. It is not limited to monetary valuation and does not focus solely on provisioning functions.

- True
- False
- True
- False

4 Social Sustainability

Social sustainability is about a community's well-being, encompassing elements like inclusion, equity, and social cohesion. It's often argued that this dimension of sustainable development is harder to grasp than environmental or economic sustainability (Foladori 2005). That's because social issues may not be as immediately visible as, say, rainforest destruction or hyperinflation. What does sustainability mean when communities are unequally affected by events such as flooding, or when Indigenous peoples are displaced by infrastructure projects? These impacts reveal deep-rooted inequalities that are largely based on societies' social structures.

This is why the social dimension is so important. A society that is socially sustainable is better prepared to tackle challenges fairly and equitably (Ballet et al. 2020). Furthermore, having social acceptance is crucial for successfully implementing sustainability measures and initiatives (Assefa and Frostell 2007; Wüstenhagen et al. 2007).

In short, social sustainability addresses the following challenge: How can we create a resource-conserving society without increasing poverty and inequality? Achieving this requires balancing fundamental questions of distributive justice with the tension between human needs and wants.

This chapter begins by introducing social sustainability as a normative dimension of sustainable development. We define the concept by focusing on the following key elements: inclusion, social cohesion, resilience, and justice. Finally, we address the cultural aspect. While some scholars see culture as a component of social sustainability, others argue that it should be considered a fourth dimension in its own right – alongside the environmental, social, and economic dimensions (Sabatini 2019).

4.1 Social sustainability, a normative concept

Interpretations of social sustainability are based, on the one hand, on empirical knowledge about how societies function and, on the other, on ethical judgements about what is valuable and should be pursued. In order to define appropriate sustainability goals, we need to know what ethical principles and values a particular social system is based on. In other words, what does a society or community perceive as “good”, desirable, and just?

As a normative concept, social sustainability aims for a fair distribution of benefits and costs within society. It encompasses various dimensions of equality (see Section 4.1.1, *Dimensions of justice*) and justice (see Section 4.6, *Justice*), and it examines how the decisions made within a particular society influence the distribution of resources.

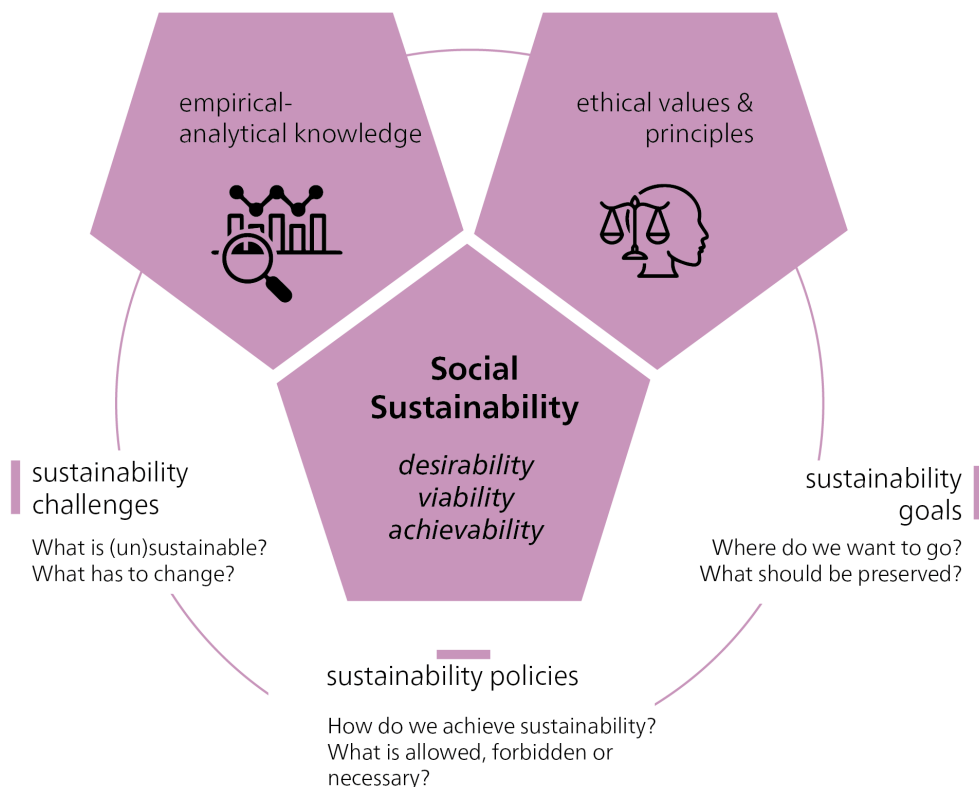


Figure 4.1: Social Sustainability. Source: Own illustration.

These are aspects exemplified by Kate Raworth's doughnut economics (2017a), a framework which views the SDGs and human rights as the basis for socially sustainable and equitable human activities. In doughnut economics, meeting the basic needs of all people is key. These basic needs, which form the framework's social foundation, include access to clean water, sanitation, healthcare, education, food, clean cooking facilities, electricity, housing, and information and support networks, as well as the opportunity to work and earn an income. Additional factors which form the social foundation are gender equality, social equality, peace, and justice, as well as political participation – i.e. the ability to exert social influence.

In doughnut economics, economic policies could contribute to achieving social justice by distributing benefits and costs without exceeding planetary boundaries. And therein lies the challenge: Countries that perform well in terms of social sustainability indicators – e.g. access to food, education, or housing – often exceed ecological boundaries. Conversely, countries that stay within these ecological limits often score poorly in social areas (see Figure 4.2) (O'Neill et al. 2018). Striking a balance between social and environmental requirements is therefore key.

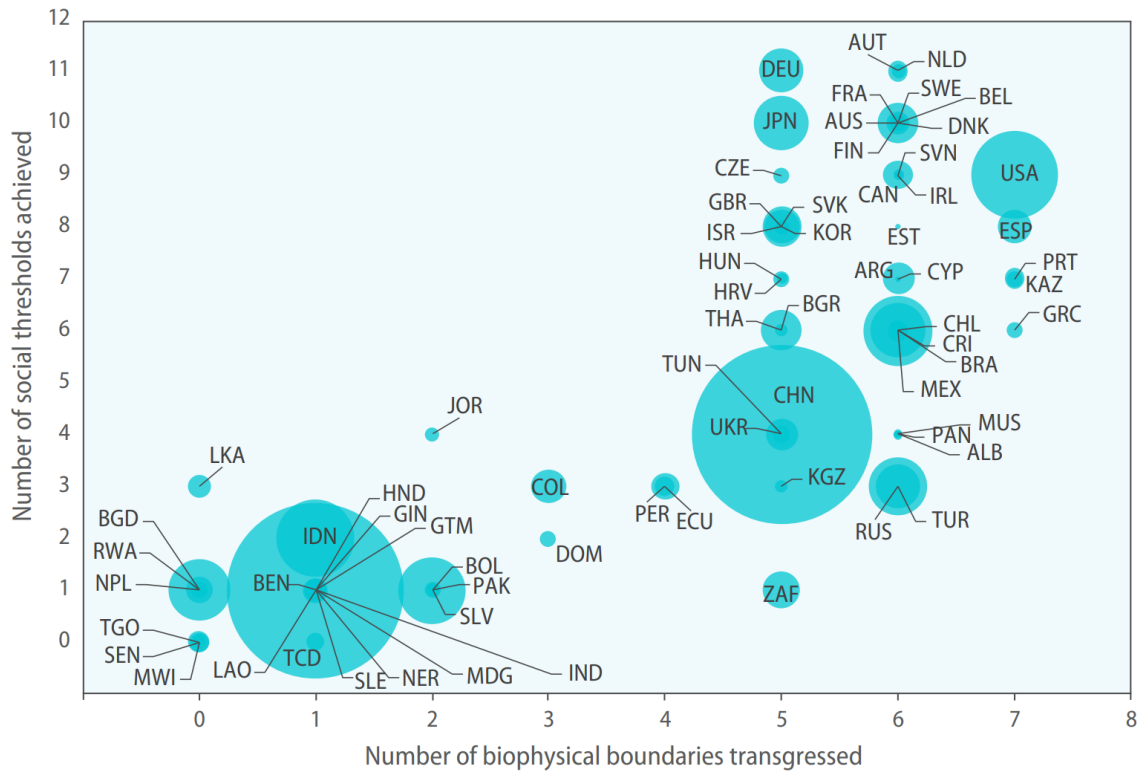


Figure 4.2: Social thresholds achieved versus biophysical boundaries transgressed for different countries, results weighted by each country's population. Source: UN/DESA (2018) based on O'Neill et al. (2018), Figure 2.

To achieve a sustainable future, Raworth (2017a) argues that we must distribute the Earth's resources more fairly. This would allow everyone to meet their basic needs while respecting the planet's limits. It would require changes in lifestyle – especially by the wealthy, who are responsible for the majority of emissions. At present, those who generate more emissions tend to have better access to essential goods such as food, water, energy, and education – the very things that people in lower-income countries often lack. In wealthier countries, people's basic needs are generally met more comprehensively, leading to more prosperity, equality, and justice. At the same time, higher living standards are often linked to increased greenhouse gas emissions and consumption – as seen in Western industrialized countries (e.g. the US, Europe) and China.

Another approach to social sustainability is found in the United Nations Human Development Reports, which use a model of human development rooted in the capability approach (see Section 4.3.1, *(In)equality*). The capability approach asks what a person needs to lead a good and fulfilling life. Because different people need different resources to achieve this, giving everyone the same thing does not foster true equality. Instead, we should provide each person with what they need to live a life they can value. This model of human development promotes respect for human dignity as an inalienable right and advocates providing realistic opportunities for everyone to flourish.

4.1.1 Dimensions of justice

Doughnut economics and the capability approach are two examples of pithy and influential normative frameworks. In this chapter, we cover a range of concepts and approaches that – through their social dimensions – inherently involve normative and ethical considerations.

The many theories and concepts of social sustainability are also linked to different ideas of justice. Some of these are described in more detail in Section 4.6 (*Justice*). Three frequently mentioned dimensions are: distributive justice, procedural justice, and recognition justice (de Bruin et al. 2024; Tribaldos and Kortetmäki 2022; Wijsman and Berbés-Blázquez 2022).

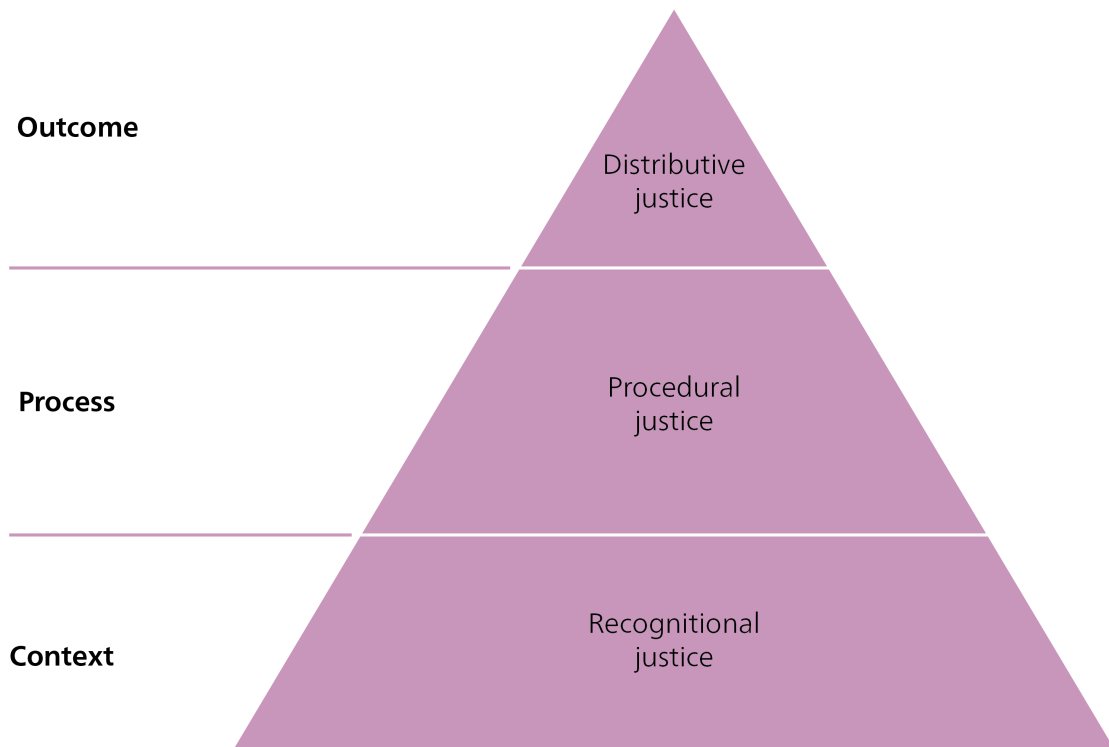


Figure 4.3: The respective focus areas of distributive justice, procedural justice, and recognition justice. Source: Own illustration adapted from See and Wilmsen (2022).

Distributive justice is a results-oriented concept that aims at a fair distribution of goods and burdens, regardless of social differences. This principle applies to issues such as access to clean water, the unequal burden of environmental pollution, or equal pay.

Procedural justice focuses on the inclusivity of decision-making processes. It calls for underrepresented groups to be fairly represented in institutions so that their perspectives can be heard and incorporated into decision-making processes.

Recognition justice emphasizes the importance of the context, and of respect for social and cultural differences. It involves recognizing structural injustices, respecting the dignities and

values of different people and social groups, and understanding that diverse needs and preferences are rooted in different historical experiences, identities, and cultural backgrounds.

Table 4.1 shows how these three dimensions influence each other (Wijsman and Berbés-Blázquez 2022). A more detailed discussion of distributive and procedural justice follows in Section 4.6 (*Justice*), as these are the dimensions that most affect social sustainability debates.

Table 4.1: Links between distributive justice, procedural justice, and recognition justice. Source: Based on Wijsman and Berbés-Blázquez (2022).

Dimension	Distributive justice	Procedural justice	Recognition justice
Distributive justice	—	The distribution of resources can influence how easy or difficult it is for individuals or communities to participate in decision-making processes. This is because participation requires time, energy, and financial resources.	The distribution of resources can influence whose interests are taken into account in decision-making processes – and whose identity is recognized and considered important.
Procedural justice	Participation in decision-making processes can have an impact on their outcomes – and therefore on how advantages and disadvantages are distributed between individuals and groups.	—	Participation in decision-making processes can influence whose interests are heard and which individuals or communities are recognized as important.
Recognition justice	Recognizing individuals or groups as valuable can influence how resources are distributed.	Recognizing individuals or groups as valuable can also determine who is invited to participate in decision-making processes.	—

4.2 Defining social sustainability

Over time, discussions on social sustainability have focused on different issues. These include poverty reduction, development, basic needs, livelihoods, and equity – as well as identity, belonging, community stability, and security (Glasson and Wood 2009). This thematic diversity is reflected in the numerous available definitions of social sustainability. Indeed, the literature is often described as fragmented, vague, or at times chaotic (Mehan and Soflaei 2017). Some researchers have even argued that the concept of social sustainability has changed more than the environmental and economic dimensions over the past 30 years (Foladori 2005).

Social sustainability can therefore best be understood as a dynamic concept that takes different forms, depending on the time and place (Boyer et al. 2016; Dempsey et al. 2011). Social priorities are diverse and context-specific, and they change over time. In addition, there are major cultural and local differences with regard to what is considered socially justifiable and desirable.

Despite this diversity, definitions of social sustainability generally describe a positive state that already exists or is considered achievable. It is often associated with the goal of strengthening

social cohesion and providing basic services that contribute to well-being, such as healthcare, education, mobility, housing, or leisure facilities. Social sustainability is achieved when social processes, systems, structures, and relationships actively support current and future generations in creating healthy and viable communities.

For the Brundtland Commission (1987), the main goal of the social dimension of sustainable development is to fulfil basic human needs. These include not only clean water, food, and security – they also extend to aspects such as well-being, justice, a democratic system of government, and a vibrant civil society. This comprehensive definition can be applied flexibly in different contexts.

In a bid to provide a more concrete framework, Barron et al. (2023) and Cuesta et al. (2024) propose the following structured approach.

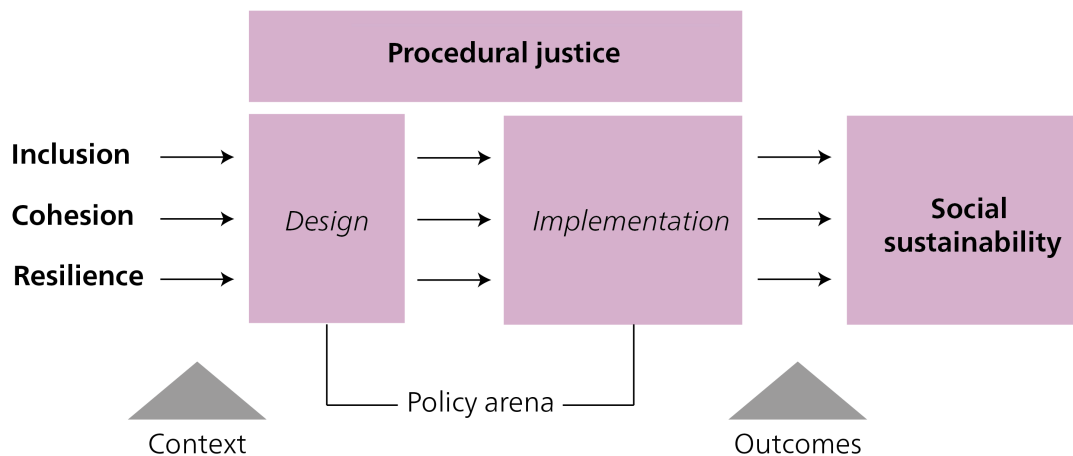


Figure 4.4: A structured approach to social sustainability. Source: Own illustration adapted from Barron et al. (2023) and Cuesta et al. (2024).

The left-hand side of Figure 4.4 depicts three dimensions of a particular context, namely: inclusion, social cohesion, and resilience. Each of these dimensions – defined below – may be found in a community or society to a greater or lesser extent:

1. **Inclusion** refers to the access of all individuals and groups to the economic, political, social, and cultural spheres.
2. **Social cohesion** is a shared sense of belonging and trust that enables communities and groups to work together without conflict.
3. **Resilience** refers to the ability to avoid internal or external shocks – such as environmental changes or political crises – or to cushion their effects.

The middle part emphasizes the importance of **procedural justice** – the fairness of decision-making processes in the **design** and **implementation** of policies and programmes. When these processes are perceived as fair, transparent, and trustworthy, they generate **process legitimacy**, meaning that people accept the authority of decisions and are more likely to support and comply with them. The process of designing and implementing policies and programmes takes place within

a so-called “policy arena”. The term “policy arena” is broadly defined and encompasses political forums at the local, national, regional, or global level, where resources are allocated, goals are set, and decisions are made.

Finally, the right-hand side of the figure depicts the outcomes of the design and implementation process. This process can foster – but also inhibit – inclusion, social cohesion, and resilience. These outcomes, in turn, influence the future level of social sustainability. In other words, social sustainability exists when inclusion, social cohesion, and resilience are gradually strengthened over time – ideally in a legitimate and transparent process.

The following sections explain the core elements of inclusion, social cohesion, resilience, and justice (both procedural and distributive) in more detail.

4.3 Inclusion

Inclusion addresses issues of social inequality. Social inclusion aims at enabling all people to participate in society – specifically, by reducing deep-rooted systemic disadvantages. Many individuals and groups face barriers that limit their socioeconomic participation. While such barriers may include poverty or having a low income, other forms of discrimination and exclusion may be based on gender, age, origin, occupation, ethnicity, religion, nationality, disability, sexual orientation, or gender identity. Such inequalities are reinforced by formal and informal norms, behaviour, laws, and institutions.

This raises normative questions: At what point is inequality considered problematic? And what level of inequality is socially unacceptable? Inclusion should not be understood as a form of charity, i.e. as something that is only “right” for ethical reasons. Reducing inequalities has considerable benefits for all of society. Greater inclusion and equal opportunities lead to better results in income, poverty reduction, and human capital (OECD 2015).

For example, a US study by Hsieh et al. (2013) found that a reduction in discrimination against female and Black workers accounted for 15 to 20% of economic growth per worker between 1960 and 2010. This growth was attributed to an increase in talented people gaining access to better opportunities. Other studies show that gender inequality can have a negative impact on economic growth (Fabrizio et al. 2018; Klasen and Lamanna 2009). There is also a link between inclusion and social stability: countries that actively promote the political participation and inclusion of disadvantaged groups tend to experience fewer social conflicts (United Nations and World Bank 2018).

In short, the costs of social inequality – such as fewer educational opportunities, lower lifetime incomes, or poorer health (Buehren et al. 2019; Wodon and de la Briere 2018) – are not only ethical issues. They also affect the overall economy, as they reduce potential, weaken productivity, and increase the risk of conflict. By contrast, more inclusive societies are more just, more stable, and more economically efficient.

4.3.1 (In)equality

As inequality is complex and diverse, there are different ways of analysing, measuring, and evaluating it. This involves asking questions about *who* is affected and *where* – and in *what* fields inequalities occur (see Figure 4.5) (Sen 1979).

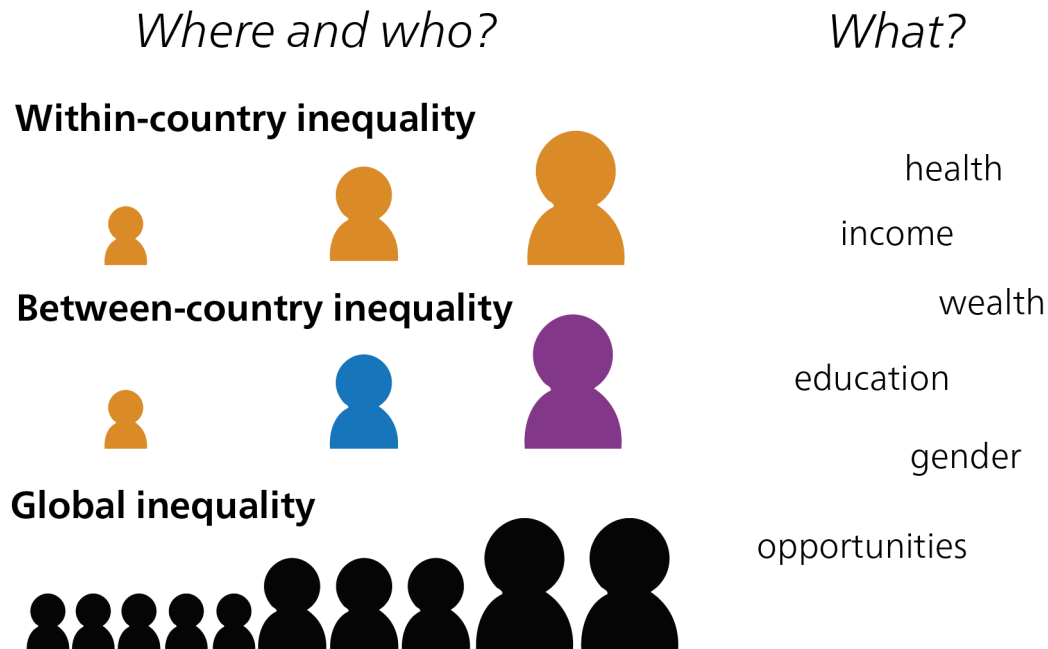


Figure 4.5: Equality of what? Source: Own illustration based on Sen (1979).

Inequality can be approached from various perspectives. Below, we provide more detail on the aforementioned capability approach.

4.3.1.1 Approaching inequality: Guided by Sen's question "Equality of what?" – the capability approach

The **capability approach** provides a normative framework for assessing inequality and human well-being that goes beyond conventional distribution principles. Traditional approaches to distributive justice often aim either at ensuring equal distribution based on performance or at guaranteeing a minimum level of resources for all. However, achieving complete equality between individuals – i.e. *equality of outcome* – is nearly impossible, as many influencing factors are distributed randomly or unfairly. It is therefore considered fairer to focus on *equality of opportunity*. This principle aims to ensure that every person has the chance to lead a meaningful and dignified life with access to at least a minimum level of prosperity. This minimum level should suffice to cover basic needs and guarantee equal civil rights, ensuring that everyone has what they need to meet their needs and participate equally in society.

Developed in the 1980s by the Indian economist, philosopher, and Nobel Prize winner **Amartya Sen** and later expanded by the philosopher **Martha Nussbaum**, the capability approach serves as an alternative to traditional welfare economics, which measures prosperity primarily through economic indicators (Roder 2020). The capability approach shifts attention away from formal entitlements or economic aggregates toward people's **real freedoms to live the life they value**. It emphasizes that genuine development depends not only on the existence of opportunities and resources but also on whether individuals can actually make use of them in practice. This, in turn, is shaped by personal factors such as health or disability as well as by social, political, and environmental conditions. For instance, for people with disabilities, well-being requires more than the formal recognition of rights; it depends on an inclusive society that removes barriers, combats discrimination, ensures access to education, and enables participation in political and social decision-making.

The capability approach defines well-being in terms of **capabilities** and **functionings**. Capabilities are real opportunities for action – i.e. what people could do or be if they were free to choose. For example, being well-nourished, getting married, being educated, or travelling without restrictions. Functionings are the capabilities that are actually converted. Whether people can convert available resources such as income, education, or public services into functionings depends on **conversion factors**. These include personal characteristics, social structures, and environmental conditions. They ultimately determine whether capabilities and functionings can actually be used.

The capability approach, therefore, shifts the focus from **equality** to **equity**. While *equality* aims to ensure that everyone receives the same resources (i.e. input), *equity* focuses on the result: what does a person need, in order to achieve a comparable level of well-being (i.e. output)? (see Figure 4.6).

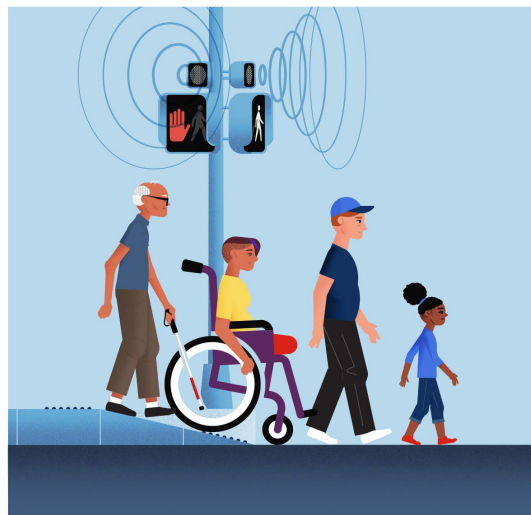
EQUALITY:

Everyone gets the same – regardless if it's needed or right for them.



EQUITY:

Everyone gets what they need – understanding the barriers, circumstances, and conditions.



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Figure 4.6: Equality vs. equity. Source: Reproduced with permission of the Robert Wood Johnson Foundation, Princeton, N.J.

4.3.1.2 Approaching inequality: Where and of whom?

Building on the capability approach, which emphasizes people's real opportunities to lead the lives they value, it is crucial to recognize that inequality can manifest in different ways and be assessed across various dimensions. Inequality can be analysed in different spaces, for example by distinguishing between **absolute** and **relative inequality** (Niño-Zarazúa et al. 2017) or between **vertical** and **horizontal** inequality (Stewart 2005).

Absolute inequality describes a situation in which people are unable to meet their most basic survival needs. This is the case, for example, when a person does not have access to clean drinking water, enough food, or safe housing, and lives below the poverty line. This form of inequality is often defined by fixed income thresholds, such as the World Bank's international poverty line.

Relative inequality measures a person's disadvantage in relation to the average standard of living within a particular society or community. For example, while a family in a rich country may have enough income to cover food and housing, they may lack the resources to participate fully in society – if, say, they can't afford internet access or their children's school trips. Relative inequality becomes evident when living conditions fall significantly below the social average. It often leads to social exclusion and limits access to resources and the opportunities necessary to lead a dignified and self-determined life in a particular social context. The key difference between the two concepts is that **absolute poverty** refers to an objective lack of essential goods, while **relative poverty** describes a person's social and economic status in relation to others in a society (Niño-Zarazúa et al. 2017).

Another concept distinguishes between **vertical** and **horizontal inequality**. Vertical inequality focuses on socioeconomic differences between individuals – such as in income, occupational status, or education. By contrast, horizontal inequality refers to differences between culturally defined groups, based for example on gender, age, marital status, nationality, region, or place of residence (Stewart 2005).

4.3.1.3 How can we analyse inequality? Individual vs. structural approaches

Analyses of inequality often distinguish between **individual inequality** and **structural inequality**. An analysis of individual inequality draws on the principles of welfare economics and the concept of absolute inequality. It focuses on socioeconomic characteristics such as differences in income, educational level, and occupational status. This approach assumes that economic inequalities are primarily the result of personal attributes and individual effort.

By contrast, analyses of structural inequality are more closely aligned with the concepts of equity, the capability approach, and relative inequality. Structural inequality is considered not as a product of an individual's characteristics, but as a result of the social framework – such as a person's position within the mode of production or the social hierarchy; their access to wages, profits, or a pension; or their nationality, all of which are key factors of global inequality (see Milanovic 2019).

This chapter has introduced various perspectives on inequality, including the widely used capability approach, which offers an overarching and normative perspective. The other perspectives may sound similar, but in fact differ in their focus (see Table 4.2).

Table 4.2: Categories and analysis dimensions of inequality.

Category	What is being compared?	Focus	Typical example	Policy relevance
relative vs. absolute	Inequality measured in relation to differences vs. absolute differences	Categorization of inequality: Who has more/less?	The wealthy receive 40\% of the income, while the poor receive 10\% (relative) vs. the income gap required to meet basic needs (absolute)	Distributive justice vs. combating poverty
vertical vs. horizontal	Differences between income levels vs. differences between social groups	Categorization of inequality: How is inequality quantified and represented?	A CEO earns 100x more than a cleaner (vertical) vs. women earn less than men for the same work (horizontal)	Redistribution vs. anti-discrimination
individual vs. structural	Differences due to personal characteristics vs. differences due to social structures	Analysis dimension of inequality: What are the causes of inequality?	Individual: someone becomes successful through talent and effort vs. structural: someone is disadvantaged because of where they come from	Promotion of equal opportunities vs. reform of social structures

4.4 Social cohesion

Social cohesion describes the feeling of togetherness, trust, and willingness to cooperate – within groups, between different groups, and towards state institutions. The concept describes the extent to which people and communities act based on interpersonal and institutional trust, what attitudes they have towards minorities, and how secure they feel. Societies with strong social cohesion can be found both in wealthy countries and in countries affected by conflict (Cuesta et al. 2024).

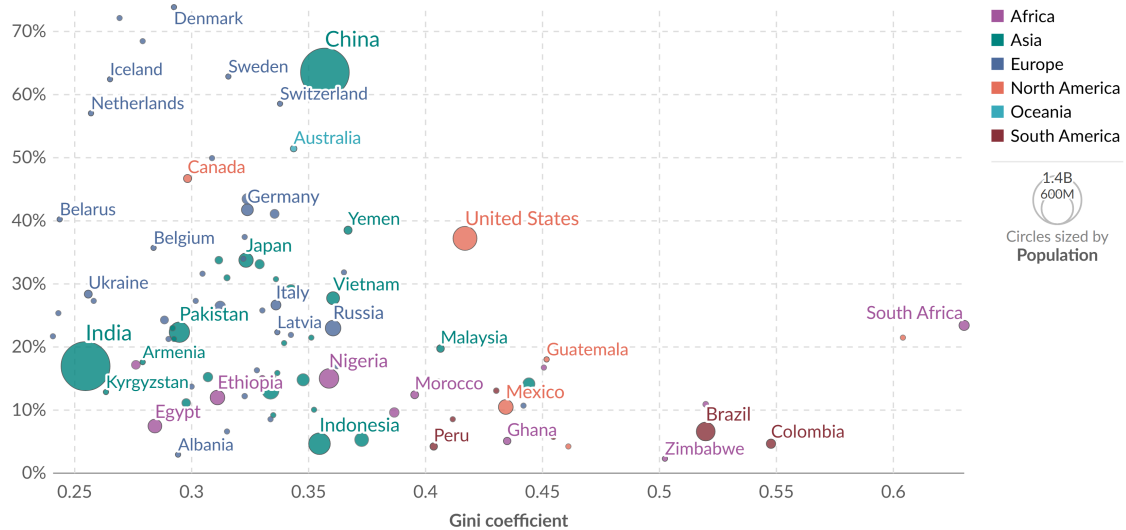
Functioning social cohesion and the well-being of a population require that all social groups can participate economically, politically, socially, and culturally. This is closely linked to inclusion and inequality. However, rather than focusing on differences between individuals or specific groups, the concept of social cohesion asks how strongly the whole of society is connected.

Social cohesion is the foundation for mutual trust and collective action. It is considered a core prerequisite for peace and prosperity (Chatterjee et al. 2023). Studies show, for example, that societies with a high level of trust experience less conflict (Putnam 2000; Rothstein and Uslaner 2005). By contrast, lower social trust is often associated with greater economic inequality (Jordahl 2007) (see Figure 4.7).

Interpersonal trust vs. income inequality

Share of respondents agreeing with the statement "Most people can be trusted". The Gini coefficient measures inequality on a scale from 0 to 1. Higher values indicate higher inequality.

Share agreeing "Most people can be trusted"



Data source: Integrated Values Surveys (2024); World Bank Poverty and Inequality Platform (2025)

OurWorldinData.org/trust | CC BY

Note: For each country, trust data is shown for the latest survey wave in the period 2009-2022. Depending on the country and year, inequality data relates to income measured after taxes and benefits, or to consumption, per capita.

Figure 4.7: Interpersonal trust vs. income inequality. Source: <https://ourworldindata.org/grapher/interpersonal-trust-vs-income-inequality>.

The above scatter diagram illustrates the relationship between interpersonal trust and income inequality. Each dot represents a country. Dot colour categorizes countries by their respective regions; dot size reflects the country's population size. The Y-axis depicts the percentage of a country's population that agrees with the statement "Most people can be trusted". The X-axis displays the Gini coefficient, which is a widely used measure of income inequality: a value of 0 indicates a state of complete income equality, while a value of 1 (or 100%) represents maximum income inequality – i.e. a single person holding all the income, while everyone else receives none.

The graph shows that the level of social trust tends to be lower in countries with a higher Gini index. One explanation for the negative correlation between inequality and trust is that people tend to place greater trust in those who are similar to them, and that greater inequality leads to tensions and distributional conflicts (Alesina and La Ferrara 2002). At the same time, some countries display low levels of trust despite having a low Gini score (e.g. Slovakia, Kyrgyzstan, and Albania). This indicates that although a general trend is recognizable, there is no causal relationship, and other factors are at play.

4.4.0.1 Key elements of social cohesion

Figure 4.7 is just one example of social cohesion. The concept is difficult to define precisely, because the scientific literature varies in regard to its exact meaning and how to measure it. Nevertheless, we can identify three key elements (Schiefer and van der Noll 2017):

1. **Social relationships:** networks, trust, acceptance of diversity, and participation
2. **Identification with society:** emotional attachment to the social community
3. **Orientation towards the common good:** a sense of responsibility, solidarity, and compliance with social rules.

Social cohesion is reflected in the attitudes and behaviour of all social groups and encompasses both normative and relational dimensions. It should be understood as a continuum along which societies can display varying degrees of cohesion.

To illustrate how these elements are operationalized, it is worth looking at how social cohesion is measured in Switzerland, where it has a special significance: promoting “internal cohesion” is enshrined in the Federal Constitution as an objective of the Swiss Confederation (Art. 2). A crucial role in connecting people in multilingual Switzerland is played by culture, which acts as “societal cement”. While social cohesion isn’t an explicit goal in the 2030 Agenda, it’s implicitly addressed in several of its targets. In Switzerland, social cohesion is defined and measured by the Federal Statistical Office, using a range of indicators (see Table 4.3) (FSO 2025).

The first element, **social relationships**, is reflected in political participation such as voting. This element also involves cultural participation, which is measured as the proportion of the population that takes part in cultural activities. Social relationships are negatively affected by factors such as discrimination, which undermines the level of trust people place in fellow human beings and institutions, as well as the sense of belonging to a social network.

The second element, **identification with society**, may be reflected in how people perceive regional equality. In Switzerland, for example, the Federal Statistical Office measures financial differences between the cantons. Multilingualism is also relevant: The ability to communicate with different language communities can strengthen one’s sense of belonging and promote intercultural understanding.

The third element, **orientation towards the common good**, encompasses aspects of social participation, particularly in the labour market. This includes, for example, the employment rate of people with disabilities or the unemployment rate according to migration status – indicators of structural inclusion and solidarity. Social inequalities are also key here, measured for example by the rate or risk of poverty among the labour force. Another important indicator is civic engagement, particularly through voluntary work, which is a direct measure of social responsibility and public spirit.

Education is a cross-cutting issue, as it affects all three elements: Education fosters social relationships, strengthens identification with society, and supports an orientation towards the common good by teaching about social norms and values. Relevant indicators include the upper secondary completion rate and the proportion of young people who are not in employment or education.

Table 4.3: Measuring social cohesion in Switzerland.

Key element [SchiefferEssentialsSocialCohesion2017]	Indicators [fsoSocialCohesion2025]
Social relationships	Political participation (e.g. voting) • Cultural participation • Experience of discrimination
Identification with society	Regional differences (e.g. financial capacity of cantons) • Multilingualism/cultural integration
Orientation towards the common good	Participation in the labour market (e.g. employment rate of people with disabilities) • Social inequality (e.g. poverty level) • Voluntary work
Cross-cutting issue	Education

4.5 Resilience

Resilience is the ability of individuals, households, communities, or entire societies to prepare for, cope with, and recover from shocks (Folke 2016). Shocks can occur at an individual (e.g. job loss, illness) or societal level (e.g. natural disasters, conflicts, food shortages). They can occur suddenly (e.g. natural disasters), intensify gradually (e.g. soil degradation), or be a constant presence (e.g. poverty, child labour, gender-based violence).

Societies with a high level of resilience can regain their quality of life and economic stability more quickly after a shock. Resilience is especially important for poor and marginalized groups, as they face more shocks, suffer greater relative losses, and receive less support (Bangalore et al. 2017). Inequalities also mean that different groups are unevenly or disproportionately affected by shocks (Hallegatte et al. 2020). Depending on social norms, groups such as women, young people, or the elderly are often more vulnerable or less adaptable (Ajibade et al. 2013; Barrett et al. 2021).

We can identify three distinct strategies for addressing risk and building resilience:

1. **Risk reduction and mitigation:** Measures that reduce the likelihood of shocks or their negative consequences (Obrist et al. 2010). Examples include immunization, income diversification, local protection infrastructure, or state social systems.
2. **Coping strategies:** Responses to shock (Imperiale and Vanclay 2021; Severi et al. 2012). Coping strategies include insurance models, recourse to savings or loans, and support from social networks. Governments can also help, for example through cash transfers or social programmes. However, without sufficient resources, people affected by shocks may resort to harmful, short-term measures, such as restricting essential consumption, overexploiting resources, or using child labour.
3. **Transformative strategies:** Far-reaching reforms or new institutions that increase the long-term resilience of societies (Mozumder et al. 2018; Pfefferbaum et al. 2017), typically combining risk reduction and coping strategies. For example, establishing a government agency for flood preparedness that improves early warning systems, fosters risk reduction research, and develops flood response plans. Or introducing measures to reduce the

risk of gender-based violence by investing in legal, institutional, and awareness-raising programmes that change incentives and social norms regarding violence.

4.5.1 Basic needs as a prerequisite for resilience

In the aforementioned doughnut economics framework, resilience is only possible if all people can meet their basic needs. Meeting these social foundations ensures that individuals and societies are better able to cope with shocks. However, in countries of the Global South, these basic needs are often unmet. Conversely, while wealthier countries are able to meet these needs, they usually exceed planetary boundaries in doing so. See Figure 4.8 comparing Switzerland and Laos.



Figure 4.8: Doughnut visualization for Switzerland (left) and Lao PDR (right), comparing each country's performance relative to its social foundation and ecological ceiling. The green doughnut represents the safe and just space where both ecological and social boundaries are respected. Red areas indicate the magnitude of ecological overshoot or shortfalls in meeting social needs, showing where each country exceeds ecological limits or falls below minimum social thresholds. Source: Kate Raworth and Christian Guthrie. CC-BY-SA 4.0, <https://doughnut-economy-fxs7576.netlify.app/>.

These minimum social standards – the “social floor” – include access to food, health, education, income, and work. They also include peace, justice, political participation, social and gender equality, housing, social networks, energy, and water. In this textbook, we use the example of **food** to show how access to one of these standards affects resilience. Difficulties with a single standard rarely occur in isolation and often affect other areas as well. Ensuring these social foundations is therefore crucial to sustainable development.

4.5.1.1 Global food security

Food security means having enough food and reliable access to it, especially staple foods. A household is considered food secure when its members are not at risk of hunger or malnutrition. Malnutrition has severe health impacts, such as emaciation and stunted growth. It also increases healthcare costs, reduces productivity, and inhibits economic growth, thereby reinforcing a cycle of poverty and disease.

Food security is closely linked to a food system's resilience – its ability to tolerate and adapt to change. Our global food system is particularly vulnerable due to its complexity and the many interdependencies among its elements. It is exposed to both external threats – such as the consequences of climate change – and internal risks, such as policy failures, unsustainable consumption patterns, or market failures.

Figure 4.9 shows that almost 2.4 billion people worldwide were affected by food insecurity in 2022. Just under half of these (1.1 billion) lived in Asia, 37 per cent (868 million) in Africa, 10.5 per cent (248 million) in Latin America and the Caribbean, and around 4 per cent (90 million) in North America and Europe. The figure also shows the proportions of severe and moderate food insecurity: severe food insecurity means that people have difficulty meeting their short-term basic food needs. Moderate food insecurity, on the other hand, refers to an inadequate supply of essential nutrients, which can result in health problems.



Figure 4.9: The concentration and distribution of food insecurity by severity differ greatly across the regions of the world.
Source: FAO et al. (2023). FAOSTAT: Suite of Food Security Indicators, www.fao.org/faostat/en/#data/FS.

This demonstrates the need for fair and resilient food systems. As Figure 4.9 shows, a food system encompasses all activities, actors, and infrastructures involved in the production, processing, transport, distribution, trade, consumption, and disposal of food. Food systems are deeply embedded in society and the environment. Changes within these systems therefore affect numerous areas: the environment (water, air, soil, biodiversity), social and cultural aspects (e.g. impact on local communities), the economy (value chains, trade, financing), and health (Tribaldos and Kortetmäki 2022).

Discussions on combating food insecurity often focus on increasing yields and reducing food waste, but these approaches are not proving to be the most effective. Current food production and consumption habits increasingly rely on highly processed meat and dairy products, as well as intensive production methods. Global consumption leads to outsourced environmental impacts like soil degradation, biodiversity loss, water overuse, and the disappearance of local plant varieties – impacts that primarily affect the producers, not the consumers.

Focusing solely on increasing production also overlooks the fact that the world already produces enough food. What's needed instead is a far-reaching transformation in how these resources are used and distributed. For example, Shepon et al. (2018) found that plant-based substitutes that are nutritionally comparable (e.g. in terms of protein content) are significantly more efficient than animal products: On one hectare of arable land, up to 20 times more food can be produced than with beef and twice as much as with eggs, which represent the most and least resource-intensive animal foods, respectively.

However, the current food system feeds large quantities of edible crops to farm animals. According to Berners-Lee et al. (2018), around 41% of human-edible crops (measured by kilocalories per person per day) are fed to farm animals. This process is highly inefficient: the animals only return around 12% of that energy in the form of meat, fish, and dairy products. In other words, a significant proportion of potential food for human consumption is lost in the process (see Figure 4.10).

a) Global food energy flow

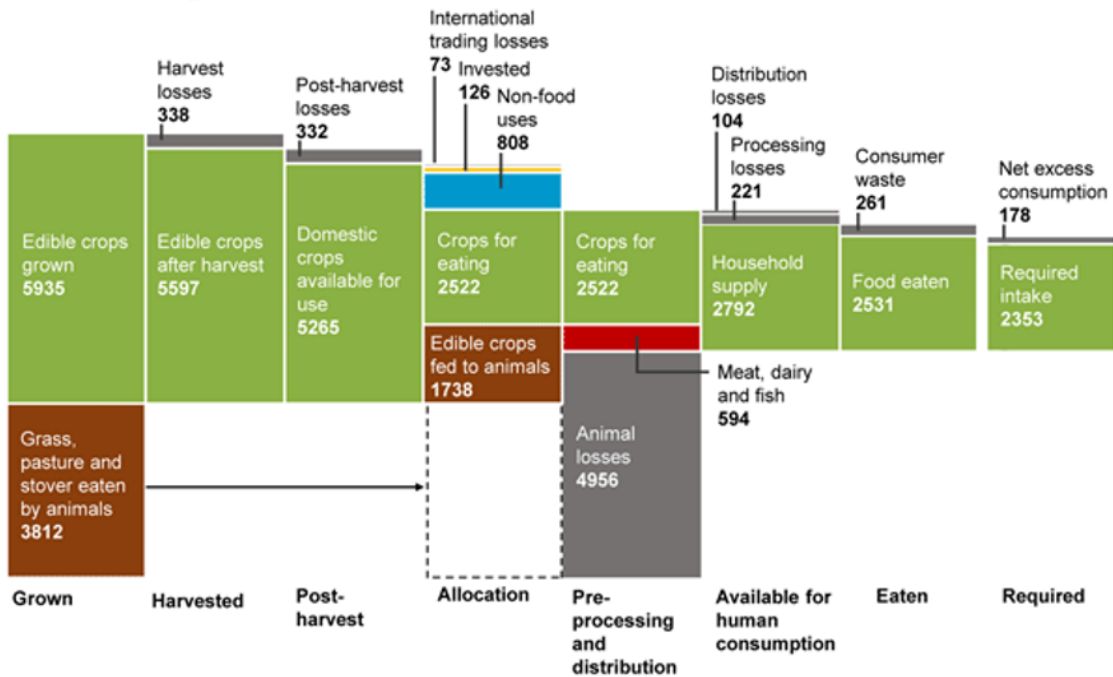


Figure 4.10: The flows of global food energy (kcal/person/day) show the path from the food produced to the food actually consumed in 2013. For crops fed to animals, the units are based on the world population, not the animal population. The left-hand bar distinguishes between crops that are directly edible for humans and fodder crops such as grass, pasture, or crop residues that can only be used by animals. Animal losses (grey bar) include all energy losses that occur in animal husbandry, for example through respiration, growth, movement, and reproduction, as well as through animal components that are not used as food. The outermost right-hand bar divides the nutrients consumed into the proportion required for a healthy human life, i.e. showing a small net increase in food consumption in 2014. Source: Berners-Lee et al. (2018), licensed under CC BY.

4.6 Justice

4.6.1 Procedural Justice

Inclusion, social cohesion, and resilience are considered the three key dimensions of social sustainability. However, whether these factors actually enable social sustainability depends on a fourth element: **procedural justice**. Procedural justice is decisive in determining whether, and to what extent, social change succeeds across the three dimensions.

Procedural justice focuses on the process. It describes *how* policies are developed, programmes designed, and measures implemented. A high degree of procedural justice is achieved when decision-making processes are perceived as fair, trustworthy, and aligned with social norms (Hinsch 2016; Wijsman and Berbés-Blázquez 2022), even when addressing conflicts and tensions. This is not an either/or situation, but rather a continuum: processes can be perceived as more legitimate or less legitimate, and different groups often perceive this differently.

Ensuring legitimacy requires the public perception that decisions are made by trustworthy and recognized actors who act in accordance with shared values and agreed rules. Legitimacy is further strengthened through transparency, participation, and tangible benefits for those affected. This is particularly relevant when decisions entail costs or burdens for certain groups, such as land redistribution, carbon taxes, or changes in working conditions. How procedural justice is achieved largely depends on the respective societal context, culture, and structures. In general, however, we can identify five key drivers:

1. **Ensuring credibility of decision-makers:** Legitimacy increases when authority stems from recognized sources such as elections, expertise, or institutional confirmation.
2. **Adhering to agreed rules:** Processes are considered legitimate if they are based on accepted standards, procedures, or traditions.
3. **Conforming with societal values:** The process takes into account moral, religious, or philosophical beliefs.
4. **Demonstrating perceived benefits:** Legitimacy increases when the affected parties believe they will benefit from the decisions.
5. **Promoting participation and transparency:** Open dialogue and co-creation foster acceptance, especially in the event of conflicts.

These drivers often interact and reinforce each other, with the relative importance of each driver depending on the social context. As shown in Figure 4.4, procedural justice encompasses both the design and the implementation of policies, programmes, and initiatives. These processes take place within a **policy arena**, the term used to describe all constellations in which policies and programmes are formulated and implemented, such as parliaments, ministries, companies, associations, or local initiatives. However, the policy arena is becoming increasingly fragmented and polarized, amid challenges like social media and disinformation. This weakens the credibility of information and hinders the ability to build broad consensus and legitimacy in decision-making processes.

In Switzerland, this is exemplified by political representation, which fails to adequately reflect the composition of the population: An analysis of the 2023 elections shows that women hold only 37.8% of seats in the National Council. Non-binary people are not yet included in statistics on political representation. In addition, the Swiss parliament is considered a “club of the highly educated”, as the proportion of councillors with a university degree is twice as high as in the general population. Only around one in six (or 16%) of the candidates had a “migration background” – compared to around 40% of the population. A quarter of the adults living in Switzerland do not have the right to vote.

Political scientist Joachim Blatter describes this exclusion from political participation as a “**democratic deficit**”, noting that Switzerland excludes more people from the political system than most other European countries (Blatter et al. 2017). In some municipalities, where only a fraction of the population is entitled to vote, this can mean that ultimately that only 10–20% of residents have a say in political decisions (Debelle 2020).

4.6.2 Distributive justice

As described in Section 4.1.1, the various dimensions of justice influence one another. While **procedural justice** focuses on the process, **distributive justice** concentrates on the outcome of the process. Distributive justice could involve ensuring the fair distribution of environmental resources and environmental impacts. Measures that aim at distributive justice include progressive taxation, social safety nets (e.g. old-age and unemployment insurance), and enforcement of equal pay regardless of gender, ethnicity, or other discriminatory factors.

Historically, discussions about distributive justice were primarily limited to political communities, such as national governments, and focused solely on the current generation. However, sustainability issues require a broader perspective. Distributive justice must be considered globally and intergenerationally, as well as in relation to our responsibility towards the environment and other affected species. Climate targets are a case in point: how should emissions reduction targets be distributed between countries, between generations, or between different social groups? Who should bear the costs of adapting to climate change?

Thus, distributive justice encompasses a wide range of topics. Take the issue of income, for example. In Switzerland, women earn on average CHF 1,364 less per month than men, despite the principle of “equal pay for work of equal value” being enshrined in the Federal Constitution since 1981. Around half of this difference can be explained by objective factors such as education or sector, but the other half – which is as of yet unexplained – could indicate wage discrimination (EBG 2023). The principle is widely accepted and rarely disputed. Consequently, opponents of measures to implement equal pay tend not to take issue with the principle itself, attempting instead to justify the gap by citing allegedly objective factors.

However, social consensus is far harder to achieve in other questions of distributive justice. Various theories (see Note 1) and concepts (see Note 2) seek to answer the question of “What is fair?”. They offer different perspectives on what individuals or groups perceive as fair, and provide guidance on how social advantages and disadvantages should be distributed. A controversial example is that of **unconditional basic income**. From a distributive justice perspective, this concept could be considered fair, as it distributes resources equally and equalizes social opportunities. However, some argue that such a model could weaken incentives to work and thus impair productivity and innovation. Depending on the theory applied (see Note 1), very different conclusions can be drawn on such topics – as shown in the reflections after Note 2.

Note 1: Theories of justice

Theories of justice describe fundamental, overarching ideas about how social structures and processes should be organized. They are therefore considered one level above concepts of justice (Note 2), representing philosophically developed models that define, justify, and explain the essence of justice.

Libertarian theories champion individual freedom and the right to property. Adherents believe the state's role should be minimal, often described as a "night watchman state".

- In *Anarchy, State, and Utopia* (1974), **Robert Nozick** develops the idea of the minimal state, one that assumes only basic protective functions such as security and contract enforcement. Nozick rejects any further redistribution (e.g. via taxes to fund welfare states) as an "encroachment on individual freedom and property rights".
- In *The Illusion of Social Justice* (1981), **Friedrich August von Hayek** argues that "social justice" is an empty term, and that no objective distributive justice is possible in complex market societies. Instead, the market should determine the allocation of resources through free pricing, as state intervention would inevitably lead to a loss of freedom.

Liberal theories also analyse societies based on individual rights and property, but they do not recognize a "natural" right to property. Unlike libertarianism, they do not necessarily view the free market as the optimal means of fair distribution and assign an active role to the state in redistribution.

- **Utilitarianism:** What is considered just is that which maximizes happiness (utility) and minimizes suffering, thereby creating the greatest possible long-term benefit for all parties involved.
- **Contractarianism:** In *A Theory of Justice* (1971), John Rawls uses a social contract thought experiment to determine fair principles for society. His two core principles are:
 1. Equal right to basic freedoms: Each person is entitled to the most comprehensive system of equal fundamental freedoms that is compatible with a similar system for everyone else.
 2. Just social and economic inequalities: Inequalities are only considered just if they meet two conditions: they benefit the least advantaged in society ("the difference principle"), and offices and positions are open to all under conditions of fair equality of opportunity.
- **Egalitarianism:** The aim is equal treatment for all, as everyone is equal in terms of dignity and moral status. One prominent approach is the capability approach. **Ronald Dworkin** offered another key approach, formulating the theory of resource equality in *What is Equality?* (1981). Dworkin posited that a society is just if resources are distributed in such a way that no redistribution would allow for a more equal distribution. The goal is not to achieve equal life satisfaction or equal prosperity, but to ensure fair starting conditions that enable individuals to pursue their own life plans.

Collective theories view societies as associations of social groups or classes that differ in their relationships to means of production and resources. They reject universalistic, abstract principles, arguing instead that each community is the product of specific historical and cultural conditions. Therefore, there are no universal norms of justice; instead, context-dependent rules emerge that are derived from social practices and traditions. Justice is understood here in terms of the common good: What is socially just is what strengthens a community's cohesion and continued existence.

- According to **Michael Walzer**, the given context leads to different "spheres of justice". What is considered just therefore varies according to time and place. In addition,

When reflecting on the introduction of an unconditional basic income, proponents of the various theories would arrive at very different conclusions – as shown in Table 4.4.

Table 4.4: Evaluation of the introduction of an unconditional basic income, based on various theories of justice.

Theories of justice	Is the idea of a basic income a good idea	Reasoning
Libertarian theories (Nozick, Hayek)	No	Social justice cannot be controlled centrally. The state should only assume minimal protective functions, and resources should be allocated via the market. Redistribution through taxes to finance a basic income is an encroachment on individual freedom and property rights.
Liberal theories (Utilitarianism, Contractarianism, Egalitarianism)	Yes/Maybe	From a utilitarian perspective, a basic income could increase overall utility (or benefit). Contractarianism (Rawls) also supports it, arguing that it strengthens equal basic freedoms. From an egalitarian perspective, a basic income can promote opportunities, but it is viewed critically because it emphasizes equality (i.e. that everyone receives the same) rather than justice in the sense of equity (i.e. that each person receives what they need). Additional measures would be necessary to ensure equity.
Collective theories of justice (Walzer, Sandel)	Depends on context	How this question is assessed depends on specific historical, cultural, and social conditions. A basic income could be considered socially just if it strengthens social cohesion and the common good.

While **theories of justice** are general frameworks or models that define what justice is and how it can be achieved in a society, **concepts of justice** focus more on specific aspects. As the overview in Note 2 shows, distributive justice issues can be viewed from a variety of perspectives.

Performance-based justice/meritocracy demands that remuneration be proportionate to an individual's contribution. Since people perform to different levels, unequal pay is considered justified. The problem is that this principle fails to recognize key services – such as care work in the family – that aren't valued on the market. The difficulty lies in determining in each individual case how high the contribution is and how this can be determined objectively.

Market justice is a special form of performance-based justice that leaves “objectification” to the market: It is based on the principle of supply and demand, meaning that income, goods, and opportunities are distributed according to market mechanisms. According to this framework, the price determines what is considered fair. Within neoclassical economic theory, market justice is measured by the marginal productivity of labour, i.e. how work contributes to maximizing profit.

Equality of opportunity aims to create equal starting conditions – i.e. without legal and cultural discrimination or barriers to market access – for example through anti-discrimination laws that prohibit discrimination based on gender, ethnicity, or religion.

Equality of needs ensures that basic needs are met – such as food, housing, or protection from poverty. It is aimed particularly at people whose needs cannot be met through market mechanisms. While commensurability with performance focuses on individual contributions, equality of needs is about the collective provision of basic needs.

Equality of participation expands on the concept of equality of needs and opportunities. It seeks to create equal opportunities to participate in and help shape social life – regardless of income. This includes access to education, health, work, culture, and political participation. Participation is seen as a prerequisite for achieving other basic needs such as health and autonomy.

Gender equality addresses the unequal starting and participation conditions that exist between genders. For example, to promote equality of opportunity, quotas are a way of redressing structural disadvantages for women. And to foster participatory justice, policies that create affordable childcare or care services can help alleviate the unequal burden of care work that disproportionately affects women.

Environmental justice emphasizes the fair distribution of environmental burdens and resources, as well as equal participation in environmental policy decisions. The term originated in the US as a response to the disproportionate placement of landfills and motorways near poor African-American neighbourhoods. The US Environmental Protection Agency (2013) defines environmental justice as the fair treatment and effective participation of all people – regardless of race, colour, gender, or income – in the design, implementation, and enforcement of environmental laws and policies.

Inter- and intragenerational equity are defined in the Brundtland Report (wcedOurCommonFuture1987a?). Intergenerational equity refers to the balance between present and future generations, while intragenerational equity refers to the balance between the Global North and South. Both concepts emphasize that opportunities for participation must be balanced across time and space.

Climate justice is a specific form of environmental justice. It addresses the unequal consequences of climate change for marginalized and vulnerable groups, and seeks to promote the fair distribution of burdens and measures aimed at reducing emissions. The Paris Climate Agreement stipulates that both the Global North and South must share responsibility for reducing CO₂ emissions. However, the North bears a greater obligation due to historically higher emissions – and must therefore reduce its emissions faster than countries in Africa, Latin America, or Asia.

4.7 Cultural sustainability

Sustainability models typically describe three dimensions of sustainable development: the environmental, the economic, and the social. Social sustainability – which was the focus of this chapter – often includes cultural aspects. However, there is growing debate as to whether culture should be understood as a fourth dimension of sustainability (Sabatini 2019).

Although the term has not yet become firmly established in the political discourse or yet been clearly defined, it is gaining traction in academic debates and in policy programmes (Zheng et al. 2021). **Cultural sustainability** refers to both tangible and intangible cultural assets. However, it remains a challenge to implement. At a global level, UNESCO plays a key role in the protection of cultural heritage, working to preserve cultural diversity through various conventions and programmes. Today, there are numerous UNESCO World Heritage Sites that preserve culturally significant places. UNESCO also protects intangible cultural assets, for example through programmes to preserve endangered languages or traditional practices.

The cultural dimension of sustainability emphasizes the role of values, traditions, beliefs, and cultural practices in shaping sustainable development. Culture influences how societies respond to environmental, economic, and social challenges – and it provides the framework within which sustainable solutions can be understood, accepted, and implemented. Sustainable development must therefore consider not just technological and economic factors, but also the cultural context in which measures are implemented. Proponents of an independent view of cultural sustainability argue that culture is more than just a part of the social sphere: it shapes how people interpret the world, form identities, and understand their relationship to society and nature. Culture shapes values, traditions, language, and knowledge systems that extend beyond social structures. In addition, the preservation of cultural diversity is viewed as a goal in its own right, because culture significantly influences all other dimensions of sustainability.

Links to the other three dimensions of sustainability:

- **Culture and social sustainability:** Culture is a key means of shaping values, behaviours, and social assumptions. It influences the openness, inclusivity, and cohesion of societies, as well as respect for human rights, health, and quality of life. Culture provides orientation and conveys shared values such as trust, solidarity, the rule of law, and democracy – and thus creates the basis for individual and social development.
- **Culture and economic sustainability:** Culture and the economy are closely connected. As UNESCO emphasizes, cultural heritage, creative industries, sustainable cultural tourism, and cultural infrastructure not only contribute to economic development and poverty reduction – they also create social and identity-forming values, especially in countries with a rich cultural heritage and great cultural diversity.
- **Culture and environmental sustainability:** Cultural factors shape lifestyles, consumption patterns, and our relationship with the environment. Local and indigenous knowledge systems and traditional environmental practices offer valuable approaches to dealing with environmental challenges. They help to prevent biodiversity loss, reduce land degradation, and mitigate climate change.

4.8 Conclusion

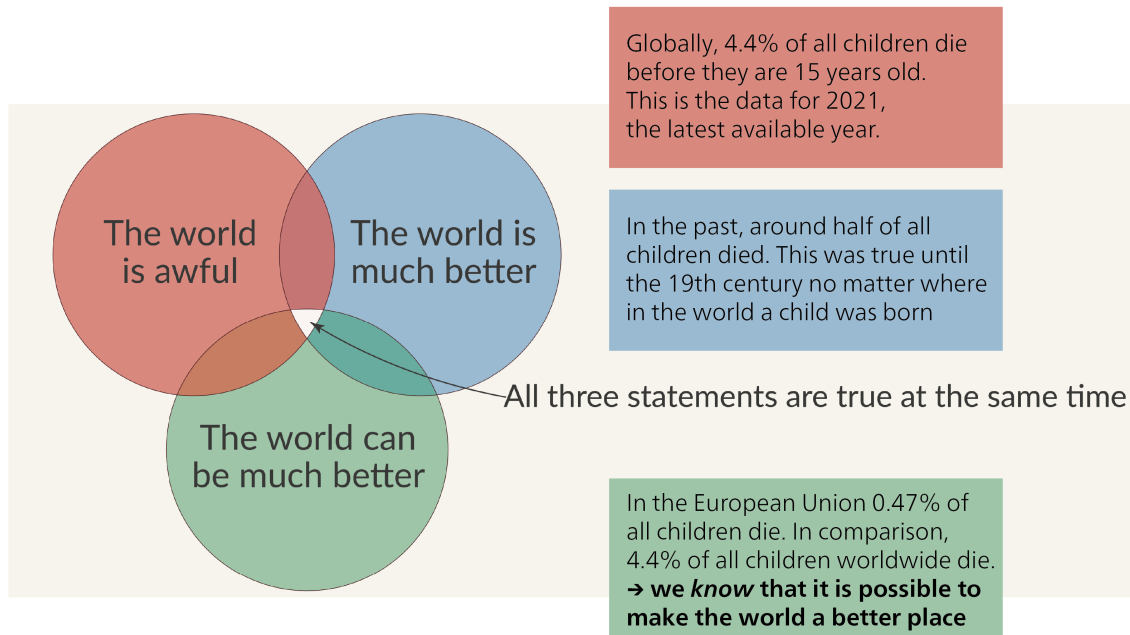


Figure 4.11: The world is awful. The world is much better. The world can be much better. All three statements are true at the same time. Source: Own illustration adapted based on Roser (2022), <https://ourworldindata.org/much-better-awful-can-be-better>.

As Figure 4.11 shows, three seemingly contradictory statements can be true at the same time: In many respects, the world is awful. Nonetheless, it's become much better. And finally – it can be much better still. This perspective also provides a good basis for understanding social sustainability strategies.

1. **“The world is awful”: Social ills persist.**

Millions of people around the world are affected by poverty, hunger, discrimination, and a lack of participation. Inequalities between and within societies jeopardize cohesion, resilience, and justice. Strategies for social sustainability must clearly identify and address these problems.

2. **“The world is much better”: Progress has been made.**

Historical comparisons show that key social indicators have improved: lower infant mortality, higher literacy, growing social rights. These developments are proof that political action, social movements, and institutional reforms are working. Strategies for social sustainability can build on these successes.

3. **“The world can be much better”: The potential is huge.**

Inclusion, social cohesion, resilience, and procedural justice are the key levers for overcoming existing grievances. Strategies range from combating discrimination, expanding social

security systems, and implementing participatory governance – to promoting education, health, and cultural diversity.

This perspective shows that social sustainability strategies must clearly identify grievances, build on progress made, and create new opportunities for greater justice.

It's true that the challenges are immense. But we are not doomed. The progress we've already made demonstrates that change is possible. The key now is to act rapidly and on a large scale. As Hannah Ritchie (2024) notes, we often waste energy on internal conflicts among sustainability proponents, thereby benefiting opponents of the cause. We need systemic change and a meliorist ethic – a belief that the world can be improved step by step through conscious action. Instead of getting bogged down in internal disagreements, we need joint effort in the same direction – even if our paths and priorities vary.

In the end, however, it's important to remember that strategies for social sustainability always require discussion. Who defines the boundaries of needs and desires, whose future vision of sustainable development should be pursued, what sustainability paths are favoured, and who is making these decisions? Also – who benefits from the chosen course, who bears the costs, and what distribution patterns result from this? The difference between needs and desires in particular shows how complex such discussions are: Needs are finite, universal, and indispensable for human well-being, like housing or food. Wants, on the other hand, are potentially infinite, subjective, and dispensable, like the choice of a particular brand of shoe. As Swiss politician Jaqueline Badran said in an interview: “You can't not have a place to live. It's different from trainers, where you can say: Adidas is too expensive for me, so I'll buy another brand. You can't do that with housing.” (Albrecht 2023). This example demonstrates that the debate about social sustainability is not just about abstract principles; it's about the concrete, existential questions of life.

Such questions cannot be answered technocratically. Instead, they require open, fair, and inclusive negotiation processes. Strategies for social sustainability must therefore not only develop concrete solutions to injustices – they must also create spaces in which these discussions can be held.

4.9 Sustainability Basics: The Quiz – Chapter 4

Which of the following statements correctly describe the three dimensions of justice commonly discussed in social sustainability? (More than one statement may be correct.)

- ☐ Recognition justice centres on respecting social and cultural differences and acknowledging structural injustices.
- ☐ Distributive justice focuses on the fair distribution of goods and burdens, such as access to clean water or exposure to environmental pollution.
- ☐ Procedural justice emphasizes the inclusivity and fairness of decision-making processes and representation.
- ☐ Distributive justice primarily concerns the recognition of cultural identities and historical experiences.
- ☐ Procedural justice is mainly concerned with the outcomes of policies rather than how decisions are made.

In social sustainability debates, distributive justice addresses the fair allocation of benefits and burdens. Procedural justice focuses on who is included in decision-making processes and how decisions are made. Recognition justice highlights the importance of respecting diverse identities, cultural contexts, and historical experiences, and of addressing structural injustices. The remaining statements incorrectly attribute the focus of one justice dimension to another.

- True
- True
- True
- False
- False

Which of the following statements correctly describe key dimensions of social sustainability? (More than one statement may be correct.)

- ☐ Inclusion refers to the access of all individuals and groups to economic, political, social, and cultural spheres.
- ☐ Resilience refers to the ability of a community or society to avoid, absorb, or cope with internal or external shocks.
- ☐ Social cohesion describes a shared sense of belonging and trust that enables cooperation within communities.
- ☐ Inclusion focuses primarily on economic growth rather than participation in social and political life.
- ☐ Social cohesion is defined mainly by the absence of diversity within a society.

Inclusion concerns equal access and participation across economic, political, social, and cultural domains. Social cohesion emphasizes trust, belonging, and the ability of groups to work together constructively. Resilience describes the capacity of societies or communities to withstand or adapt to shocks such as environmental changes or political crises. The remaining statements misrepresent these concepts by narrowing their scope or attributing incorrect characteristics.

- True
- True
- True
- False
- False

According to the capabilities approach, which of the following statements is correct?

- ☐ Development must be measured with GDP.
- ☐ Equal legal rights are sufficient for a society to be fair.
- ☐ Resources must be divided so that everyone has access to the functionings and states of being required for a good life.
- ☐ Resources, such as clean drinking water, must be evenly distributed.

The capabilities approach focuses on what people are actually able to be and to do. It therefore emphasizes that resources should be distributed in ways that enable all individuals to achieve

essential functionings and states of being necessary for a good life, rather than focusing solely on formal rights, equal resource distribution, or GDP growth.

- False
- False
- True
- False

Which of the following statements correctly distinguish between individual and structural approaches to analysing inequality? (More than one statement may be correct.)

- ☐ Structural approaches to inequality are closely aligned with welfare economics and the concept of absolute inequality.
- ☐ Structural inequality analyses are linked to concepts such as equity, the capability approach, and relative inequality.
- ☐ Individual inequality analyses assume that economic differences mainly result from personal attributes and individual effort.
- ☐ Analyses of individual inequality focus on socioeconomic characteristics such as income, education, and occupational status.
- ☐ Structural inequality is understood as the outcome of social frameworks such as positions in the mode of production, social hierarchies, or nationality.

Individual approaches to inequality emphasize differences between individuals and typically draw on welfare economics and absolute inequality, attributing inequalities largely to personal characteristics and effort. Structural approaches, in contrast, focus on the social, economic, and political frameworks that shape unequal outcomes, and are closely connected to concepts such as equity, the capability approach, and relative inequality.

- False
- True
- True
- True
- True

Which of the following best describes resilience in a social sustainability context?

- ☐ The short-term use of resources to stabilize economic growth after a crisis.
- ☐ The equal distribution of income to prevent social conflict.
- ☐ The capacity of individuals and societies to prepare for, cope with, and recover from shocks.
- ☐ The ability of societies to avoid all internal and external shocks entirely.

Resilience refers to the ability of individuals, households, communities, or entire societies to prepare for, cope with, and recover from shocks such as economic crises, natural disasters, or social conflicts. It does not imply the complete avoidance of shocks, but rather the capacity to deal with them and regain quality of life and stability.

- False
- False

- True
- False

Which strategy for building resilience focuses on long-term systemic change rather than immediate responses to shocks?

- ☐ Emergency relief measures provided immediately after a disaster.
- ☐ Risk reduction and mitigation, such as immunization or protective infrastructure.
- ☐ Coping strategies, such as relying on savings or social networks.
- ☐ Transformative strategies, such as institutional reforms and changes in social norms.

Transformative strategies aim at long-term increases in resilience by introducing far-reaching reforms or new institutions. They typically combine elements of risk reduction and coping, but go beyond short-term responses by addressing underlying vulnerabilities and changing social, institutional, or structural conditions.

- False
- False
- False
- True

Which of the following statements correctly reflect how different theories of justice assess the idea of an unconditional basic income? (More than one statement may be correct.)

- ☐ Libertarian theories generally reject an unconditional basic income because redistribution through taxation is seen as an infringement on individual freedom and property rights.
- ☐ Collective theories of justice evaluate an unconditional basic income depending on historical, cultural, and social context.
- ☐ Liberal theories may support an unconditional basic income because it can increase overall utility or strengthen equal basic freedoms.
- ☐ All theories of justice agree that an unconditional basic income is the most just form of redistribution.
- ☐ Egalitarian perspectives uncritically support an unconditional basic income because strict equality automatically guarantees justice.

Different theories of justice arrive at different conclusions regarding an unconditional basic income. Libertarian theories typically reject it due to concerns about state intervention and property rights. Liberal theories, such as utilitarianism or Rawlsian contractarianism, may support it under certain conditions, while egalitarian approaches emphasize the need to distinguish equality from equity. Collective theories stress that judgments depend on social and cultural context. There is no universal agreement across all theories.

- True
- True
- True
- False
- False

5 Economic Sustainability

Economic sustainability is a profoundly normative concept. It asks what kind of economy is socially desirable, which goals economic activity should pursue, whom it should serve, and within which social and ecological boundaries it must operate. Questions of economic growth, prosperity, work, distribution, and stability are therefore not merely technical issues, but central societal and political concerns that shape everyday life. At the same time, it has become increasingly clear that dominant economic models significantly contribute to the transgression of planetary boundaries, rising social inequalities, and the destabilization of socio-ecological systems.

Contrary to widespread belief, economic theories are not value-neutral descriptions of how the economy “works”. They are normative and performative: they shape political decisions, institutional arrangements, corporate practices, and individual action. As a result, answers to the core questions of economics – who produces and consumes what and why, how profits are distributed, and what they are used for – vary fundamentally depending on the underlying economic theory and its implicit assumptions about humans, society, and nature.

“The produce of the earth – all that is derived from its surface by the united application of labour, machinery, and capital, is divided among three classes of the community, namely, the proprietor of the land [**Rent**], the owner of the stock or capital [**Profit**] necessary for its cultivation, and the labourers by whose industry it is cultivated [**Wages**]. [...] To determine the laws which regulate this distribution, is the principal problem in Political Economy.” David Ricardo: *The Principles of Political Economy and Taxation* (1821, Preface)

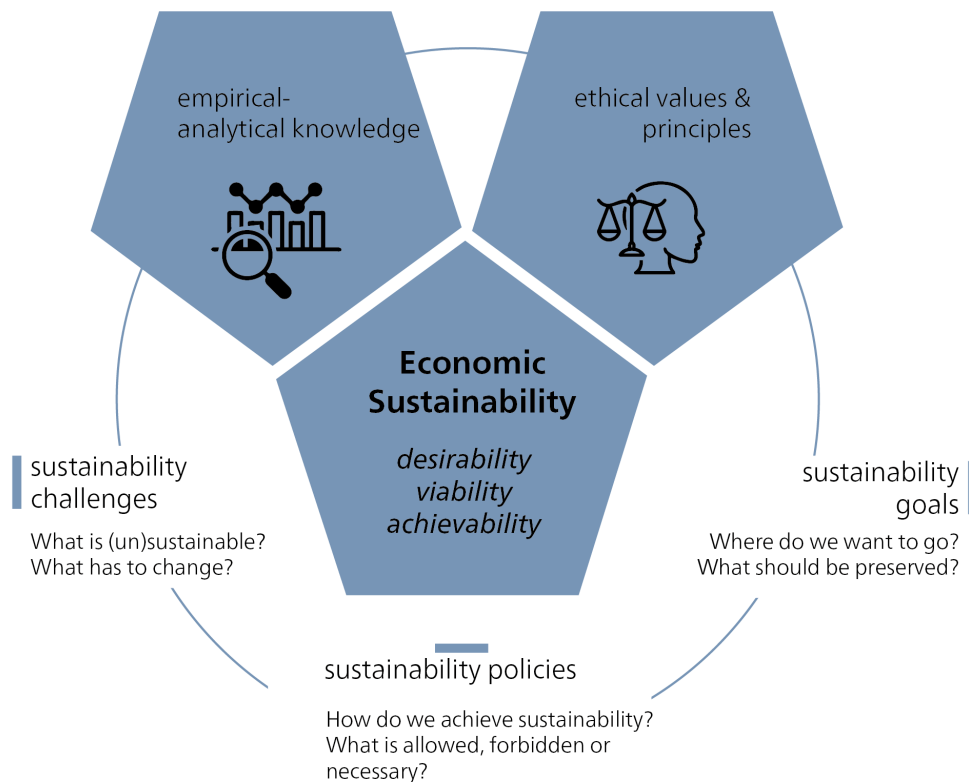


Figure 5.1: Economic Sustainability. Source: Own illustration.

Against this background, this chapter places capitalism as a historically specific mode of production and organization at the centre of analysis. Many sustainability challenges – including climate change, structural growth dependency, and financial instability – cannot be adequately understood without considering the systemic dynamics of contemporary capitalist economies. Rather than opposing “the market” and “sustainability” in a simplistic way, the chapter aims to develop a differentiated understanding of tensions, constraints, and transformation pathways towards a more sustainable economy.

5.1 Economics as a performative science

From the very beginning, modern economics has not just observed and described social reality – it has also helped shape it. This discipline has formed our thinking about what is considered “economic” – e.g. through the ideal of the rationally deciding *homo economicus*, the norm of profit maximization, the focus on efficiency, or the idea that growth is synonymous with social progress (cf. Bontrup and Marquardt 2021, 1–40).

These interpretations never remained purely theoretical. They continue to influence policy and practice to this day: from structural change to the market liberalization that took hold in the 1980s – to the shaping of climate policy through emissions trading systems and the construction of highly complex financial products. The history of economics is therefore also a history of its impact (Schneidewind et al. 2016). The most widely accepted definition of economics, originally coined by Lionel Robbins, focuses on the relationship between resource scarcity and the satisfaction of people's needs:

“Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses.” (Robbins 1932, 15).

Robbins's definition is influenced by his theoretical orientation towards neoclassical economics, the school of thought that came to be considered mainstream economics. Extending this concept, most textbooks define economics as a science of (rational) decision-making that explores how people decide to use scarce resources to achieve their aims.

Homo economicus as the foundational model of human behaviour

Homo economicus, or “economic man”, is a foundational model in neoclassical economics. It describes humans as rational, utility-maximizing actors who have access to comprehensive information and consistently make decisions to enhance their material well-being. This view of humans was decisive for the development of mathematical models in economics and the idea that markets can efficiently lead to optimal outcomes.

Historically, this concept developed over several stages. While early economists like Adam Smith still assumed a more complex view of humans – one that also included moral and social dimensions – this was increasingly reduced to purely rational, calculating behaviour in the Marginal Revolution of the 19th century. This reduction made it possible to model economic processes mathematically, but it also led to a problematic abstraction of actual social, environmental, and psychological conditions.

Critics emphasize that homo economicus is a normative construct, and not a purely analytical one. It conveys a specific idea of humans as portrayed in social models and institutions, one that regards competition, personal responsibility, and increased efficiency as natural principles. In the real world, however, people often don't behave purely rationally. Instead, factors like cooperation, fairness, trust, and even cognitive biases shape their decisions.

The homo economicus model is particularly relevant to the discussion on sustainable development. It contributes to obscuring the systemic causes of environmental destruction and social inequality, as it systematically ignores structural power relations, external effects, and long-term planetary boundaries.

The heterodox economist Ha-Joon Chang criticizes Robbins's definition as being too specific and as emerging from a theoretical approach, which subsequently causes it to prescribe a particular direction (Chang 2014). Chang defines economics not through a theoretical approach, but by its object of study: the economy. Economics is therefore the study of everything related to (re-)production, exchange, and distribution of goods and services for the satisfaction of human needs. There are of course many other definitions of economics. Building on these diverse understandings, various schools of thought consequently differ in their focus of study, which may

include power structures, institutions, or macro-economic and social connections and structures (cf. Miyamura 2020; Newman 2025).

Economic sustainability requires that we systematically consider the consequences of economic activity on humans and nature. This includes asking questions about distribution. Who bears the costs of growth, and who benefits? How much material accumulation is enough? And who decides?

An understanding of sustainability that takes these questions seriously focuses on concepts like sufficiency, minimum social standards, and planetary boundaries. It requires a departure from the dogma of a focus on growth (which primarily views growth as measured by Gross Domestic Product, see Section 5.2.4, *Economic growth*) towards a normatively grounded, reflexive, and pluralistic economy. Such an orientation cannot be achieved by economics alone – it must emerge as a democratic practice in dialogue with ethics, other social sciences, and environmental sciences.

Plural economics

Contemporary economics is largely dominated by one school of thought: neoclassical economics. At most universities and in most contexts, the term “economics” implicitly refers to this school of thought. Neoclassical economics forms the core of mainstream economics, based on the understanding of economics as formulated by Lionel Robbins. Accordingly, this school of thought examines the allocation of scarce resources and goods based on decisions made by rational, utility-maximizing individuals. The price mechanism serves as the key instrument of allocation. Supply and demand determine the optimal market price, which then ensures the optimal allocation of resources and goods.

Plural economics emphasizes that there isn't just “one” economics, but a variety of schools of thought. These include feminist, post-Keynesian, ecological, or Marxist economics, all of which differ in their theoretical assumptions, methods, and objectives. As a result, they often reach different, sometimes complementary, and at other times contradictory conclusions.

However, this is not about a wholesale rejection of neoclassical economics. Instead, the key lies in discerning which perspective best suits a given question. Neoclassical analysis provides useful insights into markets, price formation, and incentive structures. However, its utility is limited when it comes to non-monetary processes, power relations, or social reproduction work. For example, its focus on market-based exchange relationships and price mechanisms makes it difficult to take unpaid work into account. This is where alternative approaches such as feminist economics come in, as they systematically incorporate care work, social relations, and power structures into economic analysis. Consequently, plural economics advocates a methodologically open, problem-oriented approach, viewing the diversity of theoretical approaches as an enrichment – especially given the complex challenges of sustainable development.

While a detailed exploration of neoclassical economics and other schools of thought is beyond the scope of this discussion, you can delve deeper into this topic in our [Introduction to Sustainable Economics](#) course. More information on plural economics is also available on the [Exploring Economics](#) e-learning platform.

In this video – *Economics is for Everyone* – Ha-Joon Chang provides a clear and concise introduction to the idea of plural economics. He explains why economic thinking is never neutral – and why we urgently need more economic pluralism.

< <https://youtu.be/NdbbcO35arw?si=VlUrTTLGlvQTN1vQ> >

5.2 Key elements of economic sustainability

Sustainability in the economic dimension means designing economic activities to meet human needs in the long term – for current and future generations – while respecting planetary boundaries and maintaining fair social structures. Economic activities are processes that convert natural resources into goods and services for consumption, using additional production factors such as labour and capital.

In economic theory, the result of production represents a *value*. But what is the understanding of value on which this assumption is based? Historically, the concept of value in economics

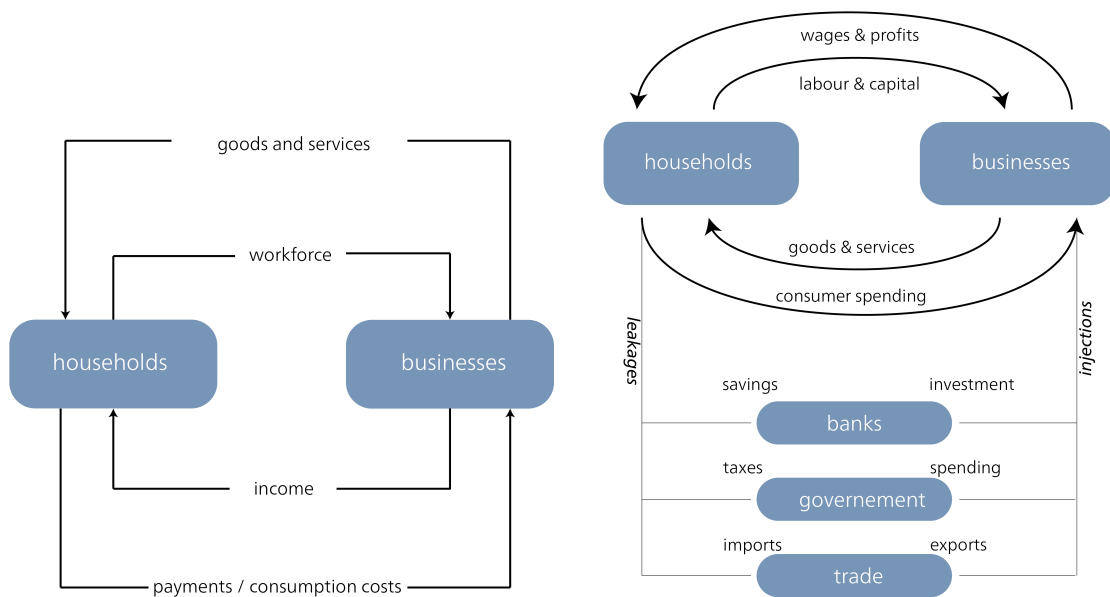
has undergone several fundamental changes (cf. Mazzucato 2018). As early as the 17th century, economic activity was often understood as serving the common good, especially when it contributed to security of supply or social stability. This understanding was closely linked to the idea of a “moral economy” (cf. Thompson 1971), which saw economic activity as embedded in traditional norms, expectations of justice, and social obligations.

The rise of merchant capitalism and colonial expansion shifted the focus of economic value attribution. During this time, wealth was increasingly epitomized by gold, silver, and other precious metals. Proponents of “mercantilist” policies – a term coined retrospectively to describe the prevailing economic practices of the 16th to 18th centuries – regarded trade as the main source of national prosperity. Traders and merchants were therefore considered “productive”, while workers and soldiers were classified as “unproductive”.

The Industrial Revolution in 18th and 19th-century Britain again reshaped the economic understanding of value: now, *labour* was considered the main source of value creation. Classical economists such as Adam Smith, David Ricardo, and Karl Marx advocated an objective theory of value, asserting that a good’s value is fundamentally determined by the labour invested in its production. These thinkers distinguished between two key concepts: use value (the practical utility of a good) and exchange value (the value that a good achieves in exchange on the market). While use value relates to the satisfaction of needs, only labour was considered the source of economic value. Adam Smith notably highlighted the division of labour as a key factor in economic progress: he argued that specialization and the functional breakdown of production processes allowed the same amount of labour to generate significantly greater output. This, in turn, spurred productivity, innovation, and ultimately, economic growth – dynamics which were further amplified by mechanization and technological innovations.

Then, in the late 19th century, yet another understanding emerged. This was the subjective theory of value, which posits that the value of a good is derived from the individual utility consumers attribute to it – irrespective of production costs or working hours. This view continues to shape neoclassical economics today, modelling economic processes as rational exchanges between producers and consumers.

This is also the basis for the simple economic cycle, which reduces the complex processes of an economy to two main actors – households and businesses – and two main flows – goods/services and money. Households provide businesses with their labour in exchange for income; businesses in turn use this labour and other factors of production to produce goods and services, which are then consumed by households. Expanded economic cycles include other actors such as the state, banks, and foreign trade.



(a) **Basic economic cycle** – Flows of goods and money between households and businesses: labour in exchange for income, consumer spending in exchange for goods.
 (b) **Extended economic cycle** – Includes banks, government, and foreign trade, showing how savings, taxes, and imports act as leakages, while loans, government spending, and exports function as *injections*.

Figure 5.2: Source: Own illustrations.

5.2.1 Critique of conventional economic cycle analysis

Conventional analyses of the economic cycle typically depict the market economy as a value-creating unit, with households portrayed primarily as consumers. Large parts of society – especially unpaid work and processes of (re-)production outside of the market logic – are largely ignored and thus implicitly devalued. Feminist economics, therefore, complements its depiction of the paid economy by including unpaid areas (see e.g. Elson 2000). Nonetheless, Mariana Mazzucato (2018) reminds us to always consider the social context in which such models emerge:

“It is crucial to remember that all types of accounting methods are evolving social conventions, defined not by physical laws and definite ‘realities’, but reflecting ideas, theories, and ideologies of the age in which they are devised.” (Mazzucato 2018, 76)

This allows us to understand why Paul Samuelson’s famous circular flow diagram of the late 1940s – influenced by the experiences of the Great Depression and the Second World War – focused largely on the uninterrupted flow of income to promote economic growth. Subsequent models of the economic cycle therefore focused almost exclusively on the flow of money between households and companies.

In contrast, Kate Raworth (2017b) illustrates the economy as embedded in society and the environment. She depicts households, the market (companies), states, and the commons as

interacting with one another, and explicitly considers ecological boundaries. This more holistic understanding is key to rethinking economic activities in terms of sustainable development.

Work is more than paid work

Work is a central production factor of each economy and necessary for transforming natural resources into goods and services. How this work is organized has varied throughout history, depending on the available tools and technologies as well as on the existing forms of organization and governance. Work is therefore context dependent.

Neoclassical economics typically focuses on paid work, which only attained its key position in industrial capitalism in the 19th century and consolidated it in the 20th. Unpaid work (e.g. domestic, care, nursing, voluntary), which is typically carried out by women, is largely ignored or taken for granted. The number of unpaid hours has been recorded in Switzerland since 2000, revealing that the amount of time spent on unpaid work exceeds that of paid work (FSO, n.d.).

From the perspective of economic sustainability, this narrow approach is problematic. After all, work is not just an input factor but constitutes economic activity itself – it enables production, reproduction, and social participation. A sustainable economy therefore requires a wider understanding of work, one that gives equal consideration to both paid work and unpaid reproductive work.

5.2.2 States, markets, and networks

Economic systems differ according to how they answer fundamental questions such as this: “Who produces what, why, and for whom?”. The two key dimensions here are ownership (private or public?) and coordination (centralized or decentralized?). From an “ideal-typical” perspective, the following basic forms emerge (Basseler 2002):

- **Capitalist market economy:** Here, coordination is decentralized through markets, driven by the interplay of supply and demand (“bottom up”). The means of production are predominantly privately owned.
- **Socialist, centralized economy:** In this model, production and distribution decisions are centrally controlled through state planning (“top down”). The means of production are generally publicly owned.
- **Mixed economy (e.g. welfare state):** These systems combine private ownership of the means of production with state coordination in key areas. While resource allocation is largely decentralized via markets, the state intervenes to regulate and distribute them – for example through social insurance, public services, or progressive taxation policies. This model aims to correct market failures and promote social justice. Coordination is therefore partly decentralized via markets and partly centralized via political institutions.
- **Community-based networks and commons economies:** These focus on self-organization, decentralized cooperation, and collective ownership. Examples include solidarity economies, cooperatives, and communally managed resources (commons).

Historically, industrial capitalism was – at least initially – the predominant model following the Industrial Revolution. It took on various forms in the 20th century, from the strong national welfare state of the postwar period to the market-oriented globalization models that have prevailed since the 1980s. Hall and Soskice (2001) describe these as “varieties of capitalism”.

Today it’s easier to imagine the end of the world than the end of capitalism, as Slavoj Žižek (1994, 1) puts it. And yet, increasingly, alternatives are emerging that show that a solidarity-based economy does not have to be limited to local niches. Instead, such an economy can be imagined and shaped globally, and move beyond market-centred or state-dominated paradigms. Networks, commons, and alternative economic forms are gaining attention, as they open up new ways of organizing economic processes in a more democratic, resilient, and environmentally sustainable way. For example, in *The Zero Marginal Cost Society*, Jeremy Rifkin (2014) envisions collaborative commons – jointly managed resources made possible by digital platforms and decentralized cooperation.

5.2.3 Capitalism as the starting point for economic sustainability issues

Many sustainability challenges – e.g. the climate crisis, social inequality, resource overuse, financial market turbulence – are not isolated phenomena. They’re an expression of an overarching context, i.e. the structural dynamics of modern global capitalism. This textbook therefore takes a systemic perspective, as adopted by academics like Polanyi (1944), Brand and Wissen (2017), Hickel (2020), Seidl and Zahrnt (2010), Schneidewind et al. (2016), and Binswanger (2006). From this perspective, moving towards sustainability requires an examination of our underlying economic structures and their political and cultural prerequisites – and thus of the capitalist system of today.

In this textbook, we view capitalism not just as an economic system, but as encompassing the current economic and social system in its entirety. This includes economic, social, legal, cultural, and environmental structures and dynamics, all of which interact. Capitalism is not a natural state of economic activity, but rather the outcome of a specific historical development. Its roots go back to late medieval Europe, particularly England, where key institutional foundations were laid through the appropriation of land (a practice known as “enclosure”), the emergence of a free labour market, and early forms of capitalist agriculture (see Wood 2017).

The Industrial Revolution in the late 18th and early 19th centuries ultimately enabled the historic breakthrough of capitalism: technological innovations, fossil fuels, and extensive capital investment gave rise to a new production system based on mass production and mass consumption. This process consolidated earlier developments into a dominant economic and social system that still characterizes key global structures today.

This system is known as the capitalist mode of production, and its underlying logic is characterized by three basic features:

- the separation of capital and labour, so that workers generally don’t own the means of production and are therefore forced to sell their labour for wages
- private ownership of the means of production
- an orientation towards profit maximization and profit utilization

Capitalists – as owners of the means of production – bring together capital, land, and labour to produce goods and services in a production process. The aim is not primarily to satisfy needs, but to generate an exchange value that can be achieved on the market – with the main purpose of increasing capital. The accumulation of capital is the key driver and structural logic of the capitalist mode of production. A key characteristic of this process is the generation of surplus value, which is the economic value workers create in production that exceeds their wages. This surplus value forms the basis for profits and is distributed in various ways:

- One part goes towards workers' wages
- One part goes to owners of land or natural resources as ground/land rent
- One part goes to the financial sector as interest and financial rents, especially where the capital is funded through debt
- The remaining surplus value comprises the profits, which can also be complemented through monopoly rents (e.g. through market power) or brand rents (e.g. through intangible assets such as reputation)

Reducing unit costs plays a key role in making production profitable and ensuring competitiveness. Technological innovations and investments in machinery, factories, and infrastructure significantly increase productivity. Energy is increasingly replacing human labour; standardized production processes like the assembly line boost efficiency and enable the mass production of inexpensive consumer goods. As Henry Ford is reported to have pointedly said: "Any customer can have a car painted any color that he wants, so long as it is black."

But achieving a profit requires more than just technological efficiency. Other critical factors include control over markets and value chains, access to cheap resources, the ability to discipline and control labour, and the design of institutional and policy frameworks. This means that profits are created on the back of unequal power relations between companies and workers, between global centres and peripheries, along supply chains, and within ownership structures.

Ensuring that the vast quantities of goods produced can also be sold requires homogenous mass demand. This is fostered through rising wages, a culture of consumption, and a social model that equates the possession of standardized consumer goods with prosperity and social advancement.

What is capitalism?

Capitalism is a historically developed form of society where production is primarily for the market, private ownership of the means of production predominates, and the accumulation of capital is essential. Its fundamental dynamic is the use of capital: as described by Karl Marx, money (M) is invested in goods or commodities (C), which are then sold for a profit, resulting in more money (M').

The capitalist mode of production is based on the separation of labour and the means of production. This means that people have to sell their labour to gain access to resources. This logic creates structural dependencies that extend beyond the economic system into all areas of society.

Historically, capitalism did not establish itself “naturally” but was enforced through processes of political expropriation, colonial expansion, and legal transformations (e.g. abolition of the commons, monetization of labour, compulsory schooling for discipline).

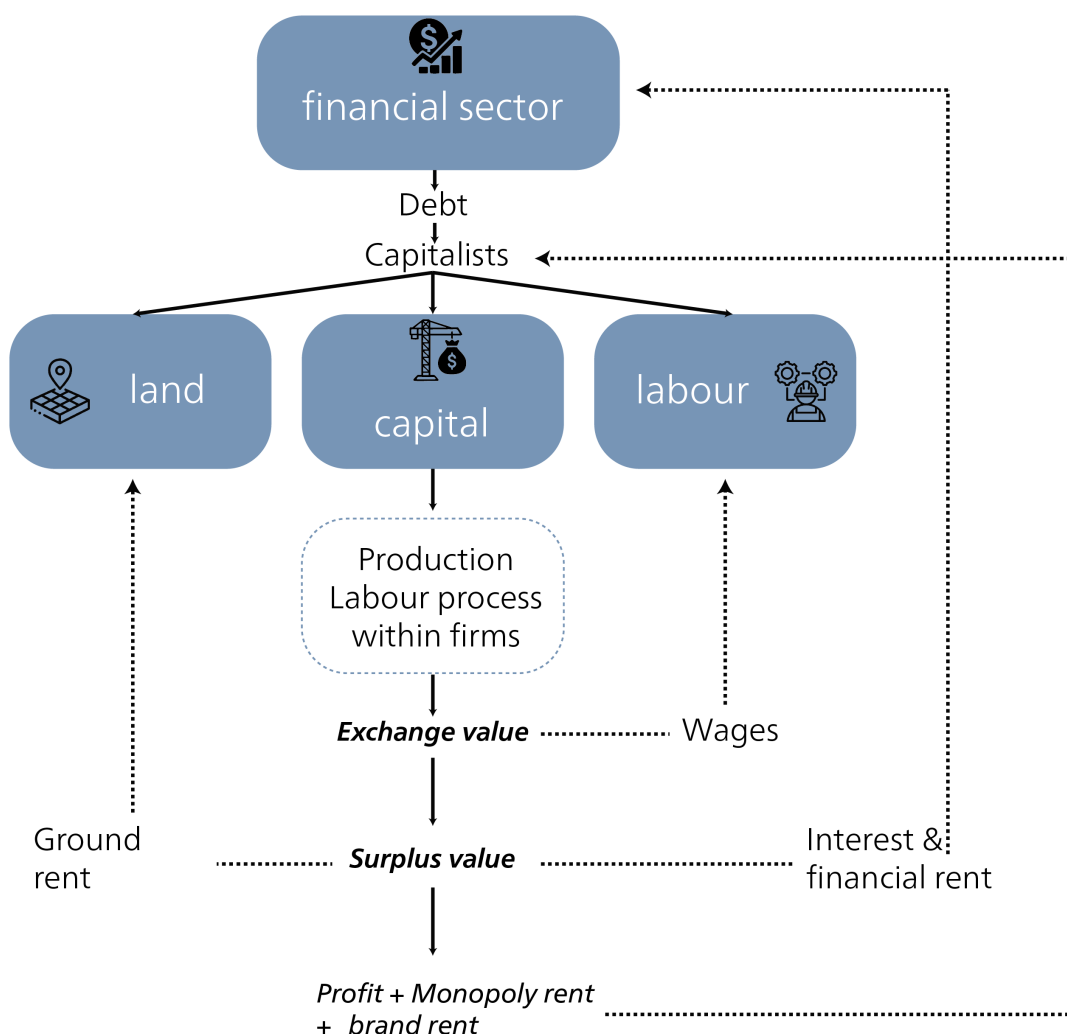


Figure 5.3: Capitalist mode of production and distribution of surplus value. The surplus value created by workers is generated in the production process and skimmed off as profit, rent, or interest. Capitalists pursue the aim of increasing capital by combining capital, land, and labour, and with the support of lending from the financial sector. Source: Own illustration, adapted from Varoufakis (2024).

Figure 5.3 illustrates how the capitalist production system produces not only goods but also social relations. In other words, those with access to capital or land can skim off surplus value, while those who only own their labour usually only receive wages. The financial sector plays an increasingly significant role in this by organizing capital flows and gaining additional access to the production process through debt (see Section 5.2.6, *Financialization and financial crises*). This mode of production therefore has an impact far beyond the economic sphere. The Hungarian economic

sociologist Karl Polanyi described this process as the transition to a market society: a society in which the markets are no longer embedded in social and political relationships but increasingly determine these themselves. Labour, land, and money became “fictitious commodities” (Polanyi 1944) – i.e. goods that were not originally intended for the market but are now subject to its logic. In a market society, markets are not part of society – society is part of the market. According to Polanyi, this leads to profound social upheavals – e.g. in the form of social insecurity, ecological degradation, or growing inequality – and repeatedly gives rise to counter-movements. This dynamic is closely linked to two key mechanisms of capitalist expansion: internalization and externalization. On the one hand, resources such as nature, labour, or knowledge are appropriated without a market-mediated exchange (internalization). On the other, social costs are outsourced to third parties, such as future generations or countries in the Global South (externalization). A more detailed explanation of these two terms can be found in the box titled “Internalizing and externalizing”.

In the following sections, we analyse how the capitalist mode of production, embedded in a market society, necessitates structural growth and thus leads to significant conflicts with sustainability.

Internalizing and externalizing

The price of a good doesn't always reflect its actual social costs. If a good is produced or sold at a market price that ignores these costs, they are often passed on to third parties – such as taxpayers, future generations, or the environment. These are known as external effects or externalities, and they can be either positive or negative. Externalities influence the production and consumption options of uninvolved parties.

Neoclassical economics generally views external costs as an anomaly within an otherwise efficient market. The usual remedy is to “internalize” these costs – e.g. through mechanisms like a carbon taxes. In contrast, William Kapp argued as early as 1950 that the more a system prioritizes private profit maximization, the greater the incentives to pass costs onto people, society, and nature (Kapp 1950). Therefore, the current capitalist system has an inherent tendency to externalize costs. Stephan Lessenich describes the affluent societies in the Global North as externalization societies that outsource costs to the Global South (Lessenich 2020). At the same time, the capitalist mode of production relies on processes of internalization: nature, labour, or knowledge are often appropriated without compensation, based on structural power relations. Examples include land grabbing, unpaid care work, and the commodification of traditional knowledge (see e.g. Saave 2022).

While liberal market theories emphasize the importance of price signals, other schools of thought question whether we really need to assign a price to everything, even nature. These fundamental debates are key to achieving a sustainable economy and are examined in depth in the *Introduction to Sustainable Economics* course.

Further reading

- Polanyi, Karl. (1944) *The Great Transformation: The Political and Economic Origins of Our Time*. Beacon Press.
- Wood, Ellen Meiksins. 2017. *The Origin of Capitalism: A Longer View*. Verso.
- Göpel, Maja. 2016. 'Why the Mainstream Economic Paradigm Cannot Inform Sustainability Transformations'. In *The Great Mindshift: How a New Economic Paradigm and Sustainability Transformations Go Hand in Hand*, edited by Maja Göpel. Springer International Publishing. https://doi.org/10.1007/978-3-319-43766-8_3.

5.2.4 Economic growth: emergence, normalization, and criticism

Today, we take economic growth for granted, yet from a historical point of view, it is a recent phenomenon. For centuries, the level of economic production remained largely stable. It was not until Britain's Industrial Revolution at the end of the 18th century that a slow increase began, picking up speed significantly from the 1950s onwards. Figure 5.4 impressively illustrates how sudden and historically unique this acceleration was. For many centuries, per capita income stagnated before it literally exploded with the rise of industrial and fossil-fuelled production methods. However, this growth was concentrated mainly in some countries of the Global North. Three factors were key to this dynamic:

1. Reconstruction after the Second World War enabled massive public and private investment.
2. Access to cheap fossil fuel – especially oil – from the Middle East dramatically reduced production costs.
3. The Fordist consumer model combined high wages with cheap mass production: Henry Ford's vision of "cars for everyone" became the guiding principle of a new production and consumption logic.

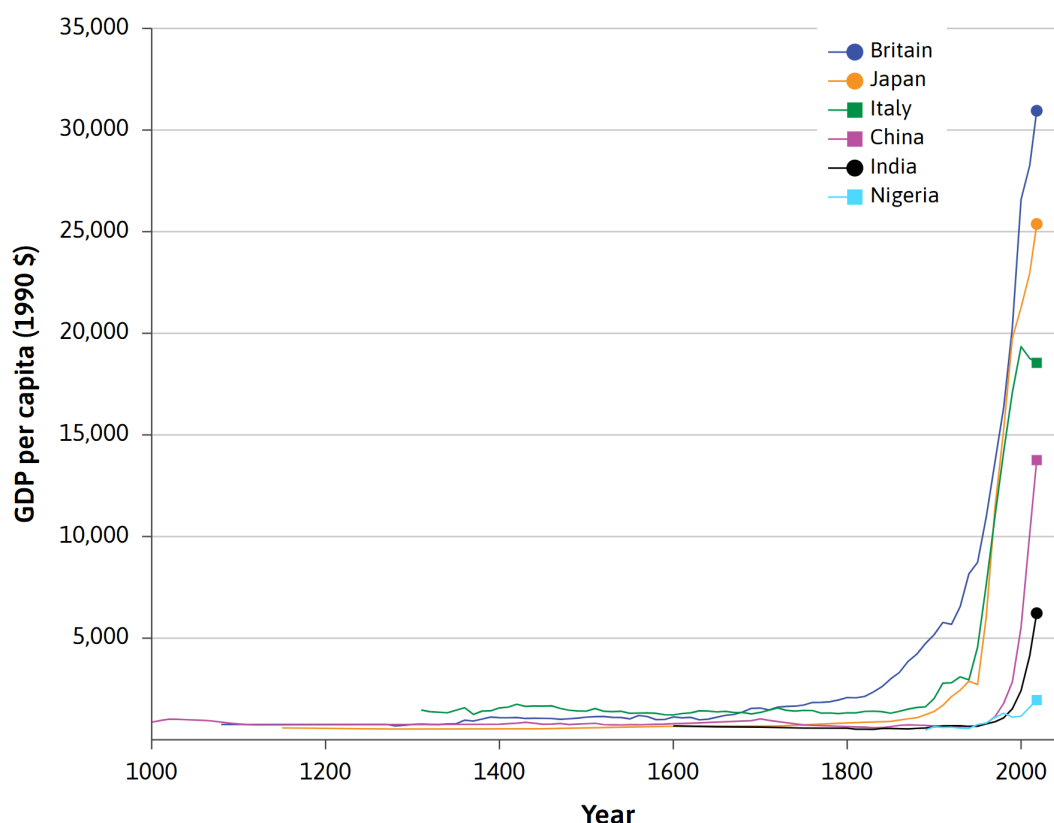


Figure 5.4: History's hockey stick: gross domestic product per capita in five countries (1000-2018). Per capita income stagnated for centuries. It was only with the Industrial Revolution and fossil capitalism that income and production volumes rose exponentially – albeit very unevenly from region to region. Source: <https://books.core-econ.org/the-economy/microeconomics/01-prosperity-inequality-02-historys-hockey-stick.html#figure-1-1>.

A key turning point was that capital replaced land as the central factor of production. While land is naturally limited, capital – understood as machines, technology, and infrastructure – initially had no obvious limits. For economists like Malthus, who still assumed that natural resources were scarce, long-term growth was almost unthinkable. Industrial production under capitalist conditions fundamentally altered this paradigm.

Economic growth led to **gross domestic product (GDP)** taking centre stage as the predominant measure of prosperity. GDP measures the monetary value of final goods and services produced in a country in a given period of time (e.g. a year), but it doesn't measure unpaid labour, ecological damage, or social justice. Since the 1980s, there has also been a divergence between rising GDP and stagnating subjective life satisfaction. GDP is well-suited to describing market activity – but not as a comprehensive indicator of prosperity or well-being.

After World War II – during the *trente glorieuses*, or “glorious thirty” (1945-1975) – steady economic growth became the new normal. During this time, a frequent topic of discussion was whether increasing prosperity could, in the long term, lead to environmental improvement. The Environmental Kuznets Curve (Stern 2004) suggests that environmental degradation initially increases in the early phases of economic growth but begins to decrease once a certain income level

is reached, as societies invest more in environmental protection. However, empirical evidence for this theory is mixed. While some local environmental issues have been managed, global environmental impacts such as CO₂ emissions continue to increase. Simply continuing to grow the economy therefore by no means guarantees reduced pressure on the environment and can further exacerbate existing planetary boundaries (Wiedmann et al. 2020). Many social subsystems, such as social security systems or labour market policy, were designed and structured to expect continuously growing government revenue, investment, and employment. This structural focus on growth continues to have an impact today, making economic sustainability a systemic issue.

5.2.5 Growth dependency and structural growth constraints

As the current capitalist economic system is structurally dependent on growth, a decline in economic activity (e.g. stagnation, recession, or even depression) would lead to an economic crisis with far-reaching consequences for the population (e.g. unemployment, poverty, social benefit cuts) (Schmelzer and Vetter 2019, 26). Without growth, the system would enter a state of crisis. Currently, there is either growth or shrinkage, but nothing in between. Matthias Binswanger (2019) demonstrates that the need for growth results from the dynamics of market competition: to remain competitive, companies must continuously expand and boost productivity – otherwise, they risk losing market share and profits, or even risk going out of business. This growth imperative is evident on a macroeconomic, corporate, and cultural level:

The macroeconomic growth imperative describes the structural dependence of modern societies and economic systems on economic growth. This growth is currently necessary because many core institutions and systems – such as labour markets, social systems, and public finances – can only function stably if there is continuous growth. For example, it is argued that without constant growth, unemployment would rise, as companies reduce production and, consequently, their workforce. Public finances would also be affected: state budgets, pensions, and social security systems are often based on continuously rising revenues generated by taxes, which in turn rely on economic growth. Last but not least, growth prevents economic and political crises, as stagnating or shrinking markets can be accompanied by uncertainty, capital flight, and social dissatisfaction, all of which facilitate political instability (Seidl and Zahrnt 2010).

The corporate logic of investment, profit, and expansion: Companies operate within a capitalist mode of production whose logic is based on continuous growth. This growth is not only desirable – it is also structurally enforced. Companies typically invest only where they expect a profit – a principle reflected in the M–C–M' formula (money–commodities–more money). This dynamic is intensified by market pressure: to remain competitive, companies must expand – or risk collapse. This “grow or die” logic is further exacerbated by the financial system. Invested capital is expected to yield continuous returns, earning interest. To fulfil these expectations, companies must constantly generate demand and invest in new production cycles (see Binswanger 2019). Companies that fail to generate profits in the long term disappear from the market. If average profits across the economy fall short, this can trigger a chain reaction of business closures – with potentially severe implications for employment and social stability.

A concise introduction to the logic of capitalism is provided by David Graeber in this short video: <https://www.youtube.com/watch?v=QOVnb3y8hm4>

Mass consumption and social norms: The growth spiral of the capitalist production system is driven not only by technological and economic factors, but to a significant extent also by cultural dynamics. Today, consumption is no longer just about satisfying immediate needs – it also fulfils a variety of social functions and has become an expression of status, self-realization, and belonging. As described in the previous section – and according to a key insight of Keynesian economists – mass demand stimulates investment, which in turn creates the basis for capitalist profits at a macroeconomic level. These connections illustrate the key role of consumption in the capitalist production system: In other words, strong consumer demand is essential for sustained economic growth. A lack of demand – e.g. if real wages stagnate or fall – would destabilize the capitalist system. In recent decades, various strategies were implemented to compensate for such shortfalls in demand:

- **Integrating women into the labour market:** In many countries, particularly in the US, women's access to the labour market increased significantly in the 1970s and 80s. This not only served to promote equality – it also helped maintain household purchasing power – as reflected in popular cultural representations of the situation, such as Dolly Parton's hit song, "9 to 5".
- **Increasing consumption on credit:** As there are natural limits to both workers and daily working hours, consumption on credit increased. Just as private households borrowed money to make up for a lack of purchasing power, so too did the state: public spending on e.g. infrastructure, social benefits, or tax relief in the 1980s was often financed through state debt to stabilize overall economic demand. The deregulation of financial markets further fuelled these developments. Since the 2008 financial crisis, central banks have also played an increasing role in stabilizing demand through monetary policy measures like low interest rates and bond purchases.
- **Combating consumer saturation:** In affluent societies, where many basic needs have already been met, the advertising industry is faced with the challenge of creating new areas of consumption. It's increasingly about creating needs rather than fulfilling them. As the economist Mathias Binswanger (2019) sums up: "We have evolved from a need fulfilment society to a need creation society," adding: "Until a few years ago, people didn't realize that they needed to record their daily heart rate, activities, calories burned, and sleep. But thanks to the efforts of health watch manufacturers, this need has been 'awakened' in them and health watches and fitness trackers are now sold in large quantities."

5.2.6 Financialization and financial crises

Initially, the financial – both money and capital – markets were closely tied to the real economy, providing liquidity for the production of goods and services. It was therefore assumed that the financial markets and the real economy would develop roughly in tandem. However, this stopped being the case several decades ago. From the 1980s onward, the money and capital markets have grown much faster, losing touch with the real economy.

One reason for this development was the deregulation of international financial markets. These deregulatory measures were introduced in the 1980s and 1990s to stimulate stagnating economic development in countries like the UK (then led by Conservative Prime Minister Margaret Thatcher)

and the US (under President Ronald Reagan, the original proponent of the “Make America Great Again” slogan). The measures were legitimized by the argument that previous crises stemmed from inefficient state regulation and intervention, which were said to have prevented the rationality of individual market participants from achieving market equilibrium. The extensive deregulation of the financial markets was intended to enable self-regulating mechanisms for economic stabilization. Deregulation involved repealing the 1933 Glass-Steagall Act, which had mandated a strict separation of credit and securities transactions. This allowed banks to conduct both businesses simultaneously, which strongly facilitated speculative activities. Speculative trading volumes subsequently skyrocketed after the turn of the millennium, amid the advent of computerized high-speed trading.

What we need to understand about money and debt

Money is far more than just a neutral medium of exchange. It fulfils four central functions in modern economic systems:

Medium of exchange: Money simplifies economic transactions by eliminating the need for simultaneous exchange of goods and services.

Store of value: Money can be saved and used later, which enables long-term planning and investment.

Unit of account (value meter): Money determines the “worth” of labour. However, this valuation is not an objective measure, but an expression of social power dynamics and ideologies.

Enabler of capital accumulation: In capitalism, money is invested to generate more money. This dynamic, if unchecked, can lead to an increasing concentration of wealth.

Through this last function, money is closely linked to debt: in today’s financial system, most money is not created through government coinage or central bank activities, but through the granting of loans by private commercial banks. This means that whenever a loan is granted, new money is created – along with a corresponding debt. Thus, every asset on the credit side has a debt on the debit side (see Graeber 2012). Without ongoing debt, there would be no money growth – and therefore no asset growth. This applies to both private households and governments. In a system of growing inequality, debtors are often structurally disadvantaged – they are dependent on lenders and subject to austerity measures and political constraints. Today’s financial system thus creates not only economic but also political dependencies. Those who have money can not only consume, but they also have power – and can exercise it, e.g. through political influence, media ownership, or strategic investments. Financialization, speculation, and global capital flows further exacerbate this development.

“The flipside of private wealth accumulation is public and private debt. Debt can only be reduced if assets are reduced at the same time.” (Holzinger 2024, 89, own translation)

Deregulation and its consequences enabled banks to significantly expand their lending – increasingly also targeting households with low credit ratings (i.e. limited repayment ability or low creditworthiness). To mitigate the associated risks, the loans were bundled and resold as seemingly safe “subprime” bonds. Such mechanisms ultimately laid the foundations for the financial crisis of 2008, when it emerged that a large proportion of these financial products were

based on high-risk lending.

This shift in economic dynamics and power towards the financial sector is known as financialization. It created a power complex comprising central banks, commercial banks, other financial institutions, private pension funds, and the owners of various assets. In this process, financial markets grow disproportionately compared to the real economy, a trend facilitated by national deregulation measures. Financialization intensifies the growth imperative, increasing not just inequality but also ecological fragility, for example through speculative investments in resources, raw materials, or land. Examples such as derivatives trading in food or resources show how financialized markets can exacerbate social vulnerabilities and ecological risks.

What are derivatives and how do they contribute to financialization?

Derivatives are financial instruments whose value is derived from the future rates or prices of other goods, assets, or reference values such as interest rates, commodity prices, or stock market indices. Common derivatives include options, futures, and swaps. Derivatives trading does not involve real economic production – no goods or services are created. Instead, derivatives were originally intended to “hedge” against price fluctuations, e.g. in commodities trading. However, the increasing deregulation and technologization of the financial markets led to the widespread use of derivatives as speculative investment tools. This significantly contributed to the decoupling of the financial markets from the real economy, fostering a massive market for financial bets often vastly exceeding the real economic values on which the derivatives are based. This dynamic is a key component of “financialization” and increases economic instability as well as social and ecological risks.

Further reading

- Blakeley, Grace. 2019. *Stolen: How to Save the World from Financialisation*. Repeater.

5.2.7 Companies as key players in economic sustainability

Companies play a key role in economic processes: they produce goods and services, create jobs, and drive innovation – all the while being significant sources of environmental harm. In the context of economic sustainability, therefore, we should view companies not just as part of the problem, but also as part of a possible solution (Hoffman 2018).

Companies are organizations that combine production factors (labour, capital, land) to produce goods and services, typically for sale in the market. Traditionally, their overriding goal has been to generate more income than expenses, thereby making a profit. This classical understanding was heavily influenced by the “shareholder doctrine”, notably articulated by Milton Friedman in his much-cited 1970 essay in the New York Times, “The Social Responsibility of Business is to Increase its Profits”. Friedman argued that the only social responsibility of a company was to maximize profits for its owners – the shareholders – within the bounds of applicable laws and market rules. Managers, therefore, owe primary allegiance to shareholders, but not to social or ecological objectives. This view formed the basis for many economic policy models and management practices

in the latter half of the 20th century. It contributed significantly to companies' strong focus on short-term returns, increased efficiency, and cost reduction – often at the expense of social justice or ecological sustainability. Critics contend that this logic systematically ignores crucial social and planetary boundaries, shifting responsibility primarily to individuals or states.

In the context of sustainability, this narrow focus is under increased scrutiny. Alternatives like the “stakeholder approach” are gaining importance, viewing companies not only as profit maximizers for owners but also as socially embedded actors responsible to diverse stakeholders, including employees, customers, suppliers, local communities, and the environment. Companies also vary significantly in size and organizational form, encompassing small and medium-sized enterprises (SMEs), large corporations, social enterprises, cooperatives, and other alternative business models. This diversity profoundly influences how sustainability is strategically embedded and practically implemented.

Dyllick and Muff (2016) categorize companies into three stages of development, based on their position on sustainability. The first stage, “business-as-usual”, describes companies that largely ignore sustainability and focus their economic activities exclusively on financial objectives. In the second stage, “Corporate Sustainability 1.0 and 2.0”, companies begin to address sustainability through efficiency improvements, compliance (i.e. adherence to legal and regulatory requirements), and the integration of corporate social responsibility (CSR) into their business strategies. However, this typically occurs without fundamentally changing the core business logic. Only in the third stage, “Sustainability 3.0”, does systemic transformation of the company take centre stage. Here, companies follow a broader purpose: they explicitly commit to social well-being and to respecting planetary boundaries, comprehensively integrating these goals into their core business model.

The third stage requires a fundamental change in perspective: while companies in the earlier stages primarily acted from the inside out by slightly adapting their existing structures, Sustainability 3.0 demands an “outside-in” approach: companies must align their identity and strategic core processes with pressing social and ecological challenges.

And still, many sustainability efforts only scratch the surface. Typical challenges include “cherry-picking”, which involves celebrating easily achievable sustainability goals while ignoring the fact that systemic changes have failed to materialize; rebound effects, in which efficiency increases lead to rising consumption; and carbon tunnel vision, a narrow focus on reducing CO₂ emissions as the sole solution to sustainability issues.

5.3 Sustainability strategies and policies for a sustainable economy

An economic transformation towards sustainable development requires targeted strategies that question and rearrange existing growth logics, power relations, and market mechanisms. Sustainability strategies in the economic dimension typically focus on three fundamental principles: efficiency, consistency, and sufficiency. Efficiency aims for a more resource-efficient use of existing technologies, while consistency aims to gear economic processes more strongly to natural biogeochemical cycles, for example through a circular economy or renewable energy. Sufficiency goes beyond that and calls for a limit to consumption and production as well as a

new understanding of wealth, one that is based on “enough” instead of “more”. While efficiency and consistency are largely realizable within existing market logics, sufficiency requires a deeper rethinking about social values and justice.

More recently, many specific reform proposals have been formulated, addressing different leverage points in the economy, such as production, consumption, work, or financing. Holzinger (2024) discusses the need for an economic paradigm shift, outlined through various thematic “transformations”:

- **transformation of work**, e.g. through working time reduction, expansion of the care economy, and recognition of unpaid work;
- **transformation of taxation**, aimed at environmental incentive impacts and a socially just redistribution;
- **financial transformation**, focusing on the common good, democratic control, and public investments;
- **corporate transformation**, focusing on transformative companies that put social goals before profit;
- **consumption transformation**, focusing on the reduction of resource use through cultural transformation and infrastructure for less (e.g. sharing, repair); and
- reforms in specific sectors: e.g. **mobility transformation, energy transformation, food transformation**.

These reform proposals make clear that sustainability requires a combination of different sectoral levers: rather than purely technical solutions, there is a need for new institutional, social, and cultural guardrails. In the economic sustainability debate, these diverse proposals can be categorized into three overarching macrostrategies with different basic assumptions and goals:

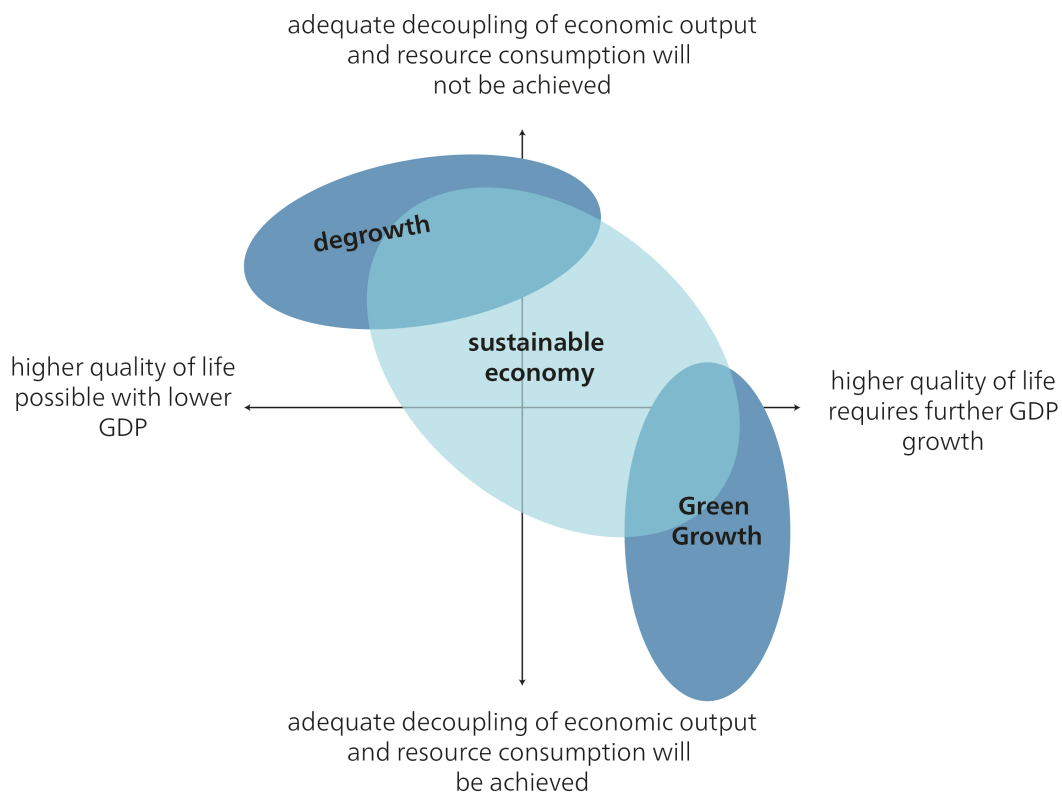


Figure 5.5: Positioning of Green Growth, Degrowth, and Sustainable Economy based on their assumptions about decoupling and the potential for high quality of life with lower GDP. Source: Own illustration based on Petschow et al. (2018).

- **Green Growth** focuses on a relative or absolute decoupling of economic growth and environmental use. Green technologies, efficiency increases, and environmental innovation management serve to respect environmental boundaries – without fundamentally calling into question the growth paradigm.
- By contrast, **degrowth (also called post-growth¹)** specifically aims to reduce economic activities in sectors that cause particular environmental harm. It seeks not only to undo resource-intensive production and consumption patterns, but also to achieve a far-reaching social transformation towards new models of wealth, time wealth, justice, and sufficiency. It focuses on the democratic negotiation of collective needs and the fair distribution of work, income, and resources.
- **A sustainable economy** strikes a balance between the two predominant approaches of green growth and degrowth. It strives for social well-being within planetary boundaries without

¹The term post-growth is also frequently used. Although “degrowth” and “post-growth” overlap in many requirements and analyses, the term post-growth more strongly emphasizes the need for institutions that are independent of growth, rather than primarily focusing on economic shrinkage.

dogmatically committing to either economic growth or shrinkage. Petschow et al. (2020), for example, describe this position as a “precautionary post-growth approach”.

5.3.1 Sustainable economy – growth-independent economic sectors

Many of modern society’s core institutions – including pension and social systems, labour markets, and public finances – are currently closely linked to economic growth. As Seidl and Zahrnt (2010) emphasize, these structures are typically based on implicit growth assumptions that are neither sustainable nor socially just in the long term. Designing growth-independent institutions is therefore key. This would involve not only financially decoupling social systems and public finances from GDP growth, but also overcoming the political fixation on growth that currently characterizes many economic policy discourses. Reforms can be implemented, for example, by designing new tax or financial policy frameworks that enable stability even without growth. In parallel, we need new types of production and consumption:

- **A sufficient lifestyle.** A cultural and infrastructural transformation towards sufficient lifestyles is not just about refraining from consumption at an individual level, but also about supporting new production and consumption patterns – e.g. through sharing models, a repair culture, regional supply chains, or the collective use of resources (see, among others, Paech 2012).
- **Collective action.** The development of alternative infrastructure, solidarity-based forms of cooperation, and non-capitalist means of productions – e.g. commons, cooperatives, or cooperative networks – are described as viable prospects for the future by authors such as Muraca, Rifkin, and Ostrom. Such forms of economic self-organization strengthen resilience, social cohesion, and environmental responsibility.
- **A caring economy.** Work in households, in care, and in the community – in other words, reproductive and care activities – must be systematically upvalued. Instead of considering production and reproduction separately, approaches like those of Biesecker and Hofmeister emphasize the need to understand care work as the foundation of economic activity,
- **Overcoming the “imperial mode of living”.** From a global perspective, the sustainability discussion is further complemented by criticism of the “imperial mode of living”. Authors such as Brand or Hickel demonstrate that the growth-based way of life in high-income societies is based on the externalization of environmental and social costs to the Global South. A sustainable transformation must therefore also critically question international power relations and access to resources.

The development of growth-independent economic sectors requires not just new practices and cultural models, but also a targeted policy framework that supports and consolidates this transformation. Without suitable tax-related, regulatory, and investment-related frameworks, sufficient lifestyles, caring economies, or commons-based production often remain limited to niches. The task of sustainable economic policies, therefore, is to create favourable structures, overcome destructive path dependencies, and systematically enable new economic logics. The following key policy instruments and macroeconomic levers could accompany this transformation at the national and international level:

- **Shifting taxation from work to resources and capital.** While the tax burden in many countries is currently strongly focused on the taxation of income from paid employment, natural resource use and massive wealth accumulation are largely undertaxed. Ecological incentive taxes (e.g. CO₂ levies or resource taxes) can be used to internalize environmental costs and disincentivize environmentally harmful behaviour. At the same time, progressive taxation of income, inheritance, and wealth can create financial scope for social and environmental investments. This would also strengthen distributive justice, incentive structures, and the state's capacity to act.
- **Reforming the money creation system.** This could take the form of a state monopoly on money creation ("Positive" or "100% Money") or mission-oriented lending ("Mission Economy", see Mazzucato 2022). The purpose of such measures would be to link money creation to social priorities – such as climate protection or care investments – and to avoid speculative misallocation.
- **Supporting social innovation in a targeted way.** The aim of this measure is to embed sufficiency and resilient structures in the economy and society. This would allow new institutions, forms of organization, and business models to emerge that don't rely on constant growth, focusing instead on moderation, justice, and environmental limits.

5.3.2 Corporate transformation – transformative companies and a sustainable economy

Given that the sustainability challenges we face are systemic, we need companies that go far beyond efficiency increases: in other words, companies that are transformative. Hug et al. (2022) describe transformative companies based on a synthesis of the nine key dimensions that resulted from their literature review:

"Transformative enterprises are pioneering SMEs who strive for fundamental changes towards sustainability. They have a social and/or ecological (1) driving mission and are oriented along the values of (2) stability and autonomy. Inside the enterprise, they implement these values through minimizing their (3) ecological footprint, assuming (4) social obligations, introducing (5) participatory governance structures, and offering (6) alternative products and services. The enterprise's core values define how it interacts with stakeholders: transformative enterprises put (7) people before profit, emphasize (8) regional embeddedness and act as (9) change agents. By spreading their vision and taking initiative for industry changes, they trigger or facilitate transformation processes and thereby contribute to sustainable, future-proof economic practices." (Hug et al. 2022, 11)

However, the path to becoming a transformative company presents many obstacles. These include the current economic system's systemic growth imperative, social discourses that individualize responsibility and conceal structural causes, and individual cognitive biases such as loss aversion or habitual inertia. Uncertainties in supply chains and among consumers also make it more difficult for companies to act sustainably.

At the same time, there are factors that can promote transformative entrepreneurship. These include a strong inner conviction, committed leaders, networks of like-minded actors, the creation

of suitable policy frameworks, and a growing social demand for sustainable products and services (Niessen and Bocken 2021).

Kate Raworth's (2017b) concept of Doughnut Economics offers an innovative framework for designing sustainable companies. It posits that companies should operate within the planetary boundaries while simultaneously securing a social foundation. To achieve this, they must critically examine and transform their design across five specific fields: their purpose, networks, governance structures, ownership, and financing logic (DEAL, n.d.). Only when fundamental change is achieved in these fields can companies truly become regenerative and distributive.

Transformative companies are no longer a peripheral phenomenon. Instead, they are necessary actors for a sustainable economy. However, their success depends not only on internal reforms, but also on social support and an appropriate policy framework.

5.4 Sustainability Basics: The Quiz – Chapter 5

Which of the following statements correctly describe economics as a performative science? (More than one statement may be correct.)

- ☐ Economics is a purely descriptive science that has no influence on political or economic decision-making.
- ☐ The history of economics is also a history of its practical impacts on markets, policies, and societal structures.
- ☐ Concepts such as homo economicus, profit maximization, and efficiency have influenced real-world economic institutions and policies.
- ☐ All definitions of economics are theory-neutral and independent of underlying schools of thought.
- ☐ Economics not only describes social reality but also shapes economic thinking, policy, and practice.

Economics has a performative character in that it actively shapes how societies understand and organize economic activity. Core concepts such as rational decision-making, efficiency, and growth have influenced policies and practices, from market liberalization to climate policy instruments. Different definitions of economics reflect distinct theoretical orientations, and the discipline's history is closely intertwined with its real-world effects.

- False
- True
- True
- False
- True

Which of the following statements correctly describe different types of economic systems and their modes of ownership and coordination? (More than one statement may be correct.)

- ☐ Community-based networks and commons economies rely primarily on centralized state control and private ownership.

- ☐ Mixed economies combine private ownership with state intervention to regulate markets and promote social justice.
- ☐ The concept of “varieties of capitalism” highlights that capitalist systems have taken different institutional forms over time and across countries.
- ☐ In a capitalist market economy, coordination is largely decentralized through markets, and the means of production are predominantly privately owned.
- ☐ In a socialist, centralized economy, production and distribution are mainly coordinated through state planning, and the means of production are generally publicly owned.

Economic systems differ in how ownership and coordination are organized. Capitalist market economies rely on decentralized market coordination and private ownership, while socialist economies are characterized by centralized state planning and public ownership. Mixed economies combine market mechanisms with state intervention to address market failures and promote social goals. Community-based and commons-based economies emphasize self-organization and common property rather than centralized control. The notion of “varieties of capitalism” captures the diversity of capitalist models that have evolved historically.

- False
- True
- True
- True
- True

Which of the following statements correctly describe capitalism as a starting point for analysing economic sustainability issues? (More than one statement may be correct.)

- ☐ Capitalism is viewed as a historically developed economic and social system rather than a natural or universal form of economic organization.
- ☐ Many contemporary sustainability challenges, such as climate change and social inequality, are linked to the structural dynamics of modern global capitalism.
- ☐ The Industrial Revolution played a key role in consolidating capitalism through technological innovation, fossil energy use, and mass production.
- ☐ The capitalist mode of production is characterized by the separation of capital and labour, private ownership of the means of production, and profit maximization.
- ☐ Sustainability transformations can be achieved without examining underlying economic structures, as environmental problems are largely isolated phenomena.

From a systemic perspective, sustainability challenges are understood as expressions of the structural dynamics of global capitalism rather than isolated problems. Capitalism is seen as a historically specific economic and social system that emerged through particular institutional developments and was consolidated during the Industrial Revolution. Its core features include the separation of capital and labour, private ownership of the means of production, and an orientation towards profit maximization. Ignoring these structures limits the ability to address sustainability challenges effectively.

- True
- True

- True
- True
- False

Why did economies (measured by GDP per capita) not grow significantly before the 19th century? (More than one answer may be correct.)

- ☐ Governments deliberately restricted economic growth to avoid inflation.
- ☐ Most societies were agrarian, with low productivity and little technological innovation.
- ☐ Economic production was largely limited by land and natural resource availability.
- ☐ Global trade and industrialization had not yet developed on a large scale.

Before the 19th century, most economies were characterized by agrarian production systems with low productivity and slow technological change. Economic output was strongly constrained by land and natural resource availability, and large-scale industrialization and global trade had not yet emerged. There is no historical evidence that governments systematically restricted growth to prevent inflation.

- False
- True
- True
- True

What is the main critique of GDP as a measure of prosperity?

- ☐ It includes unpaid work and environmental degradation.
- ☐ It measures only financial market performance.
- ☐ It ignores unpaid work, inequality, and environmental damage.
- ☐ It accounts for both well-being and happiness.

GDP measures the monetary value of goods and services produced in an economy, but it does not account for unpaid work, social inequalities, or environmental degradation. As a result, it provides a limited and potentially misleading picture of overall prosperity and well-being.

- False
- False
- True
- False

What is meant by profit maximization?

- ☐ Firms aim to sell at the lowest possible price.
- ☐ Firms seek to achieve the highest possible return on investment.
- ☐ Firms focus on social well-being.
- ☐ Firms invest only in public goods.

Profit maximization refers to the objective of firms to maximize the difference between revenues and costs, thereby achieving the highest possible return on investment. It does not primarily aim at minimizing prices, promoting social well-being, or investing exclusively in public goods.

- False
- True
- False
- False

What is meant by the term financialization?

- ☐ The growing importance of financial markets, institutions, and motives in the economy.
- ☐ The total elimination of private credit.
- ☐ The replacement of banking systems by cooperatives.
- ☐ The withdrawal of finance from the real economy.

Financialization refers to the increasing dominance of financial markets, financial institutions, and financial motives in shaping economic activities, corporate behavior, and public policy. It does not imply the disappearance of banking systems, a retreat of finance from the real economy, or the elimination of private credit.

- True
- False
- False
- False

Which of the following statements correctly describe growth dependency and structural growth constraints in capitalist economies? (More than one statement may be correct.)

- ☐ Economic stagnation or decline in growth can trigger crises such as unemployment, poverty, and social benefit cuts.
- ☐ Companies can remain profitable and competitive in the long term without generating profits or expanding production.
- ☐ The logic of capitalist investment is often described by the formula $M-C-M'$, reflecting the aim of turning money into more money.
- ☐ Modern capitalist economies are structurally dependent on continuous economic growth to maintain employment, social security systems, and public finances.
- ☐ The growth imperative results partly from market competition, as firms must expand and increase productivity to remain competitive.

In capitalist economies, continuous economic growth is structurally required to ensure the stability of labour markets, social systems, and public finances. Market competition and financial pressures reinforce a “grow or die” logic at the corporate level, captured by the investment formula $M-C-M'$. When growth stalls or reverses, economic crises can emerge with significant social and political consequences. The remaining statement contradicts the core logic of capitalist competition and investment.

- True
- False
- True
- True
- True

Which of the following statements correctly describe the idea of a sustainable economy based on growth-independent economic sectors? (More than one statement may be correct.)

- ☐ Commons, cooperatives, and solidarity-based forms of production are seen as ways to strengthen resilience, social cohesion, and environmental responsibility.
- ☐ Many core institutions such as pension systems, labour markets, and public finances are currently structurally dependent on economic growth.
- ☐ A caring economy emphasizes the systematic upvaluation of care and reproductive work as the foundation of economic activity.
- ☐ Sufficient lifestyles focus only on individual consumer restraint and do not involve changes in infrastructure or production systems.
- ☐ Designing growth-independent institutions requires decoupling social systems and public finances from GDP growth and questioning political growth fixation.
- ☐ Sustainable economic transformation can be achieved without supportive tax, regulatory, or investment-related policy frameworks.
- ☐ Overcoming the “imperial mode of living” involves questioning global power relations and the externalization of social and environmental costs to the Global South.

Growth-independent economic sectors aim to reduce the structural dependence of societies on continuous economic growth. This involves redesigning institutions, promoting sufficient lifestyles supported by appropriate infrastructure, strengthening commons-based and cooperative forms of production, and recognizing care work as a central pillar of economic activity. From a global perspective, this also requires addressing unequal power relations and cost externalization. Such transformations depend on supportive policy frameworks; without them, alternative economic practices often remain niche phenomena.

- True
- True
- True
- False
- True
- False
- True

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